Reference textbook: Introduction to Linear Algebra, 5th Edition
Gilbert Strang

Linear Algebra

Full course handout

A4 · Two slides per page · Lin Cing

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The questions of this chapter

Linear algebra starts with two operations.

$$\text{Addition: } v + w, \qquad \text{Scalar multiplication: } cv.$$

Together they produce a linear combination

$$cv + dw.$$

A matrix records which coefficients multiply which column vectors.

$$Ax = x_1 a_1 + \cdots + x_n a_n.$$

The geometric question is which targets the combinations can reach. The algebraic question is which coefficients produce a specified target.

Manipulating vector coefficients and multiplying a matrix perform the same operation. Matrix multiplication is compact notation for a family of vector combinations.

Vectors, components, and equality

A real vector with n components can be written as a column vector

$$v = (v_1, \ldots, v_n)^T \in \mathbb{R}^n.$$

The superscript T transposes the horizontally written tuple into a column. Addition requires equal dimensions.

$$(v + w)_i = v_i + w_i.$$

Scalar multiplication acts component by component.

$$(cv)_i = cv_i.$$

The size n describes the ambient data dimension. It does not assert that a collection contains n independent directions. Span and rank answer that question.

T denotes transpose, not a variable. If the dimensions differ, $v + w$ is undefined.

Zero vectors and zero coefficients

$$0 = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

is the identity for vector addition. The scalar 0 is the identity for scalar addition, whereas 1 is the identity for scalar multiplication. The two zero objects have different types despite sharing a symbol.

Vector subtraction also acts component by component.

$$v - w = v + (-1)w.$$

In the plane, an arrow represents a vector. For $p, q \in \mathbb{R}^n$, the displacement from p to q is $q - p$. An arrow from the origin to q represents q. Translating an arrow without changing its displacement leaves the vector unchanged.

The starting point of an arrow is not part of the free vector. The vector records its direction and magnitude.

Linear combinations and span

For $v = (2, 1)^T$ and $w = (1, 2)^T$, the combinations are

$$cv + dw = \begin{bmatrix} 2c + d \\ c + 2d \end{bmatrix}.$$

To reach $b = (3, 3)^T$, solve

$$2c + d = 3, \qquad c + 2d = 3,$$

which gives $c = d = 1$.

The set of all $cv + dw$ is

$$\text{span}\{v, w\}.$$

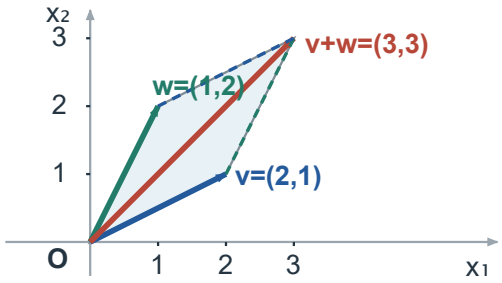
The coefficient matrix has determinant $2 \cdot 2 - 1 \cdot 1 = 3 \neq 0$. The directions are independent and their span is all of $\mathbb{R}^2$.

Geometrically, $v, w, v + w$ are two edges and the diagonal of the parallelogram with a vertex at the origin.

Linear combinations and span (continued)

Span allows every real choice of coefficients. One fixed coefficient pair produces one vector, not a span.

Parallelogram: v=(2,1), w=(1,2), v+w=(3,3)



Same vectors

v=(2,1) · w=(1,2)

c=d=1

cv+dw=(3,3)

det[[2,1],[1,2]]=3≠0

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Dependence and unreachable targets

If $w = 2v$, then $\text{span}\{v, w\} = \text{span}\{v\}$. For nonzero v in $\mathbb{R}^2$, this set is a line. If $v = 0$, the span is the zero subspace instead.

Take $v = (1, 2)^T$, $w = (2, 4)^T$, and $b = (3, 5)^T$. Every combination satisfies $y = 2x$, but $5 \neq 2 \cdot 3$. Consequently, the system

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}$$

has no solution.

Suppose scalars $c_1, \ldots, c_k$ satisfy

$$c_1 v_1 + \cdots + c_k v_k = 0$$

with at least one $c_i \neq 0$. The vectors are linearly dependent. If no such nonzero coefficient vector exists, they are linearly independent.

Inconsistency is a geometric fact about a target outside the reachable subspace, rather than a failure of calculation.

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An equation test for independence

The vectors $v_1, \dots, v_k$ are independent exactly when the homogeneous equation

$$Vc = 0, \qquad V = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$$

has only the solution $c = 0$.

For a square matrix with $k = n$, columns that form a basis make V invertible, so $Vc = 0$ has only the zero solution. If $k > n$, elimination has at most n pivot variables and therefore at least one free variable. Setting that free variable nonzero produces a nonzero solution, so more than n vectors in $\mathbb{R}^n$ cannot be independent.

For example, three vectors in $\mathbb{R}^2$ necessarily give

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$$

a nonzero coefficient solution. Existence of such a homogeneous solution differs from existence of coefficients producing a particular b.

Separate the homogeneous equation $Vc = 0$ from consistency of $Vc = b$.

Dot products and length

For $u, v \in \mathbb{R}^n$, the Euclidean inner product, or dot product, is

$$u \cdot v = u^T v = \sum_{i=1}^{n} u_i v_i.$$

The Euclidean norm is

$$\|v\|_2 = \sqrt{v^T v}.$$

A vector with $\|v\|_2 = 1$ is a unit vector. Normalize any nonzero v by forming

$$\frac{v}{\|v\|_2}.$$

For example, $v = (3, 4)^T$ has norm 5 and normalized vector $(3/5, 4/5)^T$.

In a complex vector space, the inner product $u^H v$ conjugates the left vector. Chapter 9 develops the complex formula.

Angles and orthogonality

For nonzero real vectors u, v with angle θ,

$$\cos\theta = \frac{u^T v}{\|u\|_2 \|v\|_2}.$$

For nonzero vectors, $u^T v = 0$ is equivalent to $\theta = 90^\circ$. Such vectors are orthogonal. The zero vector is orthogonal to every vector because its inner product is zero, but it has no direction, so the angle formula does not apply.

For example,

$$u = (3,4)^T, \quad v = (4,3)^T, \quad \cos\theta = \frac{24}{5 \cdot 5} = \frac{24}{25}.$$

If $w = (4,-3)^T$, then $u^T w = 12 - 12 = 0$.

Orthogonal basis vectors must be nonzero. Orthogonality itself is defined by the inner product, including at zero.

Cauchy–Schwarz and the triangle inequality

For every $u, v \in \mathbb{R}^n$,

$$|u^T v| \le \|u\|_2 \|v\|_2.$$

Proof. If $v = 0$, the result holds immediately. Otherwise let $t = (u^T v)/(v^T v)$. Then $0 \le \|u - tv\|^2 = u^T u - (u^T v)^2/(v^T v)$. Rearrangement proves the inequality. Equality holds exactly when $u = tv$. Squaring gives the equivalent statement

$$(u^T v)^2 \le (u^T u)(v^T v).$$

Equality holds exactly when u, v are linearly dependent. Applying $|u^T v| \le \|u\|\|v\|$ gives

$$\begin{aligned} \|u+v\|_2^2 &= \|u\|_2^2 + 2u^T v + \|v\|_2^2 \\ &\le (\|u\|_2 + \|v\|_2)^2. \end{aligned}$$

Thus $\|u+v\|_2 \le \|u\|_2 + \|v\|_2$. Equality requires the vectors to point in the same direction, or one to be zero. Arbitrary linear dependence does not suffice.

Taking square roots is valid because lengths are nonnegative. Preserve the direction condition when discussing equality.

Matrix multiplication combines columns

Let

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

For $x \in \mathbb{R}^n$,

$$Ax = x_1 \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}.$$

The output belongs to $\mathbb{R}^m$. Two equivalent views compute Ax.

- The column view weights column i of A by x_i and adds the results.
- The row view takes the dot product of each row of A with x.

For example,

Matrix multiplication combines columns (continued)

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

Column combinations organize later chapters. The row view explains each equation separately.

Three readings of $Ax = b$

$$Ax = b$$

The equation asks any of the following equivalent questions.

1. Which linear combination of the columns produces b?
2. Which x has dot product b_i with every row i?
3. Does the linear map $x \mapsto Ax$ send an input to b?

For $b = (0, 2)^T$ and the preceding matrix, solve

$$x_1 + 2x_2 = 0, \qquad 3x_1 + 4x_2 = 2.$$

The first equation gives $x_1 = -2x_2$. Substitution yields $-2x_2 = 2$, hence $x = (2, -1)^T$.

The shared answer $x = (2, -1)^T$ follows because the three readings describe the same equality.

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Unique and infinitely many solutions

A square matrix $A \in \mathbb{R}^{n \times n}$ has a unique solution for every b exactly when $Ax = 0$ has only $x = 0$. In that case A is invertible. Indeed, no free variables means n pivots; elimination then solves every right-hand side uniquely.

If a nonzero z satisfies $Az = 0$ and x_p solves $Ax = b$, then

$$A(x_p + cz) = b + cAz = b$$

for every scalar c. Therefore the solution is not unique.

For example,

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} x = 0$$

has solutions $x_2 = t,\ x_1 = -2t$, which form a line.

First identify the homogeneous solutions. Chapter 3 expresses the complete solution as a particular solution plus the nullspace.

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A 2×2 inverse and its conditions

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad ad - bc \neq 0,$$

the inverse is

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Direct multiplication verifies

$$\frac{1}{3} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = I_2.$$

Check the nonzero determinant condition before using this inverse. For numerical solutions, direct elimination usually costs less than constructing A^{-1}. A^{-1} is a matrix. The number $1/\det A$ is a scalar.

Coordinates in polynomial and matrix spaces

These transfer problems extend the vector operations to function and matrix spaces beyond the original Sections 1.1–1.3.

In $\mathcal{P}_2$, use the monomial basis $B = (1, t, t^2)$ and define

$$p_1 = 1 + t, \qquad p_2 = 1 - t, \qquad p_3 = 1 + t^2.$$

The coordinate map is a linear isomorphism, so

$$[p_1]_B = (1, 1, 0)^T, \quad [p_2]_B = (1, -1, 0)^T, \quad [p_3]_B = (1, 0, 1)^T.$$

Using these coordinates as columns gives

$$D = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \det D = (1 \cdot (-1) - 1 \cdot 1) \cdot 1 = -2 \neq 0.$$

Thus p_1, p_2, p_3 form a basis of $\mathcal{P}_2$. For $q(t) = 3 + 2t + t^2$, solve $Dc = [q]_B = (3, 2, 1)^T$, obtaining $c = (2, 0, 1)^T$. Hence

$$q = 2p_1 + p_3.$$

Coordinates in polynomial and matrix spaces (continued)

The same construction applies to matrices. Use column-major coordinates on $M_2(\mathbb{R})$.

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \longmapsto (a, c, b, d)^T.$$

Let $A_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$、$A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$、$A_3 = \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}$. Their coordinate columns are

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

If $c_1A_1 + c_2A_2 + c_3A_3 = 0$, comparison of entries gives $c_1 + c_2 = 0$, $c_2 + c_3 = 0$, and $c_3 = 0$. All three coefficients vanish, proving independence. An invertible coordinate map lets us test a basis even when its elements are functions or matrices.

Polynomial spectral mapping without diagonalization

Let $r(t) = \sum_{k=0}^{d} c_k t^k$, with $Av = \lambda v$ and $v \neq 0$. Since $A^k v = \lambda^k v$,

$$r(A)v = \sum_{k=0}^{d} c_k A^k v = \sum_{k=0}^{d} c_k \lambda^k v = r(\lambda)v.$$

Every $r(\lambda)$ is an eigenvalue of $r(A)$, with the same eigenvector. No diagonalizability assumption is needed. Equivalently, the polynomial identity

$$r(t) - r(\lambda) = (t - \lambda)s_\lambda(t)$$

evaluated at A gives

$$r(A) - r(\lambda)I = (A - \lambda I)s_\lambda(A),$$

so the eigenvector argument is independent of diagonalization.

Equality of the complete spectral sets also requires the reverse inclusion. Let $A \in \mathbb{C}^{n\times n}$ and let r be nonconstant. If μ is an eigenvalue of $r(A)$, factor over $\mathbb{C}$.

Polynomial spectral mapping without diagonalization (continued)

$$r(t) - \mu = c \prod_{j=1}^{d} (t - \zeta_j), \qquad c \neq 0.$$

Substitute the matrix and take determinants.

$$0 = \det(r(A) - \mu I) = c^n \prod_{j=1}^{d} \det(A - \zeta_j I).$$

At least one factor vanishes, so some ζ_j is an eigenvalue of A, and $r(\zeta_j) = \mu$. Therefore

$$\operatorname{spec}(r(A)) = r(\operatorname{spec}(A)).$$

For a constant polynomial $r = c$, use $r(A) = cI$ directly. The complete spectrum of a real matrix is also understood over $\mathbb{C}$; listing only real eigenvalues can omit conjugate pairs. This proof gives spectral sets. To count algebraic multiplicities, apply complex Schur triangularization. The diagonal entries of $r(T)$ are $r(T_{ii})$, even when T is not diagonal.

For a nondiagonalizable example, take

Polynomial spectral mapping without diagonalization (continued)

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = 2I + N, \qquad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

and $r(t) = t^2 + t + 1$. Since $N^2 = 0$,

$$r(A) = r(2)I + r'(2)N = 7I + 5N = \begin{bmatrix} 7 & 5 \\ 0 & 7 \end{bmatrix}.$$

Direct multiplication gives the same matrix, whose eigenvalue is $r(2) = 7$. The derivative term retains the off-diagonal Jordan structure and explains why evaluating a polynomial on eigenvalues alone does not determine the matrix.

Review and common misconceptions

Check that you can explain the following distinctions.

- A combination $cv + dw$ and the set $\text{span}\{v, w\}$.
- The column and row views of $Ax = b$.
- Nonzero homogeneous solutions, infinitely many solutions, and inconsistency.
- Zero dot product and linear dependence.
- Solvability for one b and solvability for every b.

Final check. If $\text{span}\{v, w\} = \mathbb{R}^2$, must v, w be independent? Yes. A nontrivial dependence would make one vector a multiple of the other, or one vector zero, leaving a span of dimension at most one. That contradicts spanning the plane.

Chapter 2 turns these decisions into systematic elimination.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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The questions of this chapter

Given $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, consider

$$Ax = b.$$

Separate four questions.

1. Does a solution exist?
2. Is the solution unique?
3. How can a finite procedure find every solution?
4. What are the arithmetic cost and numerical risks?

This chapter addresses the first three. Chapter 11 adds finite precision, stability, and condition numbers.

An absence of contradictory rows proves consistency. It does not by itself prove uniqueness.

Notation and the underlying field

- m is the output dimension, number of equations, and number of rows of A.
- n is the input dimension, number of unknowns, and number of columns of A.
- A_{ij} is the entry in row i, column j.
- e_i is standard basis vector i.
- P denotes a permutation matrix.
- $\mathbb{R}$ or $\mathbb{C}$ is the field containing the entries.

If A is $m \times n$, then x must be $n \times 1$ and b must be $m \times 1$. Check these shapes before multiplying or adding.

Dimension errors can be detected before any numerical calculation.

Two pictures of one system

Consider

$$x + 3y = 7, \qquad 2x - y = 0.$$

In the row picture, the equations are two lines in the xy plane. They intersect at $(x, y) = (1, 2)$.

Column picture:

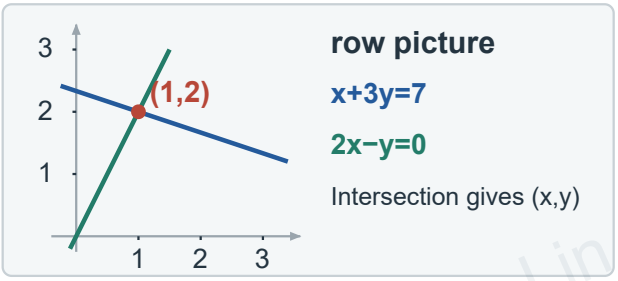
$$x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}.$$

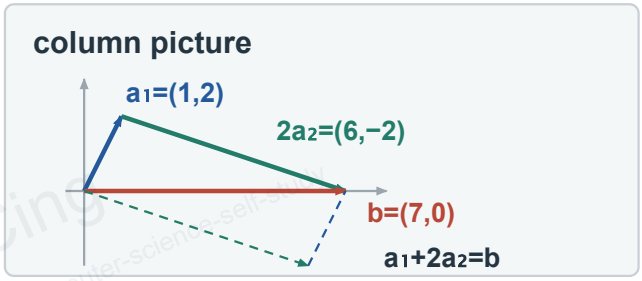
The coefficients $(1, 2)$ combine the columns into the target. Both pictures describe the same numbers. Rows express constraints, while columns express reachable directions.

In higher dimensions, the row picture uses planes or hyperplanes. The column picture still uses coefficients of vector combinations.

Two pictures of one system (continued)

One system: x+3y=7, 2x−y=0; solution (x,y)=(1,2)





Existence and uniqueness

A real linear system can have the following outcomes.

- A unique solution if every variable has a pivot and no row is contradictory.
- Infinitely many solutions if the system is consistent and has a free variable.
- No solution if a row is $[0 \ \cdots \ 0 \mid c]$ with $c \neq 0$.

For $A \in \mathbb{R}^{m \times n}$,

- If $m < n$, the homogeneous system $Ax = 0$ has a nonzero solution, but $Ax = b$ may still be inconsistent.
- Every $b \in \mathbb{R}^m$ is attainable exactly when the column space is $\mathbb{R}^m$.
- Each consistent right-hand side has a unique solution exactly when the nullspace contains only zero.

Chapter 3 formalizes rank-nullity. For now, count free variables.

A complete consistency decision

Solve

$$\begin{aligned} x &+ 2y &&= 4, \\ 2x &+ 4y &&= 9. \end{aligned}$$

Subtracting twice row 1 from row 2 gives

$$0x + 0y = 1.$$

This contradictory row proves inconsistency. In the column picture, $(4, 9)^T$ does not belong to $\mathrm{span}\{(1, 2)^T, (2, 4)^T\}$, the line $v_2 = 2v_1$.

For the right-hand side $(4, 8)^T$, elimination gives $0 = 0$. Only one independent constraint remains, giving

$$x + 2y = 4 \quad \Rightarrow \quad (x, y) = (4, 0) + t(-2, 1), \ t \in \mathbb{R}.$$

A free variable parametrizes an actual direction in the solution set.

Gaussian elimination

Elimination preserves the solution set through invertible row operations.

1. Exchange two rows.
2. Multiply a row by a nonzero scalar.
3. Add a scalar multiple of one row to another.

Choose a nonzero pivot in the first active column and eliminate entries below it. Continue with the remaining rows and columns. Skip an active column if it contains no available nonzero entry.

For example,

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 3 & 8 & 4 & 24 \\ 0 & 4 & 3 & 17 \end{array}\right]$$

Use row 1 to clear the first entry of row 2, then process the second column. Every operation preserves the equation system's solution set.

Apply an operation to both sides of an equation, rather than changing an isolated entry.

Pivots and multipliers

To use pivot p at step k to clear position (i, k), the multiplier is

$$\ell_{ik} = \frac{A_{ik}}{p},$$

and the operation is $R_i \leftarrow R_i - \ell_{ik} R_k$.

The preceding example has first multiplier $3/1 = 3$, giving

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 2 & 1 & 0 \\ 0 & 4 & 3 & 17 \end{array}\right].$$

The second pivot is 2 and its multiplier is $4/2 = 2$.

A small nonzero pivot is valid in exact arithmetic but may amplify errors in finite precision. Chapter 11 treats partial pivoting.

Back substitution and rank

Completing the preceding elimination gives

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 17 \end{array}\right].$$

Solve upward from the bottom row.

$$z = 17, \quad 2y + 17 = 0 \Rightarrow y = -17/2,$$
$$x + 2(-17/2) + 17 = 8 \Rightarrow x = 8.$$

There are three nonzero pivots, so this square matrix has rank 3 and the solution is unique. A zero coefficient row with zero right-hand side imposes no new constraint. A nonpivot variable remains free in a consistent system.

Rank is the number of nonzero pivots. Ordinary row echelon form already identifies pivot column indices. A basis for the original column space uses the corresponding original columns, not the changed columns after elimination.

A complete elimination decision procedure

For $A \in \mathbb{F}^{m \times n}$ and $b \in \mathbb{F}^m$, proceed as follows.

1. Fix the field and variable order. In exact arithmetic, choose any nonzero pivot. In floating point, usually choose the largest available absolute value in the current column.
2. Scan columns from left to right. If the current column is zero in all available pivot rows, skip it; its variable is a free variable.
3. Move a nonzero entry to the pivot row, eliminate entries below it in that column, and continue to the right. Each step left-multiplies by an invertible elementary matrix.
4. After reaching row echelon form, look for $[0 \ \cdots \ 0 \mid c]$. A nonzero c makes the system inconsistent. Otherwise it is consistent.

A complete elimination decision procedure (continued)

5. To find the nullspace, solve the homogeneous system separately. Keep the echelon coefficient matrix but replace the entire right-hand side by 0. For each free column, set its free variable to 1, all other free variables to 0, and back-substitute the pivot variables. This produces special solutions $As_j = 0$. Keeping the nonhomogeneous right-hand side instead produces solutions of the original system, not nullspace vectors.
6. For a particular solution, set all free variables to 0 and back-substitute using the transformed original right-hand side. The complete solution is

$$x = x_p + \sum_{\text{free } j} t_j s_j.$$

The chosen field matters. A pivot that is zero in one field cannot be used merely because it is nonzero in another. Apply every row operation to the augmented column when testing consistency.

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A complete elimination decision procedure (continued)

For $x_1 + 2x_2 = 4$, a particular solution is $x_p = (4, 0)^T$. The homogeneous equation $x_1 + 2x_2 = 0$, with $x_2 = 1$, gives $s = (-2, 1)^T$. Thus $x = (4, 0)^T + t(-2, 1)^T$. Keeping the original right-hand side gives $(2, 1)^T$, whose image is 4, not zero.

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Row operations as matrix products

Adding twice row 1 to row 2 of the identity produces the elementary matrix

$$E = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \qquad EI_2 = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

Left multiplication acts on rows. Right multiplication acts on columns.

$$EA \text{ changes rows}, \qquad AF \text{ changes columns}.$$

An elementary matrix E is invertible. Its inverse undoes the associated row operation. Here,

$$E^{-1} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}.$$

The side of multiplication determines the operation, so it cannot be interchanged.

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Order of elimination matrices

Applying E_1 first and E_2 second gives

$$E_2(E_1A) = (E_2E_1)A.$$

The rightmost factor acts first. Combining the matrices gives E_2E_1; exchanging them can change the row-operation result.

For a square matrix, suppose elimination uses no row swaps or pivot-row scalings, only subtraction of pivot-row multiples from lower rows. Then

$$E_k \cdots E_2E_1A = U.$$

Consequently,

$$A = (E_k \cdots E_1)^{-1}U = LU.$$

This becomes the LU factorization developed below. The first-column multipliers ℓ_{ik} become entries of L.

The inverse reverses the factor order, $(E_k \cdots E_1)^{-1} = E_1^{-1} \cdots E_k^{-1}$.

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When a row exchange is needed

If a prospective pivot is zero, exchange it with an available nonzero entry below it.

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix}.$$

If a final row has the form

$$[0 \ \cdots \ 0 \mid c],$$

with $c \neq 0$, the system is inconsistent. Continue elimination if an exchange can provide a pivot. If all available entries in that column are zero, move to the next column.

Pivots need not lie on the diagonal, particularly for rectangular matrices.

Matrix multiplication and dimensions

If $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times p}$, then

$$(AB)_{ij} = \sum_{k=1}^{n} A_{ik} B_{kj}, \qquad AB \in \mathbb{F}^{m \times p}.$$

The inner dimension n must match. The output has the row count of the left matrix and column count of the right matrix.

The product AB is defined only for compatible shapes. Its existence does not imply that BA exists. Even when both exist with the same shape, generally

$$AB \neq BA.$$

A product applies the right transformation first and the left one second. Noncommutativity records the significance of order.

Column and row combinations

$$AB = A\begin{bmatrix} b_1 & \cdots & b_p \end{bmatrix} = \begin{bmatrix} Ab_1 & \cdots & Ab_p \end{bmatrix}.$$

Each column of AB combines columns of A using the corresponding column of B as coefficients. Alternatively, decompose by rows.

$$AB = \begin{bmatrix} a_1^T \\ \vdots \\ a_m^T \end{bmatrix} B = \begin{bmatrix} a_1^T B \\ \vdots \\ a_m^T B \end{bmatrix}.$$

For example,

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ 3 & -4 \end{bmatrix},$$

Right multiplication by this diagonal sign matrix negates column 2. It does not exchange columns.

Block multiplication requires compatible dimensions at every block interface.

Noncommutativity and distribution

Take

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \qquad BA = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

Hence $AB \neq BA$. The distributive laws still hold.

$$A(B + C) = AB + AC, \quad (A + B)C = AC + BC,$$

Associativity $A(BC) = (AB)C$ also holds for compatible sizes. If A is invertible, multiply $AB = AC$ on the left by A^{-1} to obtain $B = C$. Cancellation through a general singular factor is invalid.

Do not transfer commutativity of real multiplication to matrix multiplication.

Inverses and one-sided inverses

For $A \in \mathbb{F}^{n \times n}$, if a matrix B satisfies

$$AB = BA = I_n,$$

then A is invertible and $B = A^{-1}$. The definition requires both products to be the identity.

The inverse is unique. If $AB = I$ and $CA = I$, then

$$B = IB = (CA)B = C(AB) = CI = C.$$

so a left inverse and a right inverse must coincide when both exist.

Rectangular matrices require separate one-sided inverse conditions.

If $A \in \mathbb{R}^{m \times n}$, $m \neq n$, it cannot have both $AB = I_m$ and $BA = I_n$. If it has full column rank, $n \leq m$, then $A^T A$ is invertible and

$$A_L = (A^T A)^{-1} A^T, \qquad A_L A = I_n,$$

is a left inverse, not a two-sided inverse. Other left inverses may exist. For full row rank, $m \leq n$, similarly

Inverses and one-sided inverses (continued)

$$A_R = A^T (A A^T)^{-1}, \qquad A A_R = I_m.$$

Both formulas require their stated dimensions and ranks. Determinants apply only to square matrices. A rank-deficient matrix instead requires a Moore–Penrose pseudoinverse or a least-squares formulation.

Row operations preserve the same unknown vector x. Column operations reorder or mix the unknowns and require the corresponding change of variables. This is why elimination solves equations using row operations.

Gauss–Jordan inversion

To invert A, augment it with I.

$$[A \mid I],$$

Use row operations until the left side becomes I.

$$[I \mid A^{-1}].$$

For example,

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 1 & 1 & 0 & 1 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right].$$

Thus $A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$.

Check $AA^{-1} = I$. If elimination produces a zero row on the left, A is singular.

Equivalent conditions for invertibility

For square $A \in \mathbb{F}^{n \times n}$, the following statements are equivalent over the specified field.

1. $Ax = 0$ has only the zero solution.
2. A^{-1} exists.
3. The n columns are independent.
4. The columns span $\mathbb{F}^n$.
5. Elimination produces n nonzero pivots.
6. $\det A \neq 0$.

Elimination proves the first five equivalences. A trivial nullspace means no free variables. A square matrix then has n pivots, allowing a unique solution for every right-hand side. Solve for each standard basis right-hand side to assemble the inverse. Conversely, multiplying a homogeneous equation by the inverse proves its solution is zero. Statement 6 uses the determinant developed in Chapter 5. Rectangular matrices do not satisfy all these square-matrix equivalences.

The field matters. An integer matrix may be invertible over the reals but singular modulo a prime.

From elimination to $A = LU$

If elimination needs no row exchanges and all required pivots are nonzero, then

$$A = LU,$$

Here L is unit lower triangular and U is upper triangular. Entry L_{ik} records the multiplier subtracted from row i using row k.

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}.$$

The first multiplier is $4/2 = 2$, giving

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \qquad U = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

Check that $LU = A$. Once the factors are available, solving with many right-hand sides requires only forward and back substitution for each one.

Entries of L record elimination multipliers, not the final reduced data.

Row exchanges and $PA = LU$

A zero pivot may require a row exchange. Combining the exchanges into a permutation matrix P gives

$$PA = LU.$$

For an invertible square matrix, suitable row choices always provide nonzero pivots. Left multiplication by P reorders the rows of A.

The factor order matters. $PA = LU$ factors the permuted matrix by elimination without further swaps.

In floating point, nonzero alone may be insufficient. Partial pivoting chooses the largest available absolute value in the column.

For example,

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 7 \end{bmatrix}, \quad P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Row exchanges and $PA = LU$ (continued)

Swapping the rows gives $PA = \begin{bmatrix} 1 & 3 \\ 0 & 2 \end{bmatrix} = U$, so $L = I$. Also swap the right-hand side to $Pb = (7, 4)^T$. Back substitution gives $y = 2, x = 1$, and the original system returns $(4, 7)^T$. Swapping only A changes the system. A late swap in a multistep factorization must also swap the already stored multipliers in L; otherwise $PA = LU$ fails even with the correct U.

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Cost and reuse

For a dense $n \times n$ matrix,

- Gaussian elimination uses $O(n^3)$ arithmetic operations.
- Forward and back substitution use $O(n^2)$.
- Explicit inversion followed by multiplication adds work and is usually unnecessary.

A common strategy factors

$$A \longrightarrow PA = LU$$

once, then substitutes for each new right-hand side. Elimination also tests consistency directly.

Determinants and symbolic inverses can be useful in exact algebra. Numerical computation usually solves through factors.

The leading cost follows by summation. Pivot k updates about $(n - k)^2$ trailing entries with one multiplication and one subtraction each, for a total of

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Cost and reuse (continued)

$$2\sum_{k=1}^{n-1}(n-k)^2 = \frac{n(n-1)(2n-1)}{3} = \frac{2}{3}n^3 + O(n^2)$$

floating-point additions and multiplications. Multiplier divisions and right-hand-side updates add $O(n^2)$. A triangular solve processes $n(n-1)/2$ off-diagonal terms and costs $O(n^2)$. This arithmetic model excludes data movement, sparse fill-in, and large-integer bit costs.

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Existence and uniqueness of LU without swaps

An invertible square matrix has an LU elimination with no swaps and nonzero pivots at every step exactly when every leading principal submatrix A_k is invertible. Given such a factorization, triangular structure gives $A_k = L_k U_k$, so

$$\det A_k = \prod_{j=1}^{k} u_{jj} \neq 0.$$

Conversely, induct on elimination steps. The first pivot is $a_{11} = \det A_1 \neq 0$. After the first $k-1$ steps, the determinant of the leading k-by-k block is the product of the previous pivots and the new pivot. Thus

$$u_{kk} = \frac{\det A_k}{\det A_{k-1}} \neq 0, \qquad \det A_0 = 1.$$

and every step is valid. This is stronger than invertibility of A. For example, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ is invertible but requires a row swap.

Existence and uniqueness of LU without swaps (continued)

For fixed P, suppose $PA = LU = \widetilde{L}\widetilde{U}$, with both lower-triangular factors having unit diagonal. Then $\widetilde{L}^{-1}L = \widetilde{U}U^{-1}$ is both unit lower triangular and upper triangular, hence is I. The normalized factors are unique for this P, although different pivot rules may choose different permutations.

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Transposes and permutations

The transpose is defined by

$$(A^T)_{ij} = A_{ji}.$$

If $A \in \mathbb{F}^{m\times n}$, then $A^T \in \mathbb{F}^{n\times m}$.

A permutation matrix has exactly one 1 in each row and column, with all other entries zero. For example,

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

The product PA permutes rows, while AP permutes columns. Every permutation matrix satisfies $P^TP = I$.

P^T undoes the permutation. Permuting whole equations preserves their solution set.

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Inner and outer products

For $x, y \in \mathbb{R}^n$,

$$x^T y \in \mathbb{R}$$

is an inner product with a scalar output.

For $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$,

$$uv^T \in \mathbb{R}^{m \times n},$$

is an outer product. If both vectors are nonzero, column j is $v_j u$, at least one column is nonzero, and every column lies in $\operatorname{span}\{u\}$. The rank is therefore 1. If either vector is zero, every entry is zero and the rank is 0.

For example,

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix},$$

the second row is twice the first, and the rank is 1.

Inner and outer products (continued)

Replacing u by the zero vector gives

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

with no nonzero column spanning the image, hence rank 0. The same holds for $v = 0$. Thus

$$\operatorname{rank}(uv^T) = \begin{cases} 1, & u \neq 0 \text{ and } v \neq 0, \\ 0, & u = 0 \text{ or } v = 0. \end{cases}$$

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Inner and outer products (continued)

Outer-product rank: both vectors must be nonzero for rank 1

u≠0, v≠0

u=(1,2)ᵀ, v=(3,-1)ᵀ

uvᵀ=[[3,-1],[6,-2]]

Every column follows u

rank(uvᵀ)=1

u=0, v≠0

u=(0,0)ᵀ, v=(3,-1)ᵀ

uvᵀ=[[0,0],[0,0]]

Every column is zero

rank(uvᵀ)=0

u≠0, v=0

u=(1,2)ᵀ, v=(0,0)ᵀ

uvᵀ=[[0,0],[0,0]]

All entries are zero

rank(uvᵀ)=0

The products $x^T y$ and xy^T have different output types as well as different factor order.

Integral inner products and Gram systems

To extend elimination to function spaces, use the real continuous functions $C([-1,1],\mathbb{R})$ with

$$\langle f, g\rangle = \int_{-1}^{1} f(t)g(t)\,dt.$$

Continuity ensures integrability. A nonzero continuous function has absolute value bounded below by a positive constant on some interval, so $\int f^2 > 0$. This supplies positive definiteness. An arbitrary pointwise function supported at one point can instead have zero squared integral. In L^2, functions equal almost everywhere must be identified as the same vector.

Take the nonorthogonal basis $p_1 = 1, p_2 = 1 + t$. Each Gram entry is $G_{ij} = \langle p_i, p_j\rangle$.

$$G = \begin{bmatrix} \int 1 & \int (1+t) \\ \int (1+t) & \int (1+t)^2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 8/3 \end{bmatrix}.$$

For $f = 1 + 3t$, the right-hand side is

$$b = \begin{bmatrix} \langle p_1, f\rangle \\ \langle p_2, f\rangle \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Integral inner products and Gram systems (continued)

Finding $f = c_1 p_1 + c_2 p_2$ reduces to the Gram system. The same system gives orthogonal projection when an exact expansion is unavailable.

$$G \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = b.$$

The first equation gives $c_1 + c_2 = 1$. Substitution in the second gives $c_2 = 3$, hence $c_1 = -2$. Check

$$-2(1) + 3(1 + t) = 1 + 3t.$$

Since $\det G = 2(8/3) - 2 \cdot 2 = 4/3 > 0$, the coefficients are unique. In general, $c^T G c = \| \sum_i c_i p_i \|^2$, which is positive for nonzero c when the functions are independent. With dependent functions, G may be singular, but $b_i = \langle p_i, f \rangle$ still belongs to $\mathrm{Col}(G)$. Indeed, $z \in \ker G$ gives $\sum_i z_i p_i = 0$, hence $z^T b = 0$. A projection exists and is unique as a function, although its coefficients may be nonunique. An f outside the span produces a nonzero orthogonal residual, not an inconsistent Gram system.

Use the integral inner product on $C([0, 2\pi], \mathbb{R})$. For the orthogonal basis $\{1, \cos x, \sin x\}$, projection coefficients follow by division. For $f(x) = 1 + x$ on $[0, 2\pi]$,

Integral inner products and Gram systems (continued)

$$a_0 = \frac{\int_0^{2\pi} (1 + x)\, dx}{2\pi} = 1 + \pi,$$

$$a_1 = \frac{\langle \cos x, f \rangle}{\langle \cos x, \cos x \rangle} = 0, \qquad b_1 = \frac{\langle \sin x, f \rangle}{\langle \sin x, \sin x \rangle} = -2.$$

the finite Fourier projection is $f_1(x) = 1 + \pi - 2 \sin x$. The residual $e = x - \pi + 2 \sin x$ is orthogonal to 1, $\cos x$, and $\sin x$. This is a finite-dimensional least-squares projection.

A parameterized Toeplitz matrix

For $a \in \mathbb{R}$ and a positive integer n, consider

$$T_n(a) = \begin{bmatrix} a & 1 & & 0 \\ 1 & a & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 1 & a \end{bmatrix}.$$

Let $D_n = \det T_n(a)$, with $D_0 = 1$. Expansion along the first row gives

$$D_n = aD_{n-1} - D_{n-2}.$$

The second term comes from the adjacent off-diagonal entries, leaving D_{n-2}. For $n \geq 2$, use the recurrence with $D_0 = 1, D_1 = a$. The first three values are

$$D_1 = a, \qquad D_2 = a^2 - 1, \qquad D_3 = a(a^2 - 1) - a = a(a^2 - 2).$$

An eigenvector derivation is also possible. For $k = 1, \ldots, n$, let $\theta_k = k\pi/(n+1)$, $(v_k)_j = \sin(j\theta_k)$, with boundary values $v_0 = v_{n+1} = 0$. The trigonometric identity gives

$$(v_k)_{j-1} + (v_k)_{j+1} = 2\cos(\theta_k)(v_k)_j,$$

A parameterized Toeplitz matrix (continued)

so $T_n(a)v_k = (a + 2\cos\theta_k)v_k$. These nonzero eigenvectors have distinct eigenvalues and are independent. Therefore

$$D_n(a) = \prod_{k=1}^{n} (a + 2\cos(k\pi/(n+1))).$$

For $n = 3$, the eigenvalues are $a + \sqrt{2}, a, a - \sqrt{2}$, whose product is $a(a^2 - 2)$. Over $\mathbb{R}$, $T_3(a)$ is invertible exactly when $a \notin \{-\sqrt{2}, 0, \sqrt{2}\}$. The parameter conclusion follows from the general determinant structure rather than a single numerical test.

Chapter check

You can now perform the following tasks.

- Explain the same equation $Ax = b$ using rows and columns.
- Use pivots and free variables to distinguish no solution, a unique solution, and infinitely many solutions.
- Construct the elementary matrix for a row operation.
- Explain the difference between left and right multiplication.
- Check the conditions for an inverse.
- Reconstruct L from multipliers and use $PA = LU$ when needed.
- Distinguish $x^T y$ from xy^T.

The exercises require conditions at every step, especially the underlying field and any need for row exchanges.

Chapter 3 describes the pivot structure through column spaces, nullspaces, and bases.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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From calculation to structure

For $A \in \mathbb{R}^{m \times n}$, Chapter 2 tests whether one specified b is attainable. Now consider every possible output.

$$\{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m.$$

Also consider the inputs whose output is zero.

$$\{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n.$$

The first set is the column space $C(A)$. The second is the nullspace $N(A)$. Both are vector subspaces, closed under addition and scalar multiplication.

Fix the field first. This chapter uses $\mathbb{R}$ unless stated otherwise. Span and rank also apply over $\mathbb{C}$ and finite fields, but geometric interpretations and available scalars must match the field.

Vector-space and subspace conditions

A vector space V over $\mathbb{F}$ is closed under its addition and scalar multiplication. Addition forms an abelian group, and $1v = v$, $(ab)v = a(bv)$, $a(u + v) = au + av$, and $(a + b)v = av + bv$. A subset $S \subseteq V$ inherits most axioms. It suffices to check

$$0 \in S, \qquad u, v \in S \Rightarrow u + v \in S, \qquad u \in S,\ c \in \mathbb{F} \Rightarrow cu \in S,$$

to prove that S is a subspace. The last two conditions can be combined as closure under $cu + dv$.

The sets $\{0\}$ and V are always subspaces. The empty set is not. A set containing only nonzero scalars cannot be a subspace because it omits the additive identity.

An equivalent test is nonemptiness and closure under linear combinations. A unit circle is a subset of the plane, but it fails these linear closure conditions.

Span, intersections, and column spaces

For $v_1, \dots, v_k$, their span is the set of finite linear combinations

$$\operatorname{span}\{v_1, \dots, v_k\} = \left\{ \sum_{j=1}^{k} c_j v_j : c_j \in \mathbb{F} \right\}.$$

The span is the smallest subspace containing the vectors. The intersection $S \cap T$ of subspaces is a subspace. Their union generally is not. For example, adding a nonzero vector on each coordinate axis can give a vector on neither axis.

For $A = [a_1 \ \cdots \ a_n] \in \mathbb{R}^{m \times n}$, the column view gives

$$C(A) = \{Ax : x \in \mathbb{R}^n\} = \operatorname{span}\{a_1, \dots, a_n\}.$$

Thus $Ax = b$ is consistent exactly when $b \in C(A)$, for square and rectangular matrices alike.

Adding a generator already in the span does not enlarge the span, since arbitrary finite sums are already allowed.

Three structures in one system

Consider

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix}.$$

Elimination gives

$$R = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Columns 1 and 3 contain pivots, so the rank is $r = 2$. The original pivot columns span the column space.

$$C(A) = \operatorname{span}\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Three structures in one system (continued)

The two nonzero rows of R span the row space. Both spaces have dimension 2, but they belong to $\mathbb{R}^3$ and $\mathbb{R}^4$, respectively, and are different sets.

Row reduction preserves the nullspace and row space. It generally changes the column space. A column-space basis must use original pivot columns.

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Functions and matrices as vectors

A vector need not be a column of numbers. Real polynomials of degree at most 2 form

$$P_2 = \{a + bx + cx^2 : a, b, c \in \mathbb{R}\}.$$

The coefficient map to (a, b, c) is an isomorphism with $\mathbb{R}^3$, since addition and scaling act on coefficients. Real 2×2 matrices also form a vector space. Its vectors are matrices; its scalars remain real numbers.

Distinguish two kinds of multiplication.

- Vectors in a matrix space can be added and scaled.
- Matrix-matrix multiplication is an additional bilinear operation. It is not required by the vector-space definition.

Specify a common domain before discussing a function space. All functions $f : \mathbb{R} \to \mathbb{R}$ form a vector space under pointwise operations. Continuous and differentiable functions also form subspaces because sums and scalar multiples preserve the respective properties.

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Functions and matrices as vectors (continued)

Coordinates represent vectors but do not define every possible kind of vector. Chapter 8 uses polynomial and function spaces.

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Why a nullspace is a subspace

Define

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}.$$

For $x, y \in N(A)$ and $c \in \mathbb{R}$,

$$A(x + y) = Ax + Ay = 0, \qquad A(cx) = cAx = 0.$$

Also $A0 = 0$, proving that $N(A)$ is a subspace of the domain $\mathbb{R}^n$. Its vectors have the shape of x, not of the output in $\mathbb{R}^m$.

For this chapter's example, $N(A) \subseteq \mathbb{R}^4$ whereas $C(A) \subseteq \mathbb{R}^3$. Their vectors belong to different ambient spaces.

The nullspace is the entire set of homogeneous solutions. A zero row is not itself a nullspace vector. Free columns determine the number of nullspace parameters.

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Constructing special solutions from RREF

The example's homogeneous RREF equations are

$$x_1 + 2x_2 + x_4 = 0, \qquad x_3 + x_4 = 0.$$

The pivot variables are x_1, x_3, and the free variables are x_2, x_4. Set one free variable to 1 and the other to 0 in turn.

$$x_2 = 1, x_4 = 0 \Rightarrow s_1 = (-2, 1, 0, 0)^T,$$
$$x_2 = 0, x_4 = 1 \Rightarrow s_2 = (-1, 0, -1, 1)^T.$$

Every homogeneous solution has a unique representation

$$x = x_2 s_1 + x_4 s_2.$$

Therefore $N(A) = \text{span}\{s_1, s_2\}$ and $\dim N(A) = 2$, checking $n - r = 4 - 2 = 2$.

These special solutions are not arbitrary homogeneous solutions. Their free coordinates form the identity matrix, proving independence, and the free-variable parametrization proves spanning.

Elimination preserves the nullspace

For an invertible elementary matrix E,

$$Ax = 0 \iff EAx = 0.$$

The reverse implication uses E^{-1}.

$$EAx = 0 \Rightarrow Ax = E^{-1}0 = 0.$$

A finite sequence of invertible row operations gives an invertible M, so

$$N(A) = N(MA).$$

Row exchanges, scaling by nonzero scalars, and row additions reduce any A to RREF R. Thus

$$N(A) = N(R).$$

All operations must be invertible. Without scaling pivot rows to 1, echelon form still suffices; solving for pivot variables then requires division by their nonzero pivot values.

Left multiplication transforms output vectors and generally changes the column space. Row operations preserve the span of rows and the homogeneous solution set.

Rank and freedom

Suppose the RREF of $A \in \mathbb{F}^{m \times n}$ has r pivot columns. Each supplies a pivot variable. The remaining $n - r$ columns supply free variables and one special solution per free variable.

These $n - r$ special solutions form a nullspace basis. Therefore

$$\dim N(A) = n - r.$$

If $n > r$, nonzero homogeneous solutions exist. If $r = n$, the nullspace contains only zero. Rank is at most $\min(m, n)$, since each pivot uses a distinct row and column.

A 2×4 matrix has rank at most 2 and therefore at least two free parameters. Nonzero rows alone do not imply a trivial nullspace.

Count pivots after reducing to echelon form. Counting nonzero rows of the original matrix can overestimate rank.

The complete inhomogeneous solution

Let x_p solve $Ax = b$, and let $x_n \in N(A)$. Then

$$A(x_p + x_n) = b + 0 = b.$$

Conversely, if x and x_p both solve the system,

$$A(x - x_p) = Ax - Ax_p = b - b = 0.$$

Hence the complete solution is the affine set

$$\{x_p + x_n : x_n \in N(A)\} = x_p + N(A).$$

It is a subspace exactly when it contains zero, equivalently when $b = 0$. A nontrivial nullspace gives a translated line, plane, or higher-dimensional subspace.

Differences of particular solutions belong to the nullspace. Sums generally produce right-hand side $2b$ instead of b.

Complete solution of the example

Take $b = (2, 5, 3)^T$. Row reduction of the augmented matrix gives

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

There is no contradictory row, so the system is consistent. Setting all free variables to zero gives

$$x_p = (2, 0, 1, 0)^T.$$

Using the nullspace basis yields

$$x = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + t \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix}, \quad s, t \in \mathbb{R}.$$

Direct verification gives $Ax_p = (2, 5, 3)^T$, while both special solutions map to zero.

The particular solution is not unique. Choosing $x_2 = 3$, for example, produces another representative of the same affine solution set.

Four rank cases

For an $m \times n$ matrix of rank r, there are four cases.

Condition	Structure	Consequence for $Ax = b$
$r = m = n$	Square, full rank	Exactly one solution for every b
$r = m < n$	Full row rank	Every b is consistent, with $n - m$ free parameters
$r = n < m$	Full column rank	At most one solution; some b are inconsistent
$r < m$, $r < n$	Deficient in both directions	Some b are inconsistent; consistent systems have $n - r$ free parameters

Full row rank gives $C(A) = \mathbb{R}^m$. Full column rank gives $N(A) = \{0\}$, ensuring uniqueness when a solution exists; for $n < m$, the column space is still a proper subspace of $\mathbb{R}^m$.

If $m \neq n$, full row rank and full column rank cannot both hold.

Dependence and independence

Vectors $v_1, \ldots, v_k$ are linearly independent if

$$c_1 v_1 + \cdots + c_k v_k = 0$$

holds only when every $c_i = 0$. A nonzero coefficient vector satisfying the equation proves dependence.

Equivalently, for $V \in \mathbb{F}^{m \times k}$ with these vectors as columns,

$$\text{Columns of } V \text{ are independent} \iff N(V) = \{0\}.$$

At most m vectors in $\mathbb{F}^m$ can be independent. Thus $k > m$ implies dependence. Any collection containing the zero vector is dependent, since $1 \cdot 0 = 0$ supplies a nontrivial coefficient relation.

Absence of a zero vector does not prove independence. Two nonzero multiples already form a dependent collection.

Span, bases, and dimension

Vectors $v_1, \ldots, v_k$ form a basis of a subspace S if they

1. belong to S;
2. span S;
3. are linearly independent.

Spanning guarantees existence of a coefficient representation. Independence guarantees uniqueness. If

$$w = \sum c_i v_i = \sum d_i v_i,$$

then $\sum (c_i - d_i) v_i = 0$, forcing $c_i = d_i$.

All bases of a finite-dimensional space have the same size, called its dimension. The zero space has dimension 0 and the empty basis. The standard basis of $\mathbb{R}^n$ has n vectors.

The basis itself is not unique. Its size is invariant. Extra vectors destroy independence; removing a basis vector destroys spanning.

Dimension and a function-space example

Suppose S has a basis of d vectors. The coordinate map

$$\varphi_S(c_1, \ldots, c_d) = \sum_{i=1}^{d} c_i u_i$$

is a bijection $\mathbb{F}^d \to S$. Spanning gives existence and independence gives uniqueness. Express any independent family $w_1, \ldots, w_d$ in the original basis.

$$w_j = \sum_{i=1}^{d} b_{ij} u_i.$$

Let $B = (b_{ij})$. Uniqueness of coordinates gives

$$\sum_{j=1}^{d} c_j w_j = 0 \quad \iff \quad Bc = 0.$$

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Dimension and a function-space example (continued)

Thus independence of the w_j is equivalent to $N(B) = \{0\}$. The square matrix B has no free variables and is invertible. For any $v \in S$ with old coordinates y, solve $Bc = y$ to express v in the w_j. Therefore these d independent vectors also span S. More generally, if k vectors are independent and d vectors span, the coordinate matrix is $d \times k$, and a trivial nullspace requires $k \leq d$. Comparing two bases in both directions proves that they have equal size.

The function family

$$\{1, \cos(2x), \sin^2 x\}$$

is dependent because

$$\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x).$$

In suitable function spaces, $\{1, \cos x, \sin x\}$ is independent. Proofs of function independence use identities for every x, coefficient comparisons, differentiation, or other structure.

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Dimension and a function-space example (continued)

The domain and function space matter. Different-looking formulas alone do not prove independence.

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The four fundamental subspaces

For $A \in \mathbb{R}^{m \times n}$, define

Subspace	Definition	Ambient space
column space	$C(A) = \{Ax : x \in \mathbb{R}^n\}$	$\mathbb{R}^m$
row space	$C(A^T) = \{A^T y : y \in \mathbb{R}^m\}$	$\mathbb{R}^n$
nullspace	$N(A) = \{x : Ax = 0\}$	$\mathbb{R}^n$
left nullspace	$N(A^T) = \{y : A^T y = 0\}$	$\mathbb{R}^m$

The term left nullspace comes from transposing $A^T y = 0$ into $y^T A = 0$.

Nonzero RREF rows form a row-space basis. Original pivot columns form a column-space basis. Special solutions form bases of the nullspaces of A and A^T.

Equal rank does not mean equal ambient space. Track all four spaces and the effect of row operations on each.

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Rank-nullity and the left nullspace

For $A \in \mathbb{R}^{m \times n}$, let its RREF be R, with r pivots. Combining the invertible row operations gives $R = MA$, with M invertible. First, $C(R) = MC(A)$ is an isomorphic image and preserves dimension. Each pivot column of RREF is a distinct standard coordinate vector. These columns are independent and span all other columns, so $\dim C(R) = r$, hence $\dim C(A) = r$.

Row operations also preserve the row space. Each row of $R = MA$ is a combination of rows of A, giving $C(R^T) \subseteq C(A^T)$. The inverse relation $A = M^{-1}R$ gives the reverse inclusion. Individual dependence coefficients need not stay fixed. Also $Ax = 0 \iff MAx = 0$, so $N(A) = N(R)$. The r nonzero RREF rows are independent and span the row space, giving $\dim C(A^T) = r$. This proves equality of row and column rank.

There are r pivot variables and $n - r$ free variables. Setting one free variable to 1 and the others to 0 gives special solutions whose free-coordinate block is I_{n-r}, proving independence. Every solution is determined uniquely by its free coordinates, so these vectors also span $N(A)$. Thus

$$\dim C(A) = \dim C(A^T) = r,$$

Rank-nullity and the left nullspace (continued)

$$\dim N(A) = n - r, \qquad \dim N(A^T) = m - r.$$

Apply the same proof to A^T, whose domain is $\mathbb{R}^m$. Its nullity is $m - r$. The paired dimensions satisfy

$$r + (n - r) = n, \qquad r + (m - r) = m.$$

For this chapter's 3×4 example, $r = 2$, so $\dim C(A) = 2$, $\dim N(A) = 2$, and $\dim N(A^T) = 1$. Set $y = (a, b, c)^T$ and solve $A^T y = 0$.

$$a + 2b + c = 0, \quad 2a + 4b + 2c = 0, \quad b + c = 0, \quad a + 3b + 2c = 0.$$

The second equation is twice the first; the fourth is the sum of the first and third. The independent equations give $c = -b$ and $a = -b$. Every solution is $b(-1, 1, -1)^T$, so a left-nullspace basis is

$$\left\{(-1, 1, -1)^T\right\}.$$

Each dimension identity pairs spaces in the same ambient space. It does not place all four subspaces in a single direct sum.

The fundamental decomposition

The four-subspace dimension theorem leads to

$$\mathbb{R}^n = C(A^T) \oplus N(A), \qquad \mathbb{R}^m = C(A) \oplus N(A^T).$$

Here $\oplus$ denotes a direct sum. The row space and nullspace share the domain, with dimensions r and $n - r$. The column space and left nullspace share the codomain, with dimensions r and $m - r$. Chapter 4 proves that each pair is orthogonal and therefore consists of orthogonal complements.

Every $x \in \mathbb{R}^n$ splits into a row-space component and a nullspace component. Orthogonality requires a specified inner product or dual pairing; it is not defined on an arbitrary abstract vector space without additional structure.

Dimensions summing to n alone do not prove a direct sum. We must exclude a nonzero intersection. If $z \in C(A^T) \cap N(A)$, write $z = A^T y$ with $Az = 0$. Then

$$z^T z = (A^T y)^T z = y^T A z = 0.$$

The fundamental decomposition (continued)

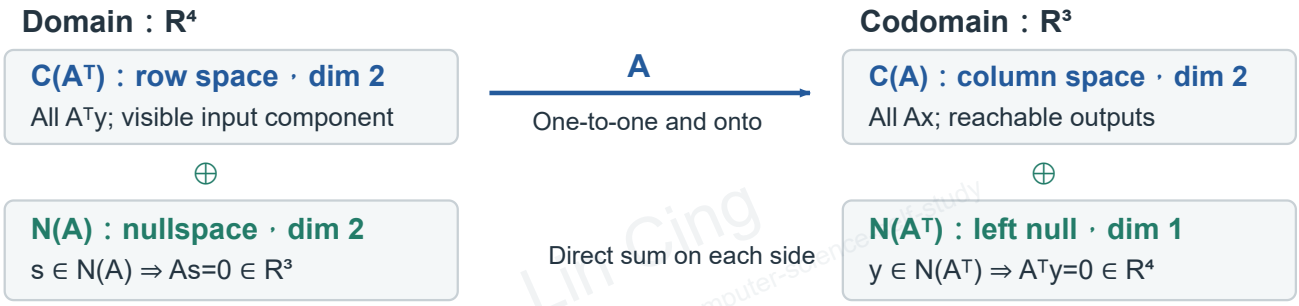
Over $\mathbb{R}$, $z^T z = \sum_i z_i^2 = 0$ forces $z = 0$. Combining bases of the two spaces gives an independent family. Indeed, a zero sum of one combination from each side makes each combination belong to the intersection, so both vanish. The combined family contains $r + (n - r) = n$ independent vectors and spans $\mathbb{R}^n$. Applying the proof to A^T gives the second direct sum. Chapter 4 develops the orthogonal-complement and projection formulas.

Positive definiteness of the real inner product is essential. Over $\mathbb{C}$, use A^* and $z^* z$. Ordinary transpose does not give the same result over every field. For $A = [1\ 1]$ over $\mathbb{F}_2$, $C(A^T) = N(A) = \operatorname{span}\{(1, 1)^T\}$. Their dimensions sum to 2 but the intersection is nonzero. Rank-nullity still holds; this transpose-based direct-sum assertion fails.

For real matrices, the restriction $A : C(A^T) \to C(A)$ is an isomorphism. Its kernel is the zero intersection just proved, so it is injective. For any $b = Ax \in C(A)$, decompose $x = x_{\text{row}} + x_{\text{null}}$; then $b = Ax_{\text{row}}$, proving surjectivity.

The fundamental decomposition (continued)

One A: 3 × 4, rank 2; boxes denote sets, not planes



Reconstructing a direct sum numerically

For the same matrix A, use the following bases.

$$u_1 = (1,2,0,1)^T, \quad u_2 = (0,0,1,1)^T,$$
$$s_1 = (-2,1,0,0)^T, \quad s_2 = (-1,0,-1,1)^T.$$

The first pair spans $C(A^T)$, and the second spans $N(A)$. If $\alpha u_1 + \beta u_2 + \gamma s_1 + \delta s_2 = 0$, coordinate comparison gives

$$\gamma = -2\alpha, \quad \delta = \beta, \quad 5\alpha - \beta = 0, \quad \alpha + 2\beta = 0.$$

Hence $\alpha = \beta = \gamma = \delta = 0$. The four vectors are independent and form a basis of $\mathbb{R}^4$.

Take

$$x = (-2,3,0,3)^T = u_1 + u_2 + s_1 + s_2,$$

Then

$$x_{\text{row}} = (1,2,1,2)^T, \qquad x_{\text{null}} = (-3,1,-1,1)^T.$$

Reconstructing a direct sum numerically (continued)

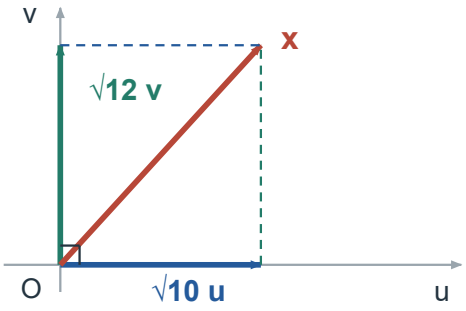
Adding the components recovers x. Independence of the combined basis proves uniqueness of this decomposition. The coordinate argument establishes the direct sum without using an inner product.

Numerically, $Ax_{\text{row}} = (7, 17, 10)^T = 7(1, 2, 1)^T + 3(0, 1, 1)^T \in C(A)$, while $Ax_{\text{null}} = 0$. Also $x_{\text{row}}^T x_{\text{null}} = -3 + 2 - 1 + 2 = 0$, so the components are orthogonal and $\|x\|^2 = 22 = 10 + 12$.

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Reconstructing a direct sum numerically (continued)

Two components of a 4D vector, drawn in their 2D span



x_row = (1,2,1,2)ᵀ

x_null = (−3,1,−1,1)ᵀ

x = (−2,3,0,3)ᵀ = x_row + x_null

u=x_row/√10 · v=x_null/√12

uᵀv=0 ; ‖x‖²=10+12=22

Ax=Ax_row=(7,17,10)ᵀ ; Ax_null=0

The axes are $u = x_{\text{row}}/\sqrt{10}$ and $v = x_{\text{null}}/\sqrt{12}$, not the original four coordinate axes. The drawing shows their actual two-dimensional span within $\mathbb{R}^4$.

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Chapter check and common errors

Use the example matrix to answer these questions.

- Why use original pivot columns for a basis of $C(A)$, rather than all columns of RREF?
- Why do the normalized free-variable choices give a basis of $N(A)$?
- Why is $x_p + N(A)$ affine rather than a subspace when $b \neq 0$?
- Why does equal rank not identify row space with column space?
- Which obstacles to existence and uniqueness do full row and full column rank remove?

Common errors put $N(A)$ in $\mathbb{R}^m$, equate column dimension with n, use RREF columns to span the original $C(A)$, or infer consistency of $Ax = b$ merely from solving $A^TAx = A^Tb$.

Ambient spaces and bases matter beyond the rank number. They support projection and least squares in Chapter 4.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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Why pair the four subspaces?

For $A \in \mathbb{R}^{m \times n}$, each pair below shares an ambient space.

$$C(A^T), N(A) \subseteq \mathbb{R}^n,$$
$$C(A), N(A^T) \subseteq \mathbb{R}^m.$$

Their dimensions sum to n and m, respectively.

$$\dim C(A^T) + \dim N(A) = r + (n - r) = n,$$
$$\dim C(A) + \dim N(A^T) = r + (m - r) = m.$$

These pairs are orthogonal as well as dimensionally complementary. Every input decomposes uniquely into row-space and nullspace components. Every output decomposes into column-space and left-nullspace components.

Fix the ambient space before taking inner products. A vector in $\mathbb{R}^m$ and one in $\mathbb{R}^n$ cannot be paired by the ordinary dot product unless the dimensions and coordinate interpretation agree.

Why row space is orthogonal to nullspace

If $x \in N(A)$, then $Ax = 0$. Expanding by rows gives

$$\begin{bmatrix} r_1^T \\ \vdots \\ r_m^T \end{bmatrix} x = \begin{bmatrix} r_1^T x \\ \vdots \\ r_m^T x \end{bmatrix} = 0.$$

Thus x is orthogonal to each row. Linearity makes it orthogonal to every linear combination of rows.

$$x^T(c_1 r_1 + \cdots + c_m r_m) = 0.$$

Conversely, orthogonality to every row gives $Ax = 0$. This proves $N(A) = C(A^T)^\perp$. The spaces are orthogonal and their dimensions sum to n, so each is the other's orthogonal complement. Hence

$$C(A^T) = N(A)^\perp.$$

Replacing A by A^T gives $C(A) = N(A^T)^\perp$.

$U^\perp$ consists of all vectors orthogonal to every vector of U. Orthogonality plus the dimension sum establishes the full complement relation.

Orthogonality and complements

Write $U \perp W$ when $u^T w = 0$ for every $u \in U, w \in W$. If $z \in U \cap W$, then $z^T z = 0$, forcing $z = 0$. Therefore $U \cap W = \{0\}$. The converse is false.

In a finite-dimensional inner-product space, if

$$U \cap W = \{0\}, \qquad \dim U + \dim W = \dim V,$$

then $V = U \oplus W$, a direct sum. The subspaces can still meet at an oblique angle. An orthogonal direct sum also requires $U \perp W$.

For example, $U = \operatorname{span}\{(1,0)^T\}$ and $W = \operatorname{span}\{(1,1)^T\}$ intersect only at zero and their dimensions sum to 2. They are not perpendicular. Replacing W by $\operatorname{span}\{(0,1)^T\}$ gives an orthogonal complement.

The inner product determines the orthogonal complement. Complex spaces require conjugate transposes.

Dimensions and a decomposition example

Since $C(A^T) \oplus N(A) = \mathbb{R}^n$, every $x \in \mathbb{R}^n$ has the unique decomposition

$$x = x_{\text{row}} + x_{\text{null}}, \qquad x_{\text{row}}^T x_{\text{null}} = 0.$$

Take

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

whose rank is 2. Its two independent rows form a row-space basis, while the nullspace is spanned by

$$(-2, 1, -1)^T$$

Check directly that

$$(1,2,0)(-2,1,-1)^T = 0, \qquad (0,1,1)(-2,1,-1)^T = 0.$$

Thus $\mathbb{R}^3 = C(A^T) \oplus N(A)$. Since A has full row rank, $N(A^T) = \{0\}$ and $C(A) = \mathbb{R}^2$.

When a subspace is the whole ambient space, its orthogonal complement is the zero subspace. The resulting decomposition with a zero component is still unique.

Projection onto a line

For nonzero $a \in \mathbb{R}^n$, remove from b its component perpendicular to $L = \operatorname{span}\{a\}$. Write the projection as

$$p = \lambda a$$

and require $e = b - p$ to be perpendicular to the whole line.

$$a^T e = a^T(b - \lambda a) = 0.$$

Since $a^T a > 0$,

$$\lambda = \frac{a^T b}{a^T a}, \qquad p = a\frac{a^T b}{a^T a}.$$

The projection matrix is

$$P = \frac{a a^T}{a^T a}, \qquad p = Pb.$$

For $a = (1,1,1)^T$, $b = (2,1,4)^T$, we have $a^T b = 7$, $a^T a = 3$, and

$$p = \frac{7}{3}(1,1,1)^T, \quad e = b - p = \frac{1}{3}(-1,-4,5)^T.$$

Projection onto a line (continued)

Direct substitution gives

$$p + e = (2,1,4)^T = b, \qquad a^T e = \frac{1(-1) + 1(-4) + 1(5)}{3} = 0.$$

Hence $p^T e = 0$, and

$$\|e\|_2 = \frac{\sqrt{(-1)^2 + (-4)^2 + 5^2}}{3} = \frac{\sqrt{42}}{3}.$$

The same values satisfy the normal equation

$$a^T a\,\lambda = a^T b, \qquad 3 \cdot \frac{7}{3} = 7.$$

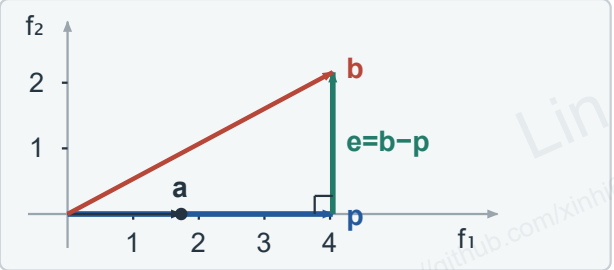
The diagram uses the orthonormal basis $f_1 = a/\sqrt{3}$, $f_2 = e/(\sqrt{42}/3)$ in $\operatorname{span}\{a, e\}$. Its two-dimensional coordinates are

$$[a] = (\sqrt{3}, 0), \quad [p] = (7/\sqrt{3}, 0), \quad [e] = (0, \sqrt{42}/3), \quad [b] = (7/\sqrt{3}, \sqrt{42}/3).$$

Projection onto a line (continued)

These coordinates give an isometry from that plane to $\mathbb{R}^2$, with equal scales on both axes. The right angle represents the original inner product. The short black vector a and longer blue vector $p = (7/3)a$ have distinct endpoints. The green residual is translated to run from p to b.

Exact projection: b=p+e in orthogonal coordinates



Orthonormal axes; equal scale

f₁=a/√3 · f₂=e/(√42/3)

a=(√3,0) · p=(7/√3,0)

e=(0,√42/3) · b=p+e

Original a=(1,1,1), b=(2,1,4)

Orthogonality also controls maximal growth directions in the SVD. Here it first characterizes the minimum-residual projection.

Projection onto a subspace

Let $A \in \mathbb{R}^{m \times n}$ have independent columns, and let

$$S = C(A).$$

The projection of b onto S has residual $e = b - Ax$ orthogonal to every basis vector of S.

$$A^T(b - Ax) = 0.$$

This gives the normal equations

$$A^T Ax = A^T b.$$

With full column rank, $A^T A$ is invertible, so

$$p = A(A^T A)^{-1} A^T b.$$

and the projection matrix is

$$P = A(A^T A)^{-1} A^T.$$

The projector P is not A^{-1}, since A need not be square. Nor may $(A^T A)^{-1}$ be replaced by $A^{-1}(A^T)^{-1}$ unless A is square and invertible.

Projection onto a subspace (continued)

Projection onto a line is the case $n = 1$. Independent columns make $A^T A$ positive definite and invertible because $z^T A^T A z = \|Az\|^2$.

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Why column independence is necessary

For every $z \in \mathbb{R}^n$,

$$z^T A^T A z = (Az)^T (Az) = \|Az\|^2 \geq 0.$$

Independent columns imply $Az \neq 0$ for nonzero z. The quadratic form is then strictly positive, so $A^T A$ is positive definite and invertible.

Conversely, dependent columns give a nonzero z with $Az = 0$. Then

$$A^T A z = A^T 0 = 0,$$

so $A^T A$ is singular. In complex spaces replace transpose by conjugate transpose.

Rank deficiency does not destroy uniqueness of the projected vector. It creates redundant parameters, so multiple coefficient vectors can produce the same projection.

Independent columns are necessary and sufficient for invertibility of $A^T A$. Full row rank does not suffice when $m < n$.

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Algebraic properties of a projector

For full column rank A, let

$$P = A(A^TA)^{-1}A^T.$$

The matrix is symmetric and idempotent.

$$P^T = A((A^TA)^{-1})^TA^T = A(A^TA)^{-1}A^T = P,$$
$$P^2 = A(A^TA)^{-1}(A^TA)(A^TA)^{-1}A^T = P.$$

Idempotence means that projecting twice changes nothing further. Symmetry distinguishes an orthogonal projector from a general oblique projector. The complementary projector

$$I - P$$

projects onto $C(A)^\perp = N(A^T)$, and $P(I - P) = 0$.

Idempotence alone does not imply orthogonality. Symmetry is essential in the real Euclidean setting.

The nearest point and its error

Let $p = Pb$ and $e = (I - P)b$. Then

$$b = p + e, \qquad p^Te = 0.$$

Pythagoras gives

$$\|b\|^2 = \|p\|^2 + \|e\|^2.$$

For any $q \in C(A)$, the difference $p - q$ lies in $C(A)$, while e is perpendicular to that space. Therefore

$$\|b - q\|^2 = \|e + (p - q)\|^2 = \|e\|^2 + \|p - q\|^2.$$

The right side is minimized uniquely at $q = p$. This proves the nearest-point property without calculus.

The cross term vanishes by orthogonality. In a complex inner-product space the corresponding norm expansion uses conjugate transposes and real parts.

Least squares

Even when $Ax = b$ is inconsistent, choose

$$x^\star \in \arg\min_x \|b - Ax\|_2^2.$$

The fitted vector $p = Ax^\star$ is the unique projection onto $C(A)$. Full column rank makes the parameter $x^\star$ unique as well. Rank deficiency gives infinitely many minimizers differing by nullspace vectors.

Minimizing the norm or its square gives the same minimizers because squaring is strictly increasing on $[0, \infty)$. The square avoids a square root and gives a linear gradient equation.

The objective minimizes a scalar residual norm, not the vector residual under an unspecified ordering. In general the fitted vector differs from b.

Least squares minimizes the sum of squared coordinate errors. The optimal residual is perpendicular to the attainable subspace; it need not vanish.

Deriving the normal equations

Define

$$E(x) = (b - Ax)^T (b - Ax).$$

For any displacement h, expand the square.

$$\begin{aligned} E(x + h) - E(x) = & -2h^T A^T (b - Ax) \\ & + h^T A^T A h. \end{aligned}$$

The gradient is therefore

$$\nabla_x E = -2A^T (b - Ax).$$

At an unconstrained minimum $x^\star$, the first-order change vanishes in every direction h, so

$$A^T (b - Ax) = 0 \iff A^T A x = A^T b.$$

With full column rank, $h^T A^T A h = \|Ah\|^2 > 0$ for every nonzero h. The quadratic remainder is strictly positive, proving the unique global minimum. Both sides of the normal equations have shape $n \times 1$.

Deriving the normal equations (continued)

For dependent columns, the normal equations may have multiple solutions. If x, z both solve them, let $q = x - z$. Subtracting gives $A^T Aq = 0$. Multiplying by q^T,

$$0 = q^T A^T Aq = \|Aq\|^2.$$

so $Ax = Az$. The parameter vectors can differ while the projection point remains unique.

A^T is not the projection matrix. The condition $A^T e = 0$ tests orthogonality to every column, equivalently $e \in N(A^T)$.

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Fitting a straight line

Fit $y = c + mt$ to

$$(t, y) = (0, 1), (1, 2), (2, 2).$$

The design matrix and vectors are

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} c \\ m \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

Compute

$$A^T A = \begin{bmatrix} 3 & 3 \\ 3 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

Solving gives

$$m = \frac{1}{2}, \qquad c = \frac{7}{6}.$$

The fitted vector is

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Fitting a straight line (continued)

$$p = \left(\frac{7}{6}, \frac{5}{3}, \frac{13}{6}\right)^T.$$

The residual is

$$e = b - p = \left(-\frac{1}{6}, \frac{1}{3}, -\frac{1}{6}\right)^T.$$

Its entries sum to zero, and $A^T e = 0$. The sum condition comes from the constant column. The full normal equation also checks orthogonality to the trend column t.

A zero sum of residuals alone does not prove least-squares optimality. Test the residual against every design column.

rank-deficient least squares

Take

$$A = \begin{bmatrix}1\\2\end{bmatrix}\begin{bmatrix}1 & 1\end{bmatrix} = \begin{bmatrix}1 & 1\\2 & 2\end{bmatrix}, \quad b = \begin{bmatrix}1\\3\end{bmatrix}.$$

The two columns of A are identical, so its rank is 1. Projection onto their line gives

$$p = \frac{a^T b}{a^T a} a = \frac{7}{5}\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}7/5\\14/5\end{bmatrix}.$$

The normal equations are

$$\begin{bmatrix}5 & 5\\5 & 5\end{bmatrix} x = \begin{bmatrix}7\\7\end{bmatrix},$$

and impose only

$$x_1 + x_2 = \frac{7}{5}.$$

Every parameter vector satisfying this relation produces the same $Ax = p$ and minimizes the error. The residual

rank-deficient least squares (continued)

$$e = \left(-\frac{2}{5}, \frac{1}{5}\right)^T$$

is orthogonal to $(1, 2)^T$. Thus projection uniqueness and parameter uniqueness are separate properties.

For deficient rank, use the pseudoinverse minimum-norm solution or explicit regularization; do not invert $A^T A$.

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Orthonormal bases

Vectors $q_1, \ldots, q_k$ satisfying

$$q_i^T q_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

are orthonormal. Placing them in $Q \in \mathbb{R}^{m \times k}$ gives

$$Q^T Q = I_k.$$

If Q is square, these independent columns form an orthonormal basis of the whole space. Then $Q^{-1} = Q^T$, and Q is an orthogonal matrix.

If Q is rectangular, $Q^T Q = I_k$ gives full column rank, not $QQ^T = I_m$. The two identities have different sizes and different meanings.

Despite the historical name, an orthogonal matrix has orthonormal columns. A rectangular matrix with orthonormal columns has no two-sided inverse.

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Gram–Schmidt construction

For independent vectors $a_1, \dots, a_n$, begin by normalizing

$$q_1 = \frac{a_1}{\|a_1\|}.$$

For a_j, subtract its projections onto the existing directions.

$$z_j = a_j - \sum_{i=1}^{j-1} (q_i^T a_j) q_i,$$

Then normalize

$$q_j = \frac{z_j}{\|z_j\|}.$$

Independence guarantees $z_j \neq 0$. Otherwise a_j would be in the span of its predecessors. For $i < j$, expand $q_i^T z_j$ to obtain

$$q_i^T z_j = q_i^T a_j - \sum_{\ell=1}^{j-1} (q_\ell^T a_j)(q_i^T q_\ell) = q_i^T a_j - q_i^T a_j = 0.$$

Gram–Schmidt construction (continued)

Inductively, the first $j-1$ vectors q are orthonormal and span $\operatorname{span}\{a_1, \dots, a_{j-1}\}$. Since a_j equals z_j plus a combination of the earlier directions, replacing z_j by its normalization q_j preserves the expanded span. At every step,

$$\operatorname{span}\{q_1, \dots, q_j\} = \operatorname{span}\{a_1, \dots, a_j\}.$$

For $a_1 = (1,1,0)^T, a_2 = (1,0,1)^T$,

$$q_1 = \frac{1}{\sqrt{2}}(1,1,0)^T,$$

$$z_2 = a_2 - (q_1^T a_2) q_1 = \left(\frac{1}{2}, -\frac{1}{2}, 1\right)^T,$$

$$q_2 = \frac{1}{\sqrt{6}}(1,-1,2)^T.$$

In floating point, a small $\|z_j\|$ signals near dependence, and division by it amplifies error. Chapter 11 develops more stable alternatives.

QR and least squares

Collect the Gram–Schmidt expansions into

$$A = QR,$$

where Q has orthonormal columns and R is upper triangular. The coefficients $R_{ij} = q_i^T a_j$ for $i < j$, and $R_{jj} = \|z_j\|$, give $A = QR$ by column multiplication. In the preceding example,

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

For full column rank, least squares reduces to

$$R^T Q^T Q R x = R^T Q^T b \Longrightarrow Rx = Q^T b.$$

Solve the invertible upper-triangular system in R. This avoids forming $A^T A$ and is more stable. For a rectangular A, the reduced Q is $m \times n$.

Do not cancel Q^T arbitrarily. In the normal-equation derivation one can cancel the invertible factor R^T, since full column rank guarantees invertible R.

QR and least squares (continued)

Classical Gram–Schmidt is convenient for proofs. Numerical implementations usually use modified Gram–Schmidt or Householder QR, with the same exact mathematical factorization structure.

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A complete QR least-squares example

Use the preceding

$$Q = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{2} & -1/\sqrt{6} \\ 0 & 2/\sqrt{6} \end{bmatrix}, \quad R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

and $b = (1, 2, 0)^T$. First compute

$$Q^T b = \left(\frac{3}{\sqrt{2}}, -\frac{1}{\sqrt{6}} \right)^T.$$

Solve $Rx = Q^T b$. The second equation gives

$$\sqrt{\frac{3}{2}}\, x_2 = -\frac{1}{\sqrt{6}} \quad \Longrightarrow \quad x_2 = -\frac{1}{3}.$$

The first equation is

$$\sqrt{2}\, x_1 + \frac{1}{\sqrt{2}} \left(-\frac{1}{3} \right) = \frac{3}{\sqrt{2}},$$

A complete QR least-squares example (continued)

so $x_1 = 5/3$. The projection is

$$p = Ax = \left(\frac{4}{3}, \frac{5}{3}, -\frac{1}{3} \right)^T,$$

and the residual is

$$e = b - p = \frac{1}{3}(-1, 1, 1)^T.$$

Check directly that

$$a_1^T e = 0, \qquad a_2^T e = 0,$$

Thus $A^T e = 0$ and $\|e\|^2 = 1/3$. The nonzero residual makes the normal-equation orthogonality condition observable.

Chapter check

Distinguish the following concepts.

- $U \perp W$, $U \cap W = \{0\}$, and $V = U \oplus W$.
- Line projection, subspace projection, and a projection matrix.
- Unique least-squares parameters at full column rank and nonunique minimizers at deficient rank.
- $P^2 = P = P^T$ and the complementary projector $I - P$.
- $Q^T Q = I$ for orthonormal columns and $QQ^T = I$ for a square orthogonal matrix in this setting.
- Gram-Schmidt, $A = QR$, and preservation of the column span.

Check projection and least-squares calculations with $A^T e = 0$. A small residual alone does not prove the required orthogonality.

Chapter 5 connects determinants to invertibility and volume. Exact invertibility alone does not guarantee a numerically well-conditioned solve.

References and attribution

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What a determinant measures

For square $A \in \mathbb{R}^{n \times n}$, $\det A$ gives a signed volume multiplier and tests invertibility.

$$A \text{ invertible} \iff \det A \neq 0.$$

2×2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

has signed area $ad - bc$. Independent columns give nonzero area with a sign determined by orientation. Collinear columns give zero area.

In n dimensions, $\det A$ is signed volume. Its absolute value gives the volume ratio and its sign records orientation reversal. The same quantity appears in inverse, pivot, and cofactor formulas.

A zero determinant tests exact singularity. Determinant magnitude alone does not measure conditioning. Chapter 11 treats scale and numerical rank.

Three defining properties

Regard the determinant as a function of its columns

$$D(a_1, \ldots, a_n).$$

with these properties.

1. Normalization:

$$D(e_1, \ldots, e_n) = 1.$$

2. Alternation. Exchanging two columns reverses the sign.

$$D(\ldots, a_j, \ldots, a_k, \ldots) = -D(\ldots, a_k, \ldots, a_j, \ldots).$$

3. Multilinearity. The function is linear in each column separately.

$$D(\ldots, \alpha a_j + \beta b_j, \ldots) = \alpha D(\ldots, a_j, \ldots) + \beta D(\ldots, b_j, \ldots).$$

Three defining properties (continued)

These properties determine the determinant uniquely. Expand each column in the standard basis and apply multilinearity. Terms with repeated basis vectors vanish. The surviving terms are permutations of the standard basis, whose values are their permutation signs. This gives the permutation formula below, which also satisfies all three properties, proving existence and uniqueness. This chapter uses $\mathbb{R}$ or $\mathbb{C}$. In characteristic 2, alternation must explicitly require zero for repeated columns, rather than relying only on sign reversal.

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Consequences for row and column operations

When two columns coincide, exchanging them leaves the matrix unchanged but reverses the determinant.

$$D = -D,$$

Thus $D = 0$ over a field of characteristic other than 2.

Adding a multiple of one column to another gives

$$D(\dots, a_j + ca_k, \dots) = D(\dots, a_j, \dots) + cD(\dots, a_k, \dots).$$

The second term has repeated columns and vanishes. Column addition preserves the determinant. Transpose invariance will give the corresponding row rule.

Dependent columns therefore give zero determinant. Express one as a combination of the others; multilinearity expands the determinant into terms with repeated columns.

Scaling, exchanges, and triangular matrices

Scaling one column by λ gives

$$\det(\ldots, \lambda a_j, \ldots) = \lambda \det(\ldots, a_j, \ldots).$$

Consequently,

$$\det(cA) = c^n \det A.$$

This agrees with scaling every edge of an n-dimensional body by c.

A row exchange multiplies the determinant by -1. For upper triangular U,

$$\det U = u_{11} u_{22} \cdots u_{nn}.$$

A zero diagonal entry of a triangular matrix makes its determinant zero and the matrix singular. This connects determinants to elimination.

Track scaling and exchanges in an implementation to preserve the determinant's value and sign.

$\det A = \det A^T$

Permutation formula：

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_{i=1}^{n} a_{i,\sigma(i)}.$$

After transposing, set $\tau = \sigma^{-1}$.

$$a_{\sigma(i),i}$$

becomes $a_{j,\tau(j)}$, and

$$\operatorname{sgn}(\tau) = \operatorname{sgn}(\sigma).$$

therefore

$$\det A^T = \det A.$$

Rows and columns obey the same determinant rules. The following nonnegativity claim assumes $A \in \mathbb{R}^{n \times n}$.

$$\det(A^T A) = (\det A)^2 \geq 0.$$

$\det A = \det A^T$ (continued)

For every real $A \in \mathbb{R}^{m \times n}$, $A^T A$ is positive semidefinite, and positive definite exactly when $\operatorname{rank} A = n$. For an invertible square matrix, both conditions hold. For complex matrices, transpose does not conjugate, and squares need not be nonnegative. For $A = [i]$, $A^T A = [-1]$, giving $z^T A^T A z = -1$ at $z = 1$.

For complex matrices, use the conjugate transpose $A^* = \overline{A}^T$.

$$\det(A^* A) = \overline{\det A} \det A = |\det A|^2 \geq 0, \qquad z^* A^* A z = \|Az\|_2^2 \geq 0.$$

Then $A^* A$ is Hermitian positive semidefinite and is positive definite exactly at full column rank. The energy identity still holds for rectangular $A \in \mathbb{C}^{m \times n}$, but $\det A$ is undefined, so the square determinant product cannot be used.

$\det A = \det A^T$ (continued)

Real Gram: A^TA is symmetric positive semidefinite

$A \in \mathbb{R}^{m \times n}$, $z^TA^TAz = \|Az\|_2^2 \geq 0$	$A = [[1,1],[1,2]]$
A^TA is always symmetric PSD	$\det A = 1$
$\operatorname{rank}(A) = n \Leftrightarrow A^TA$ is positive definite	$\det(A^TA) = 1^2 = 1$

$\det(A^TA) = \det(A)^2$ requires square A.

$\det A = \det A^T$ (continued)

Complex Gram: transpose versus conjugate transpose

A=[i]

$A^TA=[-1]$ · det=-1

$z=1$: $z^TA^TAz=-1$

Transpose Gram need not be PSD

A*=[-i] · A*A=[1]

det(A*A)=|det A|²=1

$z^*A^*Az=\|Az\|_2^2 \geq 0$

A*A is Hermitian PSD

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Products and inverses

Fix B and consider

$$F(A) = \det(AB)$$

which is multilinear and alternating in the rows of A, with

$$F(I) = \det B.$$

Uniqueness under the determinant axioms gives

$$\det(AB) = \det A \det B.$$

For invertible A,

$$1 = \det(A^{-1}A) = \det(A^{-1}) \det A,$$

so

$$\det(A^{-1}) = 1/\det A.$$

If either matrix is singular, the product has deficient rank and both determinant expressions vanish.

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Products and inverses (continued)

In general, $\det(A+B) \neq \det A + \det B$. Multilinearity in individual rows or columns is not additivity on the matrix space.

For example, let $A = I_2$, $B = \operatorname{diag}(2,3)$. Then $\det A = 1$, $\det B = 6$, but

$$\det(A+B) = \det \operatorname{diag}(3,4) = 12 \neq 1 + 6.$$

Multilinearity splits one row or column while the others stay fixed. It does not split a sum of entire matrices.

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Elimination and pivots

If elimination reaches upper-triangular U without row exchanges,

$$\det A = \det U = u_{11}u_{22}\cdots u_{nn}.$$

With q row exchanges,

$$\det(PA) = (-1)^q \det A,$$

compute the pivot product of PA and multiply by $(-1)^q$.

For example,

$$A = \begin{bmatrix} 0 & 2 \\ 3 & 4 \end{bmatrix}.$$

Swap rows：

$$PA = \begin{bmatrix} 3 & 4 \\ 0 & 2 \end{bmatrix}, \qquad \det(PA) = 6.$$

Thus $\det A = -6$, agreeing with $ad - bc = 0 \cdot 4 - 2 \cdot 3 = -6$.

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Elimination and pivots (continued)

Rank often tests invertibility more directly; signed-volume calculations require the exchange sign.

For completeness, dependent columns make one column a combination of the others. Column multilinearity then expands the determinant into terms with repeated columns, all zero. Therefore $\det A \neq 0$ forces independent columns, a basis of $\mathbb{R}^n$, and invertibility. Conversely, invertibility gives

$$\det A \det(A^{-1}) = \det(AA^{-1}) = 1.$$

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The permutation formula

The inversion count $N(\sigma)$ counts pairs with $i < j$ and $\sigma(i) > \sigma(j)$. Define

$$\operatorname{sgn}(\sigma) = (-1)^{N(\sigma)}.$$

Big formula:

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{1,\sigma(1)} \cdots a_{n,\sigma(n)}.$$

Each term selects one entry from every row and every column. In size 3×3,

$$a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}$$
$$- a_{13}a_{22}a_{31} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33}.$$

The first three permutations are even and the last three are odd. The Sarrus pattern is specific to dimension 3 and does not extend unchanged to larger matrices.

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Cofactor expansion

Delete row i and column j to form minor matrix M_{ij}. Its signed determinant is the cofactor

$$C_{ij} = (-1)^{i+j} \det M_{ij}.$$

Expansion along row i gives

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}.$$

An analogous formula holds along any column.

For example,

$$A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix}.$$

Expanding along row 1 gives

$$\det A = 2(12 - 2) - 1(-4) + 0 = 24.$$

Expanding along row 2 gives

Cofactor expansion (continued)

$$(-1)(-4) + 3(8) + 1(-4) = 24.$$

Agreement follows from the cofactor identity, rather than from a special property of this example.

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Deriving cofactor expansion

Expand row i as

$$r_i = \sum_{j=1}^{n} a_{ij} e_j^T.$$

Row multilinearity gives

$$\det A = \sum_{j=1}^{n} a_{ij} D(A_{i \leftarrow e_j}).$$

In $A_{i \leftarrow e_j}$, row i has a single 1 in column j. Move that row to the top with $i-1$ adjacent swaps and that column to the left with $j-1$ adjacent swaps. The first row is now $(1, 0, \ldots, 0)$. Clear the rest of the first column by row additions; the remaining lower-right block is M_{ij}.

Consequently,

$$D(A_{i \leftarrow e_j}) = (-1)^{i+j} \det M_{ij} = C_{ij}.$$

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Deriving cofactor expansion (continued)

Summing the terms gives cofactor expansion. The checkerboard signs record the parity of those adjacent swaps.

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Pivots and a tridiagonal recurrence

If pivoted elimination gives $PA = LU$,

$$\det(PA) = \prod_i u_{ii},$$

$$\det A = (-1)^q \prod_i u_{ii}.$$

A zero leading principal minor need not mean singularity; a row exchange may suffice.

For the tridiagonal matrix,

$$T_n = \begin{bmatrix} 2 & -1 & 0 & \cdots \\ -1 & 2 & -1 & \ddots \\ 0 & -1 & 2 & \ddots \\ \vdots & \ddots & \ddots & \ddots \end{bmatrix}.$$

expand along the first row.

$$D_n = 2D_{n-1} - D_{n-2}.$$

Pivots and a tridiagonal recurrence (continued)

The initial values $D_0 = 1, D_1 = 2$ give

$$D_n = n + 1.$$

All leading principal minors are positive, also proving that T_n is positive definite.

The adjugate

Cofactor matrix $C = [C_{ij}]$, adjugate:

$$\operatorname{adj} A = C^T.$$

Cofactor identities:

$$\sum_{j=1}^{n} a_{ij} C_{kj} = \begin{cases} \det A, & i = k, \\ 0, & i \neq k. \end{cases}$$

For $i \neq k$, the sum is the determinant after replacing row k by row i. The repeated rows make it zero.

Hence

$$A(\operatorname{adj} A) = (\det A) I.$$

If $\det A \neq 0$,

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A.$$

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The adjugate (continued)

2×2:

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

An adjugate requires n^2 minors, so it is mainly useful for identities and small matrices.

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Cramer's rule

Suppose A is invertible and $Ax = b$. Replace column j of the identity by x. Linearity and repeated-column cancellation give

$$x_j = \det(e_1, \dots, e_{j-1}, x, e_{j+1}, \dots, e_n).$$

Multiply this matrix by A and use the determinant product rule to obtain

$$(\det A)x_j = \det(a_1, \dots, a_{j-1}, b, a_{j+1}, \dots, a_n).$$

Thus

$$x_j = \frac{\det A_j}{\det A},$$

where A_j replaces column j of A by b.

For example,

$$\begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Here $\det A = -2$, $\det A_1 = -4$, and $\det A_2 = 2$, giving

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Cramer's rule (continued)

$$x_1 = 2, \qquad x_2 = -1.$$

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Limits of Cramer's rule

Cramer's rule makes each x_j linear in b, since the determinant is linear in the replaced column.

Computing n determinants separately is more costly than one pivoted LU factorization for many numerical solves. A small nonzero denominator alone gives no reliability guarantee.

When $\det A = 0$, Cramer's quotient is undefined. Use ranks and the augmented matrix to distinguish

- no solution;
- infinitely many solutions.

A zero determinant states singularity of the square coefficient matrix. It does not state incompatibility of a particular right-hand side.

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Determinants as signed areas

Let

$$u = (a, c)^T, \qquad v = (b, d)^T,$$
$$P = [u\ v] = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Parallelogram area：

$$|\det P| = |ad - bc|.$$

Without the absolute value, the sign records the orientation from u to v.

For

$$u = (2, 1)^T, \qquad v = (1, 3)^T,$$

Then

$$\det P = 6 - 1 = 5,$$

both the signed area and the area are 5.

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Determinants as signed areas (continued)

For the linear map $x \mapsto Px$,

$$\text{area}(P(S)) = |\det P| \, \text{area}(S).$$

if $\det P = 0$, the image has dimension at most one and area zero.

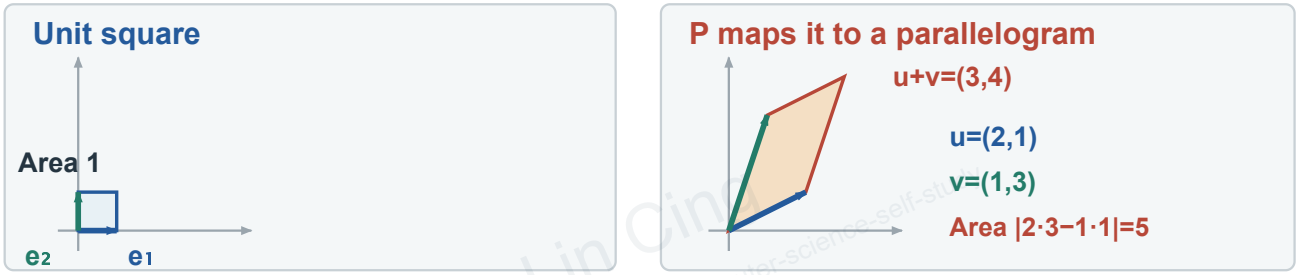
The diagram maps the unit-square edges $(1,0)^T, (0,1)^T$ to $u = (2,1)^T, v = (1,3)^T$. Their respective areas are

$$|\det I| = 1, \qquad |\det P| = |2 \cdot 3 - 1 \cdot 1| = 5.$$

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Determinants as signed areas (continued)

Signed area: det([[2,1],[1,3]])=5



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Volume and orientation in n dimensions

For parallelepiped edges $a_1, \ldots, a_n$ placed as columns,

$$\text{volume} = |\det A|.$$

This uses standard coordinate units. Convert incompatible units before interpreting physical volumes.

Exchanging edges reverses signed orientation but preserves physical volume. Adding a multiple of one edge to another is a shear and preserves both.

Elimination tracks volume through row scaling and exchanges, while the pivot product supplies the resulting signed determinant.

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Cross products and determinants

In $\mathbb{R}^3$,

$$w = u \times v$$

is perpendicular to u, v, with norm equal to their parallelogram area. Its components are

$$w_1 = u_2 v_3 - u_3 v_2,$$
$$w_2 = u_3 v_1 - u_1 v_3,$$
$$w_3 = u_1 v_2 - u_2 v_1.$$

Hence

$$w^T u = 0, \qquad w^T v = 0.$$

Antisymmetry gives

$$v \times u = -u \times v$$

the same sign change as exchanging determinant columns.

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Cross products and determinants (continued)

The formal determinant using i, j, k is a mnemonic for cofactor directions. Basis vectors are not scalar entries in the same field. The component formulas provide an unambiguous definition.

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A numerical example

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 2 & 1 & 1 \end{bmatrix}.$$

Cofactor expansion along row 1 gives

$$\det A = 1(1 - 3) - 2(0 - 6) = -2 + 12 = 10.$$

Adding twice column 1 to column 2 produces B with the same determinant, 10. The operation is a shear.

The columns span a parallelepiped with signed volume 10. Exchanging two columns reverses the sign. Dependence collapses the volume to zero.

Adjugate and Cramer computations can check small examples. Computing n^2 minors is unsuitable for a large numerical inverse.

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Chapter check

You can now do the following.

- Derive zero determinant for repeated columns from alternation and multilinearity.
- Prove the determinant product rule.
- Derive cofactor expansion from the permutation structure.
- Distinguish pivot products from leading principal minors.
- Preserve the nonzero determinant assumption in inverse formulas and Cramer's rule.
- Distinguish signed and physical area.
- Use cofactors for small calculations and stable elimination for large dense systems.

Exact singularity does not mean small entries. Determinant magnitude alone does not provide a numerical stability conclusion.

Chapter 6 uses determinants to build characteristic polynomials and studies eigenvectors and dynamics.

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The eigenproblem

For $A \in \mathbb{C}^{n \times n}$, suppose a scalar λ and nonzero vector x satisfy

$$Ax = \lambda x,$$

Then λ is an eigenvalue and x an eigenvector.

Equivalently,

$$(A - \lambda I)x = 0.$$

Since $x \neq 0$, the matrix $A - \lambda I$ is singular. Hence

$$p_A(\lambda) = \det(\lambda I - A) = 0.$$

The one-dimensional subspace $\operatorname{span}\{x\}$ is invariant. Over the reals, a negative eigenvalue reverses the arrow direction while preserving the line.

Characteristic polynomial, trace, and determinant

Using the monic convention for an $n \times n$ matrix,

$$p_A(\lambda) = \lambda^n - a_1\lambda^{n-1} + \cdots + (-1)^n \det A.$$

Counting eigenvalues with algebraic multiplicity gives

$$\sum_i \lambda_i = \operatorname{tr} A, \qquad \prod_i \lambda_i = \det A.$$

For example,

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

The trace is 6 and determinant is 8.

$$p_A(\lambda) = \lambda^2 - 6\lambda + 8 = (\lambda - 4)(\lambda - 2).$$

The eigenvalues are 4 and 2. Trace and determinant check the spectrum but do not generally determine it in dimensions above two.

How operations transform eigenvalues

If $Ax = \lambda x$, then for nonnegative integers k and polynomials q,

$$A^k x = \lambda^k x,$$

$$A^{-1}x = \lambda^{-1}x \quad (A \text{ invertible}),$$

$$q(A)x = q(\lambda)x.$$

Shift:

$$(A + cI)x = (\lambda + c)x.$$

Eigenspace:

$$E_\lambda = N(A - \lambda I).$$

The eigenspace is a subspace whose dimension is the geometric multiplicity. A triangular matrix has its diagonal entries as eigenvalues, counted with algebraic multiplicity.

Algebraic and geometric multiplicities

The multiplicity of a root in the characteristic polynomial is its algebraic multiplicity a_λ. The quantity

$$g_\lambda = \dim N(A - \lambda I)$$

is its geometric multiplicity. Extend an eigenspace basis to a basis of the whole space. The resulting matrix is block upper triangular with block λI_{g_λ}, so its characteristic polynomial contains $(t - \lambda)^{g_\lambda}$. Therefore

$$1 \le g_\lambda \le a_\lambda.$$

If any eigenvalue has $g_\lambda < a_\lambda$, the matrix is defective.

For example,

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

The algebraic multiplicity is 2. Solving

Algebraic and geometric multiplicities (continued)

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x = 0$$

gives only $x_2 = 0$, hence

$$E_2 = \text{span}\{(1,0)^T\}, \qquad g_2 = 1.$$

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A complete two-dimensional example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

$\lambda = 4$:

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} x = 0,$$

Choose $x_1 = (1,1)^T$.

$\lambda = 2$:

$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} x = 0,$$

Choose $x_2 = (1,-1)^T$.

These eigenvectors are orthogonal because A is symmetric. If

$$u_0 = c_1 x_1 + c_2 x_2,$$

Then

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A complete two-dimensional example (continued)

$$u_k = A^k u_0 = c_1 4^k x_1 + c_2 2^k x_2.$$

For $c_1 \neq 0$, the eigenvalue-4 mode determines the limiting direction.

Taking $c_1 = c_2 = 1$,

$$u_0 = x_1 + x_2 = (2, 0)^T, \quad u_1 = (6, 2)^T, \quad u_2 = (20, 12)^T.$$

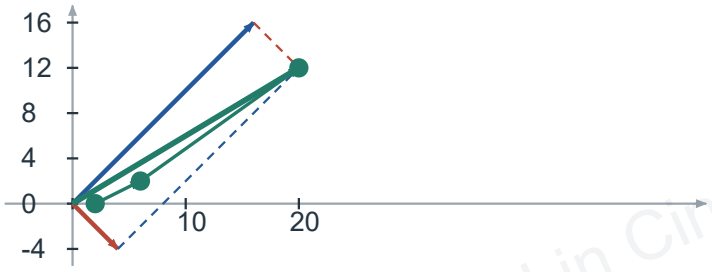
The diagram shows $c_1 4^k x_1$, $c_2 2^k x_2$, and their sum u_k. For $c_1 = 0, c_2 = 1$, instead,

$$u_0 = (1, -1)^T, \quad u_1 = (2, -2)^T, \quad u_2 = (4, -4)^T,$$

the trajectory stays in the eigenvector direction x_2. It has no component in the larger-eigenvalue mode. Real coefficients can also be used with unit eigenvectors; here the diagram retains the specified nonunit x_1, x_2, so its coefficients refer to that basis.

A complete two-dimensional example (continued)

Eigenmodes: components, sum, and trajectory



A=[[3,1],[1,3]]

$\lambda_1=4 \cdot x_1=(1,1)$

$\lambda_2=2 \cdot x_2=(1,-1)$

Blue: $4^2 x_1=(16,16)$

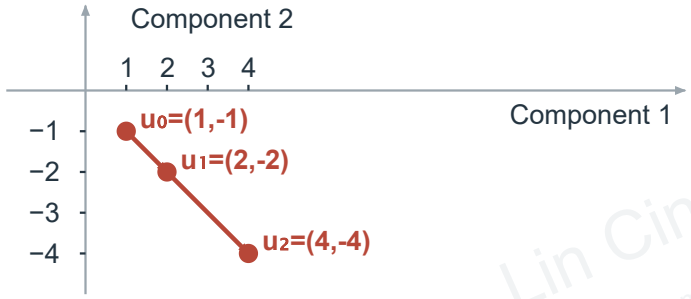
Red: $2^2 x_2=(4,-4)$

Green: $u_2=(20,12)$

$u_0,u_1,u_2=(2,0),(6,2),(20,12)$

A complete two-dimensional example (continued)

Exceptional trajectory $c_1=0$: always along $x_2=(1,-1)$



$c_1=0$ · $c_2=1$

$u_k=2^k(1,-1)$

Every u_k has the same eigenvector direction.

Fixed ratio; no 4^k dominant term.

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Rotation and complex eigenvalues

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

$$p(\lambda) = \lambda^2 - 2\cos\theta\,\lambda + 1,$$

$$\lambda = e^{\pm i\theta}.$$

Unless $\theta \equiv 0 \pmod{\pi}$, a planar rotation has no nonzero real eigenvector.

Complex eigenvalues of a real matrix come in conjugate pairs. If

$$Ax = \lambda x,$$

taking complex conjugates gives

$$A\overline{x} = \overline{\lambda}\,\overline{x}.$$

Real solutions require compatible conjugate coefficients. A single complex mode is generally not a real physical solution.

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Diagonalization

Suppose A has n independent eigenvectors x_i. Form

$$X = [x_1 \ \cdots \ x_n].$$

$$AX = X\Lambda, \qquad \Lambda = \operatorname{diag}(\lambda_i),$$

$$A = X\Lambda X^{-1}, \qquad X^{-1}AX = \Lambda.$$

Diagonalization is equivalent to the eigenspace dimensions summing to n. Eigenvectors for distinct eigenvalues are independent. Indeed, a shortest nontrivial relation $\sum c_i x_i = 0$, after applying $A - \lambda_k I$, would give a shorter relation, a contradiction. Thus n distinct eigenvalues suffice. Repeated eigenvalues can also have complete eigenspaces, as for $2I$.

Powers and difference equations

$$A^k = X\Lambda^k X^{-1}.$$

Recurrence

$$u_{k+1} = Au_k$$

has solution

$$u_k = \sum_{j=1}^{n} c_j \lambda_j^k x_j.$$

If every

$$|\lambda_j| < 1,$$

zero solution asymptotically stable.Spectral radius:

$$\rho(A) = \max_j |\lambda_j|.$$

Powers and difference equations (continued)

then every fixed initial vector decays to zero. The condition $\rho(A) < 1$ guarantees this even for defective matrices, since polynomial factors cannot overcome geometric decay. Eigenvalues on the unit circle can give bounded, periodic, or defective polynomial growth.

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Similarity invariants

If

$$C = B^{-1}AB,$$

then

$$C - \lambda I = B^{-1}(A - \lambda I)B,$$

Thus characteristic polynomial, eigenvalues, trace, determinant, rank, and Jordan structure are unchanged.

If $Ax = \lambda x$, set $y = B^{-1}x$. Then

$$Cy = \lambda y.$$

Similarity changes coordinates while preserving the intrinsic spectrum.

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Defective matrices and Jordan structure

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

There is only one eigendirection, so no similarity can diagonalize this matrix.

Write

$$A = 2I + N, \qquad N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad N^2 = 0.$$

Since $2I$ and N commute,

$$A^k = 2^k I + k2^{k-1} N = \begin{bmatrix} 2^k & k2^{k-1} \\ 0 & 2^k \end{bmatrix}.$$

Let $e_1 = (1, 0)^T, e_2 = (0, 1)^T$. Then

$$(A - 2I)e_1 = 0, \qquad (A - 2I)e_2 = e_1.$$

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Defective matrices and Jordan structure (continued)

Thus e_1 is an eigenvector, e_2 a generalized eigenvector, and (e_1, e_2) a Jordan chain of length 2. The diagonal describes scaling along e_1; the off-diagonal mixes e_1 into the image of e_2, producing the factor $k2^{k-1}$ in matrix powers. This is the Jordan block $J_2(2)$. Its missing independent eigenvector prevents a decomposition of the whole space into eigenvector lines.

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Spectral decomposition

If the rows of X^{-1} are y_j^T,

$$A = \sum_{j=1}^{n} \lambda_j x_j y_j^T.$$

Each $x_j y_j^T$ has rank one. For a real symmetric matrix choose a real orthogonal matrix Q:

$$A = Q \Lambda Q^T = \sum_j \lambda_j q_j q_j^T.$$

This scales independent orthonormal directions. Nonnormal matrices generally have nonorthogonal eigenvectors, so eigenvalues alone do not describe transient amplification.

From scalar to matrix ODEs

Scalar:

$$u' = \lambda u \quad \Longrightarrow \quad u(t) = Ce^{\lambda t}.$$

System:

$$u' = Au, \qquad u(0) = u_0.$$

If A is diagonalizable,

$$u(t) = \sum_j c_j e^{\lambda_j t} x_j.$$

$$c = X^{-1} u_0.$$

For $\lambda = \alpha + i\beta$,

$$e^{\lambda t} = e^{\alpha t}(\cos \beta t + i \sin \beta t).$$

The real part determines exponential growth or decay. The imaginary part determines oscillation.

The matrix exponential

$$e^{At} = I + At + \frac{A^2t^2}{2!} + \cdots.$$

The absolutely convergent power series can be differentiated term by term on bounded time intervals.

$$\frac{d}{dt}e^{At} = Ae^{At} = e^{At}A.$$

Diagonalizable:

$$e^{At} = Xe^{\Lambda t}X^{-1} = \sum_j e^{\lambda_j t}x_j y_j^T.$$

Defective $A = 2I + N$, $N^2 = 0$:

$$e^{At} = e^{2t}(I + tN).$$

Eigenvalue real parts determine asymptotic stability. Nonnormal transient behavior also depends on norms or more detailed information about the matrix.

Stability and its boundary cases

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}.$$

Use eigenbasis：

$$u(t) = c_1 e^{4t}(1,1)^T + c_2 e^{2t}(1,-1)^T.$$

Both eigenvalues are positive, so every nonzero initial vector produces a growing solution.

The zero solution of a linear system is asymptotically stable exactly when every eigenvalue has negative real part. A Jordan block $J = \lambda I + N$ of size m gives $e^{Jt} = e^{\lambda t}\sum_{\ell=0}^{m-1} t^\ell N^\ell/\ell!$. A negative real part makes every polynomial times the decaying exponential tend to zero. A positive real part gives an unstable mode.

Stability and its boundary cases (continued)

At the boundary, if all real parts are nonpositive and every imaginary-axis Jordan block has size one, e^{At} is bounded and the zero solution is Lyapunov stable. Any remaining imaginary-axis mode prevents asymptotic stability. For $A = 0$, solutions retain their initial values; for skew-symmetric rotation, they retain their length. In contrast, $J = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ gives $e^{Jt} = I + tJ$, taking e_2 to $(t, 1)^T$. Zero real parts can therefore still be unstable. These conclusions concern the stated linear system.

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Converting a second-order ODE

$$u'' + Bu' + Cu = 0.$$

For scalar constants B, C, set $v = u'$.

$$\frac{d}{dt}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -C & -B \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix}.$$

Characteristic polynomial:

$$\det\begin{bmatrix} -\lambda & 1 \\ -C & -B-\lambda \end{bmatrix} = \lambda^2 + B\lambda + C.$$

Complex conjugate eigenvalues produce oscillation and their real parts determine decay. The initial data $(u(0), u'(0))$ become a two-dimensional initial vector.

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Real eigenvalues of real symmetric matrices

Let $S \in \mathbb{R}^{n \times n}$, $S = S^T$, and choose a nonzero $x \in \mathbb{C}^n$ with $Sx = \lambda x$. Real entries and symmetry give $S^* = \overline{S}^T = S$. Using the conjugate transpose,

$$x^* S x = \lambda x^* x.$$

because

$$\overline{x^* S x} = x^* S^* x = x^* S x,$$

the numerator is real. Since $x^* x > 0$,

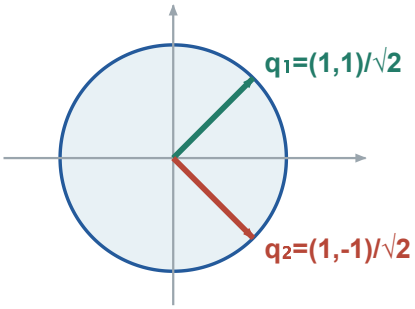
$$\lambda = \frac{x^* S x}{x^* x} \in \mathbb{R}.$$

The first equality uses conjugation of a scalar; the second uses $S^* = S$, which does not follow from $S^T = S$ for a complex matrix. The denominator $x^* x = \sum_j |x_j|^2 > 0$ establishes the conclusion.

The counterexample $S = [i]$ satisfies $S^T = S$ but $S^* = [-i] \neq S$, with eigenvalue i. The complex-space premise is Hermitian symmetry $S^* = S$. The same proof gives real eigenvalues, and diagonalization uses a unitary matrix $S = U \Lambda U^*$.

Real eigenvalues of real symmetric matrices (continued)

Orthonormal eigenbasis: $Q^T S Q = \mathrm{diag}(4,2)$



S=[[3,1],[1,3]]

Q=(1/√2)[[1,1],[1,-1]]

$Q^T S Q = \mathrm{diag}(4,2)$

$S = Q \Lambda Q^T$; the axes are perpendicular.

Normalize (1,1),(1,-1) before forming Q.

Real eigenvalues of real symmetric matrices (continued)

Complex symmetric does not imply Hermitian

S=[i] , $S^T=S$

Only eigenvalue: i

$S^*=[-i]\neq S$

The Hermitian premise is necessary.

Hermitian : $S^*=S$

$x^*Sx=\lambda x^*x$

$x^*x>0 \Rightarrow \lambda\in R$

$S=U\Lambda U^*$, with U unitary

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Orthogonality of distinct eigenspaces

Let $S \in \mathbb{R}^{n\times n}$ be real symmetric, with real eigenvectors x, y. If $Sx = \lambda x$, $Sy = \mu y$,

$$\lambda x^T y = (Sx)^T y = x^T S^T y = \mu x^T y.$$

Consequently,

$$(\lambda - \mu)x^T y = 0.$$

then $\lambda \neq \mu$ implies $x \perp y$. A complete orthonormal eigenbasis follows by induction. The continuous quadratic form $x^T Sx$ attains a maximum on the unit sphere; its extremum condition gives a real unit eigenvector q. For $v \perp q$, $q^T Sv = (Sq)^T v = 0$, so $q^\perp$ is invariant. The restriction remains symmetric. Induction gives a complete orthonormal eigenbasis; within repeated eigenspaces, use Gram–Schmidt.

Spectral theorem：

$$S = Q\Lambda Q^T.$$

For example,

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Orthogonality of distinct eigenspaces (continued)

$$S = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad Q = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Since $Q^T Q = I$ and

$$Q^T S Q = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix},$$

we obtain $S = Q \Lambda Q^T$. The unnormalized vectors $(1, 1)^T$, $(1, -1)^T$ specify directions only. Divide them by $\sqrt{2}$ before using them as columns of Q.

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Equivalent positive-definiteness conditions

For real symmetric S, the following conditions are equivalent.

1. $x^T S x > 0$ for every $x \neq 0$.
2. Every eigenvalue is positive.
3. Every leading principal minor is positive.
4. Symmetric elimination without swaps has positive pivots.
5. $S = A^T A$ for some full-column-rank A.
6. $S = R R^T$ for an invertible lower-triangular R with positive diagonal, the Cholesky factorization.

Symmetric elimination completes squares, giving $S = L D L^T$ and $x^T S x = (L^T x)^T D (L^T x)$. Positive pivots are therefore equivalent to positive energy. Leading minors satisfy $\Delta_k = d_1 \cdots d_k$, so positivity of all leading minors is equivalent to positive pivots. Taking $R = L D^{1/2}$ gives Cholesky, and $A = R^T$ gives the Gram form. Conversely, squared lengths give positive energy. A positive determinant alone is insufficient; $-I_2$ has determinant 1 but is negative definite.

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Equivalent positive-definiteness conditions (continued)

2×2：

$$S = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$$

is positive definite exactly when $a > 0, ac - b^2 > 0$.

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Energy proof and counterexamples

$$x^T S x = \sum_i \lambda_i (q_i^T x)^2.$$

If all $\lambda_i > 0$, every nonzero x has some $q_i^T x \neq 0$, making the energy positive. Conversely, an eigenvalue $\lambda_i \leq 0$ supplies a nonzero eigenvector with nonpositive energy, contradicting strict positive definiteness.

For a counterexample,

$$S_1 = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

eigenvalues $3, -1$，not positive definite。

If A has independent columns,

$$x^T A^T A x = \|Ax\|^2 > 0$$

for every $x \neq 0$, so $A^T A$ is positive definite. Full row rank cannot replace column independence.

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Positive semidefinite boundary

$$S = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

At $x = (1, -1)^T$, $x^T S x = 0$; the eigenvalues are $2, 0$. Thus the matrix is positive semidefinite but not positive definite.

For $S = A^T A$, dependent columns give nonzero x with $Ax = 0$ and zero energy. Conversely, zero energy forces $Ax = 0$.

A semidefinite matrix can have an LDL^T factorization, but Cholesky must handle zero pivots. A pseudoinverse does not make a semidefinite matrix invertible.

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Ellipsoids and conditioning

Positive definite S:

$$x^T S x = 1$$

is an origin-centered ellipsoid. In eigen-coordinates,

$$y = Q^T x,$$
$$\sum_i \lambda_i y_i^2 = 1.$$

The semiaxis length in direction i is

$$1/\sqrt{\lambda_i}.$$

The smallest eigenvalue gives the longest semiaxis. Also,

$$\kappa_2(S) = \frac{\lambda_{\max}}{\lambda_{\min}}.$$

Positive definiteness guarantees exact invertibility, but a small minimum eigenvalue relative to the largest still gives poor conditioning.

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Chapter check

You can now do the following.

- Find eigenvalues from the characteristic equation.
- Distinguish algebraic and geometric multiplicities.
- Give defective counterexamples.
- Solve powers and ODEs by diagonalization when an eigenbasis exists.
- Combine complex conjugate modes into real solutions.
- Prove the real symmetric spectral theorem and orthogonality relations.
- Connect energy, minors, pivots, Cholesky, and Gram factorizations.
- Interpret conditioning through ellipsoid semiaxes.

SVD in Chapter 7 applies to rectangular matrices and does not require a complete eigenbasis of the original matrix.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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Matrices as data

A grayscale image can be represented by $A \in \mathbb{R}^{m \times n}$, with pixel intensities as entries. Color images usually form tensors; a matrix treatment must specify how the channels are arranged.

Rank-one matrix:

$$uv^T = \begin{bmatrix} u_1 \\ \vdots \\ u_m \end{bmatrix} \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix},$$

$$(uv^T)_{ij} = u_i v_j.$$

A rank-one factorization stores $m + n$ scalars rather than mn entries. SVD expresses a general matrix as a weighted sum of such pieces. Large singular values often capture substantial reconstruction energy.

Statement of the SVD

Every

$$A \in \mathbb{R}^{m \times n}, \qquad \operatorname{rank} A = r,$$

admits a full SVD

$$A = U \Sigma V^T.$$

$$U \in \mathbb{R}^{m \times m}, \qquad V \in \mathbb{R}^{n \times n},$$

where U and V have orthonormal columns. The nonzero diagonal entries of the $m \times n$ matrix Σ are

$$\sigma_1 \geq \cdots \geq \sigma_r > 0.$$

Reduced SVD:

$$A = U_r \Sigma_r V_r^T,$$

$$U_r \in \mathbb{R}^{m \times r}, \quad \Sigma_r \in \mathbb{R}^{r \times r}, \quad V_r \in \mathbb{R}^{n \times r}.$$

The remaining entries of the full rectangular Σ are zero. The reduced SVD retains only the r positive singular values and their vectors.

Right singular vectors from A^TA

$$x^TA^TAx = \|Ax\|^2 \geq 0.$$

Thus A^TA is symmetric positive semidefinite and has spectral decomposition

$$A^TA = V\Lambda V^T.$$

Order the positive eigenvalues and set

$$\lambda_1 \geq \cdots \geq \lambda_r > 0,$$

$$\sigma_i = \sqrt{\lambda_i}.$$

For $i \leq r$,

$$\|Av_i\|^2 = v_i^TA^TAv_i = \sigma_i^2,$$

so $Av_i \neq 0$.

The vectors $v_1, \ldots, v_r$ form an orthonormal row-space basis. The vectors $v_{r+1}, \ldots, v_n$ form an orthonormal nullspace basis.

Left singular vectors

Define

$$u_i = \frac{Av_i}{\sigma_i}.$$

For $i \neq j$, orthogonality follows from

$$\sigma_i\sigma_ju_i^Tu_j = v_i^TA^TAv_j = \lambda_jv_i^Tv_j = 0.$$

The lengths satisfy

$$\|u_i\|^2 = \frac{\|Av_i\|^2}{\sigma_i^2} = 1.$$

Consequently,

$$Av_i = \sigma_iu_i.$$

The vectors $u_1, \ldots, u_r$ form an orthonormal column-space basis. Complete them with $u_{r+1}, \ldots, u_m$, an orthonormal basis of the left nullspace.

To complete the existence proof, let $V = [v_1, \ldots, v_n]$ be an orthonormal eigenbasis of A^TA. For $i > r$,

Left singular vectors (continued)

$$\|Av_i\|^2 = v_i^T A^T A v_i = 0$$

so $Av_i = 0$. Extend $u_1, \dots, u_r$ to a complete orthonormal basis of $\mathbb{R}^m$. For any $x = \sum_{j=1}^{n} c_j v_j$,

$$Ax = \sum_{j=1}^{r} c_j A v_j = \sum_{j=1}^{r} c_j \sigma_j u_j = U \Sigma V^T x.$$

This verifies the right vectors, left vectors, singular values, and factorization together.

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Comparing $A^T A$ and AA^T

$$AA^T = U \Sigma \Sigma^T U^T.$$

The two Gram matrices have the same positive eigenvalues and rank. The multiplicity of zero can differ because their sizes can differ.

This explains the general structure.

- The eigenvectors of $A^T A$ give V.
- The eigenvectors of AA^T give U.
- The positive eigenvalues are the squared singular values.
- Rectangular and rank-deficient matrices also admit SVD.

Numerical implementations usually avoid forming $A^T A$, which squares the relevant condition number and can overflow. Stable methods use bidiagonal reduction and algorithms for the resulting bidiagonal singular-value problem.

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Rank-one expansion

Reduced SVD:

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T.$$

The Frobenius inner product is $\langle B, C \rangle_F = \operatorname{tr}(B^T C) = \sum_{ij} B_{ij} C_{ij}$, and $\|B\|_F = \sqrt{\sum_{ij} B_{ij}^2}$. Distinct rank-one terms are orthogonal in this inner product.

$$\langle u_i v_i^T, u_j v_j^T \rangle_F = 0, \quad i \neq j.$$

Consequently,

$$\|A\|_F^2 = \sum_i \sigma_i^2.$$

Retaining the first k terms preserves the energy fraction

$$\frac{\sum_{i=1}^{k} \sigma_i^2}{\sum_{i=1}^{r} \sigma_i^2}.$$

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Rank-one expansion (continued)

This ratio requires $A \neq 0$. The nuclear norm is $\|A\|_* = \sum_i \sigma_i$, which differs from the Frobenius norm. Reconstruction energy need not match perceptual relevance, so an image application must specify its error metric and task.

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Truncated SVD and Eckart–Young

$$A_k = \sum_{i=1}^{k} \sigma_i u_i v_i^T.$$

Remainder：

$$A - A_k = \sum_{i=k+1}^{r} \sigma_i u_i v_i^T.$$

For every matrix B of rank at most k,

$$\|A - B\|_2 \geq \sigma_{k+1},$$

$$\|A - B\|_F \geq \sqrt{\sum_{i=k+1}^{r} \sigma_i^2}.$$

Truncated SVD and Eckart–Young (continued)

For the spectral bound, B has a nonzero kernel vector in $\operatorname{span}\{v_1, \ldots, v_{k+1}\}$. Normalize it to $\|z\| = 1$. Then $\|(A - B)z\| = \|Az\| \geq \sigma_{k+1}$. For the Frobenius bound, let P project onto $C(B)$. Then $\|A - B\|_F^2 \geq \|(I - P)A\|_F^2 = \|A\|_F^2 - \operatorname{tr}(PAA^T)$. Extend the left singular vectors to an orthonormal basis $u_1, \ldots, u_m$, setting $\sigma_i = 0$ for $i > r$. Let $\alpha_i = u_i^T P u_i$. Since $0 \leq \alpha_i \leq 1$ and $\sum_{i=1}^{m} \alpha_i = \operatorname{tr} P = \operatorname{rank} P \leq k$, we have $\operatorname{tr}(PAA^T) = \sum_{i=1}^{m} \sigma_i^2 \alpha_i \leq \sum_{i=1}^{k} \sigma_i^2$. The discarded terms of A_k attain both lower bounds. Here $0 \leq k < r$; for $k \geq r$, the optimal error is zero.

If $\sigma_k = \sigma_{k+1}$, the optimal subspace need not be unique. Individual axes may rotate within a tied singular subspace.

For singular values $9, 4, 1, 0$, the rank is 3. At $k = 1$, the optimal errors are

$$\|A - A_1\|_2 = 4, \qquad \|A - A_1\|_F = \sqrt{4^2 + 1^2} = \sqrt{17}.$$

At $k = 2$, both errors equal 1; at $k = 3$, both vanish. The spectral error is the largest discarded singular value, while the Frobenius error is the square root of their squared sum.

Storage cost of low-rank compression

Rank-k factors store

$$mk + nk + k = k(m + n + 1)$$

raw scalars, plus finite-precision metadata. Raw storage can decrease only if this count is below mn.

For a 1000×1000 image and rank 10, the factor count is 2.001×10^4, compared with 10^6 entries. Actual formats also store headers, quantization information, and other metadata.

Random full-rank data often compress poorly. Compressibility is a property of the data, not a guarantee supplied by SVD.

The data orientation in PCA

Let

$$A_0 \in \mathbb{R}^{m \times n},$$

with variables in rows and samples in columns. With row means μ_j, center the data by

$$A_{js} = (A_0)_{js} - \mu_j.$$

The first principal direction u_1 maximizes the score vector's

$$s_1 = u_1^T A$$

sum of squares,

$$\|u_1^T A\|^2 = u_1^T A A^T u_1.$$

subject to $\|u_1\| = 1$. The spectral theorem gives maximum σ_1^2, attained at a dominant eigenvector of AA^T, the first left singular vector.

Scores and sign ambiguity

From

$$A = U\Sigma V^T$$

the principal scores are

$$S = \Sigma V^T = U^T A.$$

Row i lists all samples' scores on component i. Its squared norm equals σ_i^2, with zero rows included in a full representation.

Changing the signs of both u_i and v_i preserves the factorization. Changing only one changes the reconstruction. An individual sign has no intrinsic meaning.

Repeated singular values permit rotations of the component axes within the tied subspace. Compare subspace angles across PCA fits rather than matching arbitrary column numbers.

Covariance denominators

For variables in rows and $n \geq 2$ samples, the usual sample covariance is

$$\widehat{C}_{n-1} = \frac{1}{n-1} AA^T,$$

The descriptive or population-denominator version is

$$\widehat{C}_n = \frac{1}{n} AA^T.$$

The eigenvectors coincide, while eigenvalues differ by factor $n/(n-1)$. For $n = 1$, the sample-variance denominator vanishes.

Unscaled PCA favors features with larger variance or larger numerical units. Standardization yields correlation-based PCA but changes the original geometry; it is a modeling choice.

A two-dimensional PCA example

Samples:

$$(-1,-1), \quad (0,1), \quad (1,0).$$

Centered matrix:

$$A = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

$$AA^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

$$u_1 = (1,1)^T/\sqrt{2}, \qquad u_2 = (1,-1)^T/\sqrt{2}.$$

Scores:

$$s_1 = \begin{bmatrix} -\sqrt{2} & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix},$$
$$s_2 = \begin{bmatrix} 0 & -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

The score rows are orthogonal and their squared norms are 3 and 1.

How singular vectors determine ellipse axes

The input and output each use an orthonormal basis, generally with different directions. Take

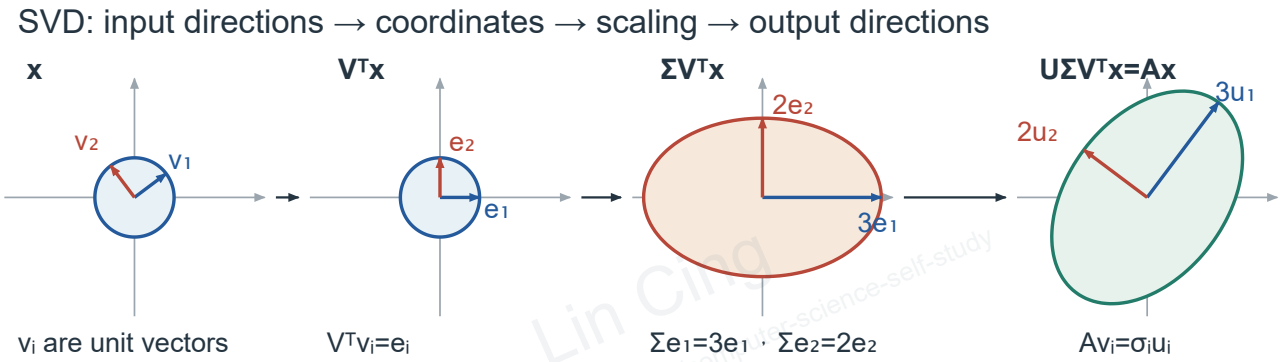
$$V = \frac{1}{5}\begin{bmatrix} 4 & -3 \\ 3 & 4 \end{bmatrix}, \qquad U = \frac{1}{5}\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}, \qquad \Sigma = \operatorname{diag}(3,2).$$

Here $V^TV = U^TU = I$, and

$$A = U\Sigma V^T = \frac{1}{5}\begin{bmatrix} 12 & -1 \\ 6 & 12 \end{bmatrix}.$$

All panels use the same coordinate scale. First V^T maps input directions v_i to e_i, then Σ scales, and finally U sets the output directions. Rotation preserves the circle as a set, but moves the marked vectors; that step must remain visible.

How singular vectors determine ellipse axes (continued)



Check the two axes individually.

$$v_1 = (4/5, 3/5)^T, \quad Av_1 = (9/5, 12/5)^T = 3u_1,$$
$$v_2 = (-3/5, 4/5)^T, \quad Av_2 = (-8/5, 6/5)^T = 2u_2.$$

The semiaxis vectors are $3u_1, 2u_2$. The unit vectors u_1, u_2 specify directions. This full-rank example maps the unit circle to an ellipse boundary and the unit disk to the filled elliptical disk.

SVD geometry with a right-angle rotation

For the unit ball $\|x\| \le 1$,

$$y = Ax = U\Sigma V^T x.$$

all three steps use the same v_i, u_i. Take

$$V = I, \quad \Sigma = \mathrm{diag}(3, 2), \quad U = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A = U\Sigma V^T = \begin{bmatrix} 0 & -2 \\ 3 & 0 \end{bmatrix}.$$

Apply the factors in order.

1. $V^T x = Ix$ leaves input coordinates unchanged; the disk retains the coordinate-axis directions.
2. $\Sigma(V^T x) = (3x_1, 2x_2)$ scales the orthogonal axes, producing semiaxes 3 and 2.
3. $U\Sigma V^T x$ rotates the ellipse, sending the standard axes to $u_1 = (0, 1)^T, u_2 = (-1, 0)^T$.

The definitions give

$$v_1 = (1, 0)^T, \quad Av_1 = (0, 3)^T = 3u_1,$$
$$v_2 = (0, 1)^T, \quad Av_2 = (-2, 0)^T = 2u_2.$$

SVD geometry with a right-angle rotation (continued)

Thus the ball maps to a filled ellipsoid in $C(A)$, with positive singular values as semiaxis lengths. In full-rank two dimensions, its boundary is an ellipse. With a nullspace, the unit sphere can map to the filled ellipsoid rather than only its surface. The maximum singular value is

$$\|Av_1\| = \sigma_1.$$

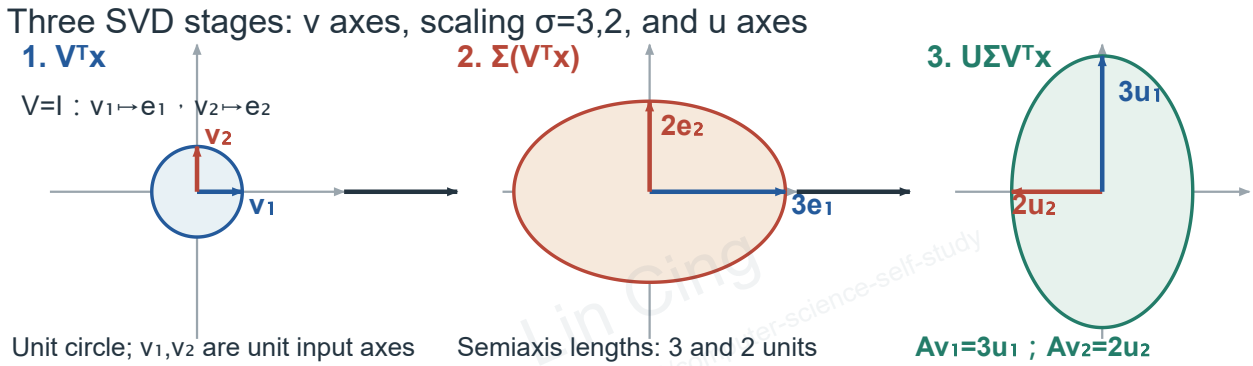
Matrix norm：

$$\|A\|_2 = \sigma_1.$$

The smallest positive σ_r controls the weakest growth from row space to column space.

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SVD geometry with a right-angle rotation (continued)



For deficient rank, take

$$B = \begin{bmatrix} 0 & 0 \\ 3 & 0 \end{bmatrix} = U\,\mathrm{diag}(3,0)V^T.$$

Then $Bv_1 = (0,3)^T$, while $Bv_2 = 0$. The v_2 direction collapses, and the unit disk maps onto the complete segment from $(0,-3)^T$ to $(0,3)^T$.

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SVD geometry with a right-angle rotation (continued)

Rank-deficient SVD: B=[[0,0],[3,0]], σ_2=0 collapses



rank(B)=1; the full image is a line, the circle image a segment

Spectral norm and condition number

$$\|A\|_2 = \max_{x \neq 0} \frac{\|Ax\|_2}{\|x\|_2}.$$

Expand

$$x = \sum_i c_i v_i,$$

$$\|Ax\|^2 = \sum_i \sigma_i^2 |c_i|^2 \leq \sigma_1^2 \|x\|^2.$$

Equality holds at $x = v_1$, so

$$\|A\|_2 = \sigma_1.$$

For full column rank, the usual singular-value condition number is

$$\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

An orthogonal square matrix has every singular value equal to 1 and condition number 1. A rank-deficient matrix has a zero smallest singular value and no finite ordinary inverse condition number.

Distance to singularity and numerical rank

For an invertible square $A \in \mathbb{R}^{n \times n}$, remove $\sigma_n u_n v_n^T$.

$$\|A - A_0\|_2 = \sigma_n.$$

Thus σ_n is the spectral-norm distance to singularity. More generally, A_k is a nearest rank-at-most-k matrix, at distance σ_{k+1}.

Numerical rank uses a tolerance τ.

$$\sigma_i > \tau \|A\|.$$

The tolerance must reflect noise, dimensions, and downstream sensitivity. Stable singular values do not imply stable individual singular vectors, especially near repeated or closely spaced values.

Perturbation radius and numerical rank

For equally sized A, B, let $\delta = \|A - B\|_2$. Eckart–Young gives $\sigma_i(A) = \min_{\operatorname{rank} C < i} \|A - C\|_2$. Use the best low-rank approximation of B and the triangle inequality.

$$\sigma_i(A) \le \|A - B\|_2 + \sigma_i(B).$$

Exchanging A, B proves Weyl's bound $|\sigma_i(A) - \sigma_i(B)| \le \delta$. This is an absolute bound; relative errors in tiny singular values may still be large.

Suppose observed A is within δ of an unknown true B, and let $r_\delta = \#\{i : \sigma_i(A) > \delta\}$. Every compatible B has rank at least r_δ, since those singular values stay positive. Conversely, A_{r_δ} has error $\sigma_{r_\delta+1} \le \delta$, so a compatible rank-$r_\delta$ model exists. If $r_\delta = q$, use $A_q = A$.

Thus r_δ is the minimum rank guaranteed within the perturbation ball, not the exact rank of every compatible model. For $A = \operatorname{diag}(1, 10^{-8}, 10^{-12})$, $\delta = 10^{-10}$, both a rank-2 truncation and the original rank-3 matrix are compatible. The data cannot identify whether the third direction is exactly zero.

Perturbation radius and numerical rank (continued)

For a fixed absolute threshold t, classifications are robust only when $\sigma_i(A) > t + \delta$ or $\sigma_i(A) < t - \delta$. Directions near the threshold are uncertain. A relative threshold $t = \tau\|A\|_2$ also requires a recorded scale and noise model.

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QR pivots are not singular values

In exact arithmetic, $A = Q_1 R$, $Q_1^T Q_1 = I$, gives $A^T A = R^T R$, so A and R share singular values. However, the diagonal of R is not its singular-value list. Take

$$R = \begin{bmatrix} 1 & K \\ 0 & 1 \end{bmatrix}, \qquad K \geq 1.$$

Both diagonal entries are 1, but $\|Re_2\|_2 = \sqrt{K^2 + 1}$, giving $\sigma_{\max} \geq K$. Since $\sigma_{\max}\sigma_{\min} = |\det R| = 1$, we obtain $\sigma_{\min} \leq 1/K$ and $\kappa_2(R) \geq K^2$. A large off-diagonal entry can conceal near dependence from the diagonal alone.

The sizes of R_{ii} can be a heuristic, not a singular-value perturbation theorem. Column-pivoted QR often estimates numerical rank, but guaranteed accuracy requires explicit rank-revealing bounds or an SVD check. A nonzero floating-point pivot need not represent a trustworthy data direction.

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Polar decomposition

$$A = U\Sigma V^T = (UV^T)(V\Sigma V^T).$$

This page concerns square $A \in \mathbb{R}^{n\times n}$, so U, V, Σ all have size $n \times n$. Set

$$Q = UV^T, \qquad S = V\Sigma V^T.$$

Then Q is orthogonal and S symmetric positive semidefinite, giving

$$A = QS.$$

If A is invertible, every singular value is positive, so S is positive definite. The sign of $\det Q$ records orientation reversal. The factors separate an orthogonal rotation or reflection from a symmetric stretch.

The pseudoinverse

Reduced SVD:

$$A = U_r\Sigma_r V_r^T.$$

Moore-Penrose pseudoinverse:

$$A^+ = V_r\Sigma_r^{-1}U_r^T.$$

$$AA^+ = U_rU_r^T, \qquad A^+A = V_rV_r^T.$$

These are the orthogonal projectors onto column space and row space, respectively.

Full column rank:

$$A^+ = (A^TA)^{-1}A^T.$$

Full row rank:

$$A^+ = A^T(AA^T)^{-1}.$$

The pseudoinverse (continued)

The vector A^+b is the minimum-norm least-squares solution. To prove this, write $x = \sum_i c_i v_i$ in the full right singular basis. Then $\|b - Ax\|^2 = \sum_{i=1}^{r}(u_i^T b - \sigma_i c_i)^2 + \sum_{i=r+1}^{m}(u_i^T b)^2$. Minimization requires $c_i = u_i^T b/\sigma_i$ for $i \le r$. Nullspace coefficients remain arbitrary. Setting them to zero uniquely minimizes $\|x\|^2 = \sum_i c_i^2$, giving A^+b.

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Rank-deficient least squares

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = (1,2)^T(1,1).$$

Normalize:

$$u_1 = (1,2)^T/\sqrt{5}, \qquad v_1 = (1,1)^T/\sqrt{2}.$$
$$A = \sqrt{10}\, u_1 v_1^T.$$

Here u_1 is an output direction and v_1 an input direction. Both have two coordinates in this example but have different values. For a rectangular matrix even their dimensions can differ. Rank one does not imply $u_1 = v_1$.

For $b = (1,3)^T$,

$$A^T A = \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}, \qquad A^T b = \begin{bmatrix} 7 \\ 7 \end{bmatrix}.$$

Every parameter satisfying

$$x_1 + x_2 = 7/5$$

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Rank-deficient least squares (continued)

is a minimizer. The normal matrix is singular and cannot be inverted.

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The minimum-norm solution

$$\sigma_1 u_1^T b = \sqrt{10}\frac{1+6}{\sqrt{5}} = 7\sqrt{2}.$$

$$x^+ = A^+ b = \frac{1}{\sigma_1} v_1 u_1^T b = \frac{7}{10}(1,1)^T.$$

Projection:

$$AA^+ b = u_1 u_1^T b = \frac{7}{5}(1,2)^T.$$

Residual:

$$e = b - AA^+ b = \left(-\frac{2}{5}, \frac{1}{5}\right)^T.$$

$$e^T(1,2) = 0,$$

The residual is therefore orthogonal to column space.

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Numerical workflow

SVD workflow:

1. Check shape, missing values, and units.
2. Center PCA data and standardize when the modeling choice requires it.
3. Use a stable SVD algorithm without forming normal equations.
4. Inspect singular-value gaps and numerical rank.
5. Choose k from the error budget.
6. Validate on held-out data or the downstream task.

Signs can flip and tied axes can rotate. Linear reconstruction alone establishes neither causal importance nor fairness.

The four subspaces

Space	Ambient space	SVD basis	Dimension
row space	$\mathbb{R}^n$	$v_1, \dots, v_r$	r
nullspace	$\mathbb{R}^n$	$v_{r+1}, \dots, v_n$	$n - r$
column space	$\mathbb{R}^m$	$u_1, \dots, u_r$	r
left nullspace	$\mathbb{R}^m$	$u_{r+1}, \dots, u_m$	$m - r$

Input and output singular vectors belong to different roles and cannot generally be paired by an inner product. Equal dimensions allow an explicit coordinate identification, but do not make the bases identical. SVD pairs them through

$$Av_i = \sigma_i u_i.$$

Chapter check

You can now do the following.

- Construct v_i from $A^T A$ and set $u_i = Av_i/\sigma_i$.
- Complete the null directions to obtain a full SVD.
- Compute spectral and Frobenius truncation errors.
- Explain PCA centering, scores, variance, and denominator choices.
- Distinguish input and output singular bases.
- Derive the operator norm, condition number, and distance to singularity.
- Construct polar factors and a pseudoinverse.
- Handle sign ambiguity, ties, and numerical rank.

Chapter 8 reinterprets transformations through bases and coordinates.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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Abstract maps and their coordinate matrices

A linear transformation is a map between vector spaces

$$T\colon V \to W,$$

that preserves addition and scalar multiplication. The map itself does not require a basis. Choosing domain and codomain bases gives its matrix representation.

Rotation matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

is the representation of rotation in standard coordinates. Changing bases changes the entries without changing the underlying rotation.

Kernel, image, rank, injectivity, and surjectivity are intrinsic to the map. Matrix entries depend on coordinates. For determinants, invariance holds under a common basis change for an endomorphism; independent domain and codomain changes alter the coordinate volume factors.

Linearity

A map $T\colon V \to W$ is linear if

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v)$$

for all vectors and scalars. Equivalently,

$$T(u+v) = T(u) + T(v), \qquad T(cu) = cT(u).$$

Taking scalar $c = 0$ gives

$$T(0) = 0.$$

An affine map that moves the origin,

$$T(x) = Ax + b, \qquad b \neq 0$$

is therefore not a vector-space linear map. Graphics can represent it linearly after adding a homogeneous coordinate.

Linearity preserves every finite linear combination.

Linearity (continued)

$$T\left(\sum_i c_i v_i\right) = \sum_i c_i T(v_i).$$

Consequently, the images of a basis determine the whole map.

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Kernel, image, and rank-nullity

Define kernel and image by

$$N(T) = \{v \in V : T(v) = 0\},$$
$$C(T) = \{T(v) : v \in V\}.$$

They are subspaces of V and W. For finite-dimensional V, rank-nullity states

$$\dim V = \dim N(T) + \dim C(T).$$

To prove it, take a kernel basis

$$v_1, \dots, v_k,$$

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and extend it to a basis of V.

$$v_1, \dots, v_k, u_1, \dots, u_r.$$

The vectors $T(u_1), \dots, T(u_r)$ span the image by applying T to the full basis expansion. If $\sum_i c_i T(u_i) = 0$, then $\sum_i c_i u_i$ lies in the kernel. Independence of the extended basis forces every $c_i = 0$. Thus these image vectors form a basis.

Injectivity, surjectivity, and dimension

A linear map is injective exactly when

$$N(T) = \{0\}.$$

If the kernel is trivial, $T(u) = T(v)$ implies $T(u - v) = 0$, hence $u = v$. Conversely, injectivity and $T(v) = T(0)$ imply $v = 0$.

Surjectivity means

$$C(T) = W.$$

For a finite-dimensional map $T\colon V \to V$, the following are equivalent.

- T invertible;
- The kernel contains only zero.
- The rank equals $\dim V$.
- coordinate matrix invertible.

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Injectivity, surjectivity, and dimension (continued)

With unequal dimensions, injectivity and surjectivity differ. A map $\mathbb{R}^2 \to \mathbb{R}^3$ can be injective, but its image has dimension at most 2 and cannot fill the codomain.

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Examples of linear maps

Rotation:

$$T(x, y) = (-y, x)$$

has zero kernel and the whole plane as its image.

Projection:

$$P(x, y) = (x, 0)$$

has the vertical axis as kernel and the horizontal axis as image.

Polynomial differentiation:

$$D(p) = p'$$

On $\mathcal{P}_k$, with $k \geq 1$, differentiation has the constant polynomials as kernel and $\mathcal{P}_{k-1}$ as image. For $k = 0$, its image is the zero space.

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Examples of linear maps (continued)

Integration:

$$I(p)(t) = \int_0^t p(s)\, ds$$

On $\mathcal{P}_k$, this integration map has trivial kernel. Its image consists of the polynomials of degree at most $k + 1$ whose constant term is zero.

Scaling by c is invertible if $c \neq 0$. For $c = 0$, the entire space maps to zero.

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Linear and affine maps

Affine map:

$$T(x) = Ax + b.$$

An affine map preserves affine combinations, $T((1-t)x + ty) = (1-t)T(x) + tT(y)$. A line can collapse to a point if its direction lies in the kernel of A; otherwise its along-line ratios are preserved. For $b \neq 0$,

$$T(0) = b \neq 0,$$

so the map is not linear on the original vector space.

A straight-line graph need not describe a linear map. For example,

$$f(x) = 2x + 1$$

fails $f(0) = 0$ and the scalar multiplication rule.

The map $T(x) = x^2$ is also nonlinear, failing addition and scalar multiplication. Testing $T(0) = 0$ can reject a map immediately, but passing that test alone does not establish linearity.

Matrix columns from basis images

Input basis:

$$B_{\text{in}} = \{v_1, \ldots, v_n\},$$

output basis:

$$B_{\text{out}} = \{w_1, \ldots, w_m\}.$$

Expand each $T(v_j)$ in the codomain basis.

$$T(v_j) = \sum_{i=1}^{m} M_{ij} w_i.$$

These coordinate vectors form the columns

$$M = \begin{bmatrix} [T(v_1)]_{B_{\text{out}}} & \cdots & [T(v_n)]_{B_{\text{out}}} \end{bmatrix}.$$

If

$$x = \sum_j c_j v_j,$$

Matrix columns from basis images (continued)

then

$$[T(x)]_{B_{\text{out}}} = M[x]_{B_{\text{in}}}.$$

The column count equals the domain dimension and the row count equals the codomain dimension. Coordinate order must remain consistent.

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A complete standard-matrix example

Let

$$T(x, y) = (x + y, x - y, 2x).$$

The images of the standard basis vectors are

$$T(e_1) = (1, 1, 2)^T, \qquad T(e_2) = (1, -1, 0)^T.$$

so

$$A = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{bmatrix}.$$

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This matrix is 3×2. Its kernel satisfies

$$x + y = 0, \quad x - y = 0, \quad 2x = 0$$

forcing $x = y = 0$, so T is injective. Its image is a rank-2 plane in $\mathbb{R}^3$, not the entire codomain.

Now let $S\colon \mathbb{R}^3 \to \mathbb{R}^2$,

A complete standard-matrix example (continued)

$$S(x,y,z)=(x+z,y-2z),\qquad N=\begin{bmatrix}1&0&1\\0&1&-2\end{bmatrix}.$$

Composition means $(S\circ T)(v)=S(T(v))$. When the output basis of T matches the input basis of S,

$$[S\circ T]_{\text{std}}=NA=\begin{bmatrix}1&0&1\\0&1&-2\end{bmatrix}\begin{bmatrix}1&1\\1&-1\\2&0\end{bmatrix}=\begin{bmatrix}3&1\\-3&-1\end{bmatrix}.$$

For $v=(1,2)^T$, $T(v)=(3,-1,2)^T$, so

$$S(T(v))=(5,-5)^T=(NA)v.$$

The rightmost factor A acts first, followed by N. Insert a coordinate change if the intermediate bases differ.

The change-of-basis formula

Let A be the standard-coordinate matrix, with new input and output basis matrices $B_{\text{in}}, B_{\text{out}}$.

$$[x]_{\text{std}}=B_{\text{in}}[x]_{B_{\text{in}}},$$
$$[y]_{B_{\text{out}}}=B_{\text{out}}^{-1}y_{\text{std}}.$$

Apply these operations in order.

1. Convert input basis coordinates to standard coordinates with B_{in}.
2. Apply A.
3. Convert the output to its new basis coordinates with B_{out}^{-1}.

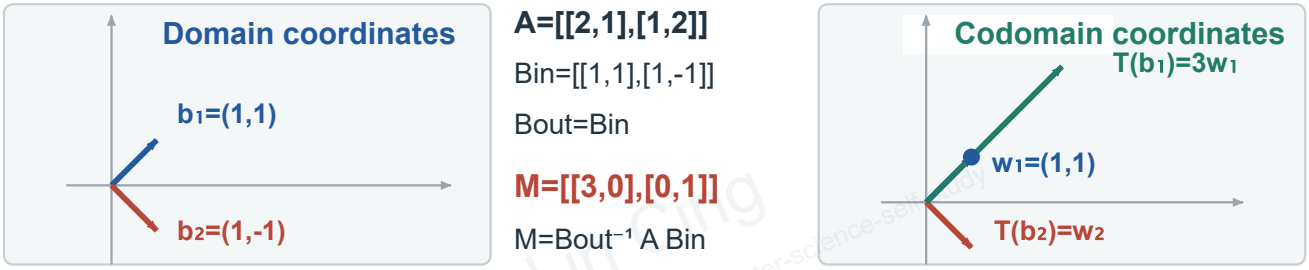
Consequently,

$$[T(x)]_{B_{\text{out}}}=B_{\text{out}}^{-1}AB_{\text{in}}[x]_{B_{\text{in}}},$$
$$M=B_{\text{out}}^{-1}AB_{\text{in}}.$$

The domain and codomain bases can have different dimensions.

The change-of-basis formula (continued)

Basis conversion: Bin at input, Bout⁻¹ at output



For a two-dimensional example, take

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad B_{\text{in}} = B_{\text{out}} = B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Since $Ab_1 = (3,3)^T = 3b_1$ 、$Ab_2 = (1,-1)^T = b_2$ ，

The change-of-basis formula (continued)

$$M = B^{-1}AB = \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}.$$

for $x = (2,0)^T$, the input coordinates are $B^{-1}x = (1,1)^T$. Applying M gives $(3,1)^T$, and converting back with B gives

$$B(3,1)^T = (4,2)^T = A(2,0)^T.$$

The input conversion uses B_{in}, while the output conversion uses B_{out}^{-1}. These directions must not be reversed.

Similarity on one space

When $V = W$ and the same new basis B is used on both sides,

$$[T]_B = B^{-1}AB.$$

If $Av = \lambda v$ and $y = B^{-1}v$,

$$B^{-1}AB\,y = B^{-1}Av = \lambda y.$$

so the eigenvalues agree. More fully, $\lambda I - B^{-1}AB = B^{-1}(\lambda I - A)B$, proving equality of characteristic polynomials. Trace and determinant are also invariant.

$$\operatorname{tr}(B^{-1}AB) = \operatorname{tr} A,$$
$$\det(B^{-1}AB) = \det A.$$

Independent input and output bases need not give similarity. The resulting matrix can be rectangular, and even square matrices related by independent left and right changes need not be similar.

An eigenbasis example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}.$$

Eigenbasis:

$$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$
$$B^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Therefore

$$B^{-1}AB = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

In this basis, the same map applies independent scalings. An ill-conditioned eigenbasis can amplify coordinate errors, so exact diagonalization does not itself guarantee a good numerical representation.

Differentiation in two bases

$D\colon \mathcal{P}_2 \to \mathcal{P}_1$, input monomial basis

$$(1, t, t^2),$$

output basis $(1, t)$.

$$D(1) = 0, \qquad D(t) = 1, \qquad D(t^2) = 2t.$$

Hence

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}.$$

If the input basis becomes $1, s, s^2$, where $s = t - 1$, then

$$D(s^2) = 2s = -2 + 2t,$$

the new coordinate matrix is

$$\begin{bmatrix} 0 & 1 & -2 \\ 0 & 0 & 2 \end{bmatrix}.$$

The representation changes, while differentiation itself does not.

Invariant subspaces

A subspace $U \subseteq V$ is invariant if

$$T(U) \subseteq U.$$

An eigenvector line is a one-dimensional nonzero invariant subspace. If

$$V = U \oplus W$$

and both summands are invariant, the matrix is block diagonal in a basis adapted to the decomposition.

If only U is known invariant, begin a basis with a basis of U. The matrix has block upper triangular form

$$\begin{bmatrix} A_1 & B \\ 0 & A_2 \end{bmatrix}.$$

Invariant subspaces reduce a large map to smaller blocks. Both Schur and Jordan forms use this structure.

Eigenbases and diagonalization

If $b_1, \dots, b_n$ form an eigenbasis, let

$$B = [b_1 \ \cdots \ b_n],$$

Then

$$B^{-1}AB = \operatorname{diag}(\lambda_1, \dots, \lambda_n).$$

The eigenbasis separates coupled equations into scalar modes, simplifying powers, exponentials, and recurrences.

For a real symmetric matrix, choose an orthonormal eigenbasis Q, giving

$$Q^{-1} = Q^T.$$

Inversion becomes transposition and the basis condition number is 1. A nonsymmetric eigenbasis can be nearly dependent despite existing exactly.

Jordan form and generalized eigenvectors

Every complex square matrix has a Jordan form

$$J = B^{-1}AB.$$

whose blocks have the form

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & 0 \\ 0 & \lambda & \ddots & \\ & \ddots & \ddots & 1 \\ 0 & & 0 & \lambda \end{bmatrix}.$$

The diagonal contains eigenvalues; superdiagonal ones encode generalized-eigenvector chains. A nonzero vector w satisfying

$$(A - \lambda I)^m w = 0$$

for some $m \geq 1$ is a generalized eigenvector. A Jordan chain satisfies $(A - \lambda I)w_1 = 0$, $(A - \lambda I)w_{j+1} = w_j$, giving the superdiagonal ones. The general Jordan existence theorem is stated without its full proof here; the exercises compute a second-order nilpotent block explicitly.

Jordan form and generalized eigenvectors (continued)

Jordan form is a theoretical classification. Numerical algorithms usually use Schur form, SVD, or structured methods because computing Jordan structure is unstable.

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Circulant matrices and a Fourier basis

For $n = 4$, the left cyclic shift

$$P = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

maps $(x_0, x_1, x_2, x_3)^T$ to $(x_1, x_2, x_3, x_0)^T$.

Choose roots of unity

$$\lambda \in \{1, i, -1, -i\},$$
$$v_\lambda = (1, \lambda, \lambda^2, \lambda^3)^T.$$

The wraparound coordinate uses $\lambda^4 = 1$, so

$$Pv_\lambda = \lambda v_\lambda.$$

The four distinct eigenvalues give four independent eigenvectors. The Fourier matrix diagonalizes P and every circulant polynomial

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Circulant matrices and a Fourier basis (continued)

$$C = c_0 I + c_1 P + c_2 P^2 + c_3 P^3.$$

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Useful bases for function spaces

Common finite bases and infinite expansion systems include the following.

- monomials $1, x, x^2, \ldots$:symbolic polynomial operations;
- trigonometric functions:periodic signals;
- Complex Fourier exponentials for translations and convolution.
- Legendre polynomials for orthogonal approximation on $[-1, 1]$.
- Chebyshev polynomials for interpolation and numerical approximation.

The task determines the useful basis. Eigenfunctions can diagonalize a differential operator. A well-conditioned orthogonal basis supports stable approximation. Rapidly decaying coefficients support compression of sufficiently smooth functions. Infinite expansions require a specified convergence sense.

Equal spans mean equal representable spaces. The choice of basis still changes sparsity, conditioning, and interpretation.

Basis conditioning

For

$$X = [x_1 \ \cdots \ x_n]$$

and

$$x = Xc,$$

A coordinate perturbation can be amplified by

$$\|X^{-1}\|_2$$

A large $\kappa_2(X)$ allows a small geometric change to cause a large coordinate change.

This does not make X singular in exact arithmetic. It can be an ill-conditioned basis. An orthonormal basis satisfies

$$\kappa_2(X) = 1.$$

High-degree monomials on large intervals can be poorly conditioned. Scaling, orthogonalization, or Chebyshev bases can help. Record the normalization and condition number of a coordinate change.

A combined example

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

$$B^{-1}AB = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}.$$

For

$$x = (2, 0)^T,$$

eigenbasis coordinates:

$$c = B^{-1}x = (1, 1)^T.$$

$$[T^5 x]_B = \begin{bmatrix} 4^5 & 0 \\ 0 & 2^5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1024 \\ 32 \end{bmatrix}.$$

return to standard coordinates by

$$T^5 x = B \begin{bmatrix} 1024 \\ 32 \end{bmatrix} = \begin{bmatrix} 1056 \\ 992 \end{bmatrix}.$$

A combined example (continued)

The eigenbasis separates the two modes. An ill-conditioned basis would require an additional coordinate-error analysis.

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Chapter check

You can now do the following.

- Test linearity and use the necessary zero-image condition.
- Prove kernel and image are subspaces and apply rank-nullity.
- Build matrix columns from basis images.
- Use the two-basis matrix formula with the correct multiplication order.
- Distinguish similarity from independent domain and codomain changes.
- Use invariant subspaces, eigenbases, Jordan chains, and Fourier vectors.
- Distinguish exact basis existence from numerical conditioning.

Chapter 9 develops complex inner products, unitary matrices, and FFT.

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References and attribution

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Why complex numbers arise

Real rotation matrix

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

has no nonzero real eigenvector when $\theta \not\equiv 0 \pmod{\pi}$. Its eigenvalues are

$$\lambda = e^{\pm i\theta}.$$

Complex eigenvectors describe oscillations, roots of unity, and periodic signals. A real matrix has conjugate-paired complex eigenvalues, and real solutions combine the corresponding modes with conjugate coefficients.

We use $i^2 = -1$. Electrical engineering often writes j for the same imaginary unit.

Arithmetic in the complex field

Write $z = a + bi$ with real a, b. Add and subtract components.

$$(a + bi) + (c + di) = (a + c) + (b + d)i.$$

Multiplication gives

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

The complex numbers form a field. For nonzero $a + bi$,

$$\frac{1}{a + bi} = \frac{a - bi}{a^2 + b^2}.$$

Multiplication commutes, and the positive squared modulus of a nonzero number supplies the reciprocal denominator. Standard basis vectors also work in $\mathbb{C}^n$, with complex coefficients.

Conjugation, modulus, and dimension

Conjugate:

$$\overline{a+bi} = a - bi.$$

Modulus:

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2+b^2}.$$

Conjugation preserves sums and products.

$$\overline{z+w} = \overline{z} + \overline{w}, \qquad \overline{zw} = \overline{z}\,\overline{w}.$$

Also $|zw| = |z||w|$, and the triangle inequality holds.

The complex dimension of $\mathbb{C}^n$ is n. Its real dimension is $2n$, because every component has a real and imaginary part. Specify the scalar field when counting dimensions or ranks.

Polar form and Euler's formula

A nonzero complex number

$$z = a + bi$$

has

$$r = |z|, \qquad \theta = \operatorname{atan2}(b, a),$$
$$z = re^{i\theta} = r(\cos\theta + i\sin\theta).$$

Euler formula:

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Multiplication multiplies moduli and adds arguments.

$$(re^{i\theta})(se^{i\phi}) = rse^{i(\theta+\phi)}.$$

Powers:

$$(re^{i\theta})^k = r^k e^{ik\theta}.$$

For example,

Polar form and Euler's formula (continued)

$$1 + i = \sqrt{2} e^{i\pi/4},$$
$$(1+i)^8 = 16 e^{2\pi i} = 16.$$

Arguments differing by 2π represent the same number. A branch choice matters for multivalued operations such as the complex logarithm.

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Roots of unity

Solve

$$z^n = 1.$$

Set

$$\omega = e^{2\pi i/n}.$$

The roots are

$$1, \omega, \omega^2, \dots, \omega^{n-1}.$$

They are equally spaced on the unit circle, forming a regular polygon for $n \geq 3$.

If $z = re^{i\theta}$ and $z^n = 1$, then

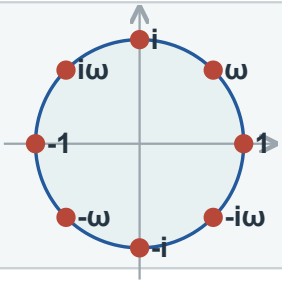
$$r = 1, \qquad n\theta = 2\pi k.$$

For $n = 4$, the roots are $1, i, -1, -i$. They also generate Fourier vectors and eigenvectors of cyclic shifts.

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Roots of unity (continued)

Eighth roots of unity: equally spaced on the unit circle



ω=e^(2πi/8)

roots：1,ω,…,ω⁷

$F_8F_8^H=8I$

Orthogonal columns; Fourier basis

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The complex inner product

Our convention is

$$\langle u, v\rangle = u^H v = \sum_j \overline{u_j} v_j,$$

where

$$u^H = \overline{u^T}.$$

For $u = (1, i)^T$,

$$\|u\|^2 = u^H u = |1|^2 + |i|^2 = 2,$$

rather than the bilinear expression $1^2 + i^2 = 0$.

Inner product properties:

$$\langle u, v\rangle = \overline{\langle v, u\rangle},$$
$$\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle,$$
$$\langle u, \alpha v\rangle = \alpha\langle u, v\rangle,$$

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The complex inner product (continued)

$$\langle \alpha u, v \rangle = \overline{\alpha} \langle u, v \rangle.$$

Conjugation belongs to the first argument throughout. A different convention requires consistent changes in every formula.

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Orthogonality and the four subspaces

Orthogonality means

$$u^H v = 0.$$

For example,

$$u = (1, i)^T, \qquad v = (1, -i)^T,$$
$$u^H v = (1, -i)(1, -i)^T = 1 + (-i)(-i) = 1 - 1 = 0.$$

The complex four-subspace relations use the conjugate transpose.

$$C(A) \perp N(A^H), \qquad C(A^H) \perp N(A).$$

Ordinary A^T does not generally give the Hermitian orthogonal complements. For instance, $(A^H y)^H x = y^H A x$, which vanishes when $Ax = 0$.

Dimensions:

$$\dim C(A) = \dim C(A^H) = r,$$
$$\dim N(A) = n - r, \qquad \dim N(A^H) = m - r.$$

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Hermitian matrices

A square matrix is Hermitian if

$$S^H = S.$$

Entrywise:

$$s_{ji} = \overline{s_{ij}}.$$

Its diagonal entries are real. For every z,

$$(z^H S z)^H = z^H S^H z = z^H S z.$$

The scalar equals its own conjugate, so

$$z^H S z \in \mathbb{R}.$$

For example,

$$S = \begin{bmatrix} 2 & 1+i \\ 1-i & 3 \end{bmatrix}$$

is Hermitian.

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Hermitian eigenvalues and eigenvectors

If nonzero z satisfies

$$Sz = \lambda z,$$

multiply on the left by z^H.

$$z^H S z = \lambda z^H z.$$

The left side is real and $z^H z > 0$, proving λ real.

If

$$Su = \lambda u, \qquad Sv = \mu v,$$

then, using reality of λ,

$$\lambda u^H v = (Su)^H v = u^H S v = \mu u^H v.$$

For distinct eigenvalues,

$$(\lambda - \mu) u^H v = 0,$$

so the eigenvectors are orthogonal.

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Hermitian eigenvalues and eigenvectors (continued)

For completeness of the eigenbasis, maximize the real-valued quadratic form $z^H S z$ on the complex unit sphere. Stationarity in real and imaginary directions gives $Sq = \lambda q$. The complement $q^\perp$ is invariant because $q^H S v = (Sq)^H v = 0$. Restriction remains Hermitian, so induction yields a complete orthonormal eigenbasis. Gram-Schmidt can be applied within repeated eigenspaces. Thus

$$S = Q \Lambda Q^H,$$

where $Q^H Q = I$ and Λ is real diagonal.

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Unitary matrices

A square matrix $Q \in \mathbb{C}^{n \times n}$ is unitary if

$$Q^H Q = I.$$

It also satisfies $QQ^H = I$, so

$$Q^{-1} = Q^H.$$

It preserves inner products.

$$\langle Qu, Qv \rangle = (Qu)^H (Qv) = u^H v.$$

Thus $\|Qu\| = \|u\|$.

If $Qu = \lambda u$,

$$\|u\| = \|Qu\| = |\lambda| \|u\|,$$

so

$$|\lambda| = 1.$$

The determinant also records length preservation.

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Unitary matrices (continued)

$$1 = \det(Q^H Q) = (\det Q)^H \det Q = |\det Q|^2,$$

Hence $|\det Q| = 1$. Real orthogonal matrices are the real special case of unitary matrices.

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Normal matrices

A square matrix is normal if

$$AA^H = A^H A.$$

A complex square matrix is unitarily diagonalizable exactly when it is normal. For sufficiency, take a unit eigenvector and extend it to a unitary basis, producing $\begin{bmatrix} \lambda & b^H \\ 0 & C \end{bmatrix}$. Equality of the $(1,1)$ entries of AA^H and $A^H A$ gives $|\lambda|^2 + \|b\|^2 = |\lambda|^2$, so $b = 0$. The remaining block is normal, and induction proves the result. Necessity follows by multiplying $A = Q\Lambda Q^H$ and its adjoint. Important classes include

- Hermitian:eigenvalues real;
- skew-Hermitian $K^H = -K$:eigenvalues pure imaginary;
- unitary:eigenvalues modulus 1.

Ordinary diagonalizability requires an eigenbasis. Unitary diagonalizability requires an orthonormal eigenbasis, a stronger condition.

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A Hermitian diagonalization example

$$S = \begin{bmatrix} 2 & i \\ -i & 2 \end{bmatrix} = S^H.$$

Characteristic polynomial:

$$\det(S - \lambda I) = (2-\lambda)^2 - |i|^2 = (\lambda - 1)(\lambda - 3).$$

Eigenvalues 1,3.

For $\lambda = 3$,

$$\begin{bmatrix} -1 & i \\ -i & -1 \end{bmatrix} z = 0$$

gives $z_1 = iz_2$. Choose

$$q_1 = (i, 1)^T / \sqrt{2}.$$

For $\lambda = 1$, choose

$$q_2 = (-i, 1)^T / \sqrt{2}.$$

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A Hermitian diagonalization example (continued)

Both vectors have unit norm and $q_1^H q_2 = 0$. Therefore

$$S = Q \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} Q^H,$$

$$Q = \frac{1}{\sqrt{2}} \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix}.$$

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The Fourier matrix

Let

$$\omega = e^{2\pi i/n}.$$

Forward Fourier matrix:

$$(F_n)_{jk} = \omega^{jk}, \qquad 0 \le j, k < n.$$

Forward transform:

$$y = F_n c.$$

The Fourier matrix is symmetric. It is generally not Hermitian for $n > 2$; sizes 1 and 2 are real symmetric exceptions. Its columns are orthogonal because

$$\sum_{j=0}^{n-1} (\omega^{jk})^H \omega^{j\ell} = \sum_{j} \omega^{j(\ell-k)} = n\delta_{k\ell}.$$

Therefore

$$F_n F_n^H = nI, \qquad F_n^{-1} = F_n^H/n.$$

Inverse DFT and Parseval

Inverse:

$$c_j = \frac{1}{n} \sum_{k=0}^{n-1} \omega^{-jk} y_k.$$

Software using

$$\omega = e^{-2\pi i/n},$$

interchanges the forward and inverse sign conventions.

Parseval:

$$\|y\|^2 = (F_n c)^H (F_n c) = n\|c\|^2.$$

Normalized unitary Fourier matrix:

$$Q = F_n/\sqrt{n}$$

satisfies $Q^H Q = I$, preserving the norm. Normalization changes scale constants without changing which information the transform represents.

DFT and cyclic convolution

Define cyclic convolution by $h_j = \sum_{k=0}^{n-1} a_k b_{(j-k) \bmod n}$. Its Fourier transform is

$$F_n h = (F_n a) \odot (F_n b)$$

Substitute this definition into $\sum_j \omega^{\ell j} h_j$ and set $s = (j-k) \bmod n$. The sum factors into $(\sum_k a_k \omega^{\ell k})(\sum_s b_s \omega^{\ell s})$. This identity uses the unnormalized forward transform. A forward transform divided by $\sqrt{n}$ requires an extra factor $\sqrt{n}$ on the right.

Two FFTs, pointwise multiplication, and an inverse FFT take $O(n \log n)$ arithmetic operations, compared with $O(n^2)$ for general direct cyclic convolution. Sequences of lengths p, q have linear convolution length $p + q - 1$; pad to at least that length to avoid wraparound.

Coefficient order, sign, and normalization must match the convolution definition. Record these conventions in FFT or finite-field transform implementations.

Even–odd decomposition in radix-2 FFT

Let $n = 2m$, $0 \le j < m$, and split the sequence into

$$c' = (c_0, c_2, \dots, c_{2m-2}),$$
$$c'' = (c_1, c_3, \dots, c_{2m-1}).$$

Separate $y_j = \sum_{k=0}^{n-1} c_k \omega_n^{jk}$ into even indices $k = 2s$ and odd indices $k = 2s+1$. Since $\omega_n^2 = \omega_m$, $\omega_n^m = -1$,

$$y_j = y'_j + \omega_n^j y''_j,$$
$$y_{j+m} = y'_j - \omega_n^j y''_j.$$

where y', y'' are DFTs of size $m = n/2$.

This butterfly operation combines two half-sized problems with m twiddle multiplications. Recursing to size 1 completes the FFT.

For $n = 4$, $c = (1, 2, 0, -1)^T$, bit reversal first gives

$$(c_0, c_2, c_1, c_3) = (1, 0, 2, -1).$$

Even–odd decomposition in radix-2 FFT (continued)

The first stage performs two two-point butterflies.

$$(1,0) \mapsto (1,1), \qquad (2,-1) \mapsto (1,3).$$

The second stage uses $\omega_4^0 = 1, \omega_4^1 = i$.

$$(1,1) \mapsto (2,0), \qquad (1,3) \mapsto (1+3i, 1-3i).$$

Consequently,

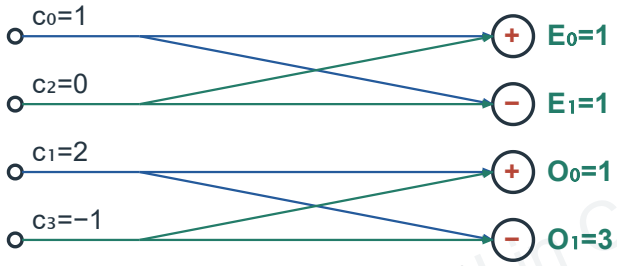
$$y = F_4 c = (2, 1+3i, 0, 1-3i)^T.$$

Direct evaluation gives $y_1 = 1 + 2i + (-1)(-i) = 1 + 3i$, $y_3 = 1 - 2i + (-1)i = 1 - 3i$. The plus and minus branches share the same twiddle product.

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Even–odd decomposition in radix-2 FFT (continued)

Four-point FFT stage 1: even and odd two-point DFTs



Even $(c_0,c_2)=(1,0)$

$E_0=c_0+c_2$ · $E_1=c_0-c_2$

Odd $(c_1,c_3)=(2,-1)$

$O_0=c_1+c_3$ · $O_1=c_1-c_3$

Upper branch adds; lower subtracts green from blue. Next, pair matching E_j and O_j.

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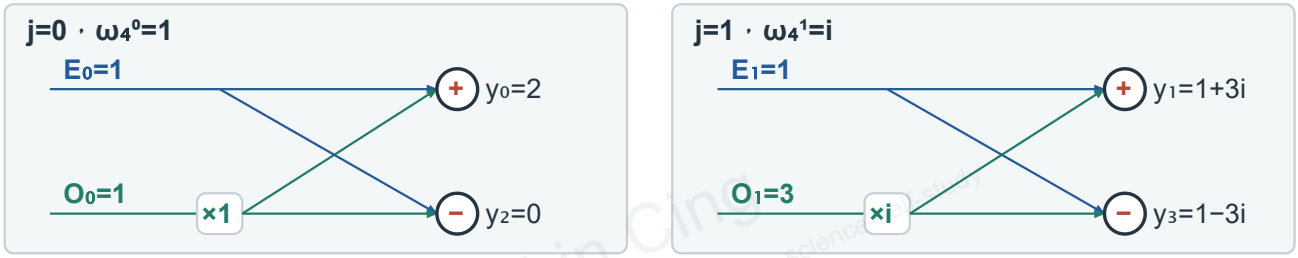
Which intermediate values combine in a four-point FFT?

The first-stage array is $(E_0, E_1, O_0, O_1) = (1, 1, 1, 3)$. The second stage pairs positions $(0, 2)$ and $(1, 3)$. Only matching E_j, O_j produce y_j, y_{j+2}. The diagram separates the two butterflies to keep those pairings visible.

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Which intermediate values combine in a four-point FFT? (continued)

Four-point FFT stage 2: combine matching E_j and O_j



Blue $u = E_j$; green $\omega v = \omega_4^j O_j$. Upper: $u + \omega v$; lower: $u - \omega v$. Output order y_0, y_1, y_2, y_3.

The first three intermediate values happen to agree in this example, concealing a wrong pairing. Instead take $c = (2, 5, 7, -1)$. The first stage is $(9, -5, 4, 6)$, and the correct output is $(13, -5 + 6i, 5, -5 - 6i)$. Pairing E_0, E_1 incorrectly would give $9 + (-5) = 4$ instead of 13. Pairing indices are part of the algorithm.

Recursion and arithmetic cost

$n = 2^{\ell}$:

$$T(n) = 2T(n/2) + \Theta(n).$$

There are $\log_2 n$ levels with linear work at each level, giving

$$T(n) = \Theta(n \log n).$$

Naive Fourier matrix multiplication:

$$n^2$$

multiplications in the direct matrix formula.

For $n = 1024 = 2^{10}$,

$$n^2 = 1,048,576,$$
$$\frac{n}{2} \log_2 n = 5,120.$$

The theoretical multiplication-count ratio is 204.8 in this model. This is not a measured runtime speedup. Additions, memory access, parallel execution, and root generation affect actual runtime.

Iterative FFT and bit reversal

The recursion can be implemented bottom-up. A radix-2 decimation-in-time arrangement for $n = 8$ begins with bit-reversed input order

$$0, 4, 2, 6, 1, 5, 3, 7.$$

and applies butterflies of lengths 2, 4, and 8.

$$(a, b) \mapsto (a + \omega b, a - \omega b).$$

An inverse transform can conjugate the input, apply the forward transform, conjugate the output, and divide by n.

Other sizes can use mixed-radix methods or Bluestein's algorithm. A real-input FFT can exploit conjugate symmetry, but a generic complex interface need not do so automatically.

Numerical interpretation

Practical issues include the following.

- Compute or store twiddle factors accurately.
- Control accumulated phase error.
- Specify the forward scale and inverse sign.
- Zero-padding changes the frequency grid but creates no new observations.
- Windowing changes the signal while reducing some leakage.
- Bin frequency depends on the sampling interval through $k/(n\Delta t)$, with the signed-frequency interpretation fixed by convention.

A spectral peak can combine signal, leakage, and noise. A nonzero numerical bin alone does not establish an exact physical frequency.

Chapter check

You can now do the following.

- Compute complex products, moduli, powers, and roots.
- Use a consistent Hermitian inner product.
- Prove the Hermitian spectral properties.
- Construct a unitary eigenbasis.
- Check unitary norm preservation and eigenvalue moduli.
- Derive Fourier orthogonality and inversion.
- Implement radix-2 butterflies and derive $\Theta(n \log n)$ cost.
- Interpret signs, scaling, padding, and numerical limits.

Chapter 10 applies these structures to networks, engineering, Markov models, optimization, Fourier series, graphics, and cryptography.

References and attribution

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Shared mathematical structures

The applications repeatedly use differences, energies and fluxes, nonnegative evolution, projection, and coordinate transformations. The projection formula below assumes that the v_i form a nonzero orthogonal basis of the target subspace.

$$\text{difference } A, \qquad \text{energy } A^T C A, \qquad u_{k+1} = A u_k,$$

$$\operatorname{proj}_U b = \sum_i \frac{\langle v_i, b \rangle}{\langle v_i, v_i \rangle} v_i.$$

A model must specify entry units and signs, the assumptions behind its equations, and the distinction between exact invertibility, physical interpretation, and numerical reliability.

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Oriented graphs and incidence matrices

Let the nodes be $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$. Choose an orientation for each edge. For a graph without self-loops, define $A \in \mathbb{R}^{m \times n}$ by

$$a_{ij} = \begin{cases} -1, & e_i \text{ leaves } v_j, \\ +1, & e_i \text{ enters } v_j, \\ 0, & \text{otherwise.} \end{cases}$$

Each edge row sums to zero. For a node-potential vector x,

$$(Ax)_i = x_{\text{head}} - x_{\text{tail}}.$$

Thus A is a discrete difference operator. Reversing an edge negates its row and difference, without changing the squared difference.

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Kernel and rank of a connected graph

Connected means that the underlying undirected graph is connected after ignoring the chosen edge directions; strong connectivity is not required. All edge differences vanish exactly when all node potentials agree.

$$N(A) = \operatorname{span}\{\mathbf{1}\}, \qquad \dim N(A) = 1.$$

Each edge row sums to zero, so $A\mathbf{1} = 0$ and the rank is less than n. This does not assert an equal-coefficient dependence among all rows. Equality of potentials along paths proves that the kernel consists exactly of constant vectors. Rank-nullity then gives rank $n - 1$; alternatively use the $n - 1$ independent edge rows of a spanning tree. Thus

$$\operatorname{rank} A = n - 1, \qquad \dim N(A^T) = m - n + 1.$$

The column space $C(A) \subseteq \mathbb{R}^m$ describes attainable edge potential differences. The row space lies in node space and is the orthogonal complement of the constant vectors. The left nullspace describes cycle flows. These dimensions encode the graph topology.

Cycles, flows, and Kirchhoff laws

The equation $Ax = b$ is solvable exactly when $b \in C(A)$, equivalently when every cycle flow $j \in N(A^T)$ satisfies

$$j^T b = 0.$$

Thus signed potential differences sum to zero around every closed loop, the linear-algebra form of Kirchhoff's voltage law.

The left nullspace consists of edge flows satisfying

$$A^T j = 0.$$

At every node, inflow minus outflow must balance, giving Kirchhoff's current law. Fix a spanning tree. Each nontree edge and the unique tree path between its endpoints form a fundamental cycle. Give its edges coefficients +1, −1, or 0 according to orientation; then $A^T j = 0$. Each such cycle contains its own nontree edge and no other nontree edge, so comparison of these coordinates proves independence. There are $m - n + 1$ cycles, equal to the left nullity, so they form a cycle-space basis. Arbitrarily chosen small cycles need not be independent or span the cycle space.

Cycles, flows, and Kirchhoff laws (continued)

For resistive networks, fix $e = Ax$ as head potential minus tail potential, and $y = Ce$ as weighted potential differences. Physical current j along the chosen arrows satisfies Ohm's law $j = -y$. Positive f denotes external injection into a node. Conservation is $-A^T j = f$, hence $A^T y = f$. Positive y must not simultaneously be called current along an arrow pointing from lower to higher potential.

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Laplacians and network equations

Put positive conductances c_e in diagonal C and define

$$L = A^T C A.$$

For potentials x,

$$x^T L x = (Ax)^T C (Ax) = \sum_e c_e (\Delta x_e)^2 \geq 0.$$

Thus L is symmetric positive semidefinite. For a connected graph with positive conductances, its kernel consists of constant potentials. Without a reference potential, every solution admits an arbitrary common offset.

Ground a node, for example $x_n = 0$, and delete its row and column to obtain L_{red}. Connectivity makes the reduced matrix positive definite.

The network relations are

$$e = Ax, \qquad y = Ce, \quad j = -y, \qquad f = -A^T j = A^T y,$$

and combine into

Laplacians and network equations (continued)

$$A^T C A x = f.$$

Reversing edge directions does not change L. Let each orientation change be $s_e \in \{-1, 1\}$, and set

$$D = \mathrm{diag}(s_1, \ldots, s_m), \qquad A' = DA.$$

Since $D^T = D$, $D^2 = I$, and C is diagonal,

$$(A')^T C A' = A^T D\, C\, D A = A^T C A.$$

The corresponding edge quantities change consistently as $e' = De$, $y' = Dy$, giving

$$(A')^T y' = A^T D^2 y = A^T y = f.$$

Orientation changes only the signed representation of edge differences and currents. It preserves the Laplacian, energy, physical flow, and node balance.

Also,

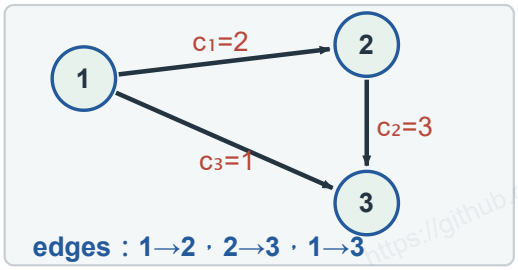
$$\mathbf{1}^T f = \mathbf{1}^T A^T C A x = (A\mathbf{1})^T C A x = 0,$$

Laplacians and network equations (continued)

so without an external cross-network source, injections must sum to zero. A nonzero total is a model or data inconsistency that changing the solver cannot repair.

The triangle uses edges $1 \to 2, 2 \to 3, 1 \to 3$, with each incidence row computing head potential minus tail potential.

Triangle network: incidence, orientation, and differences



Rows of A

[-1, 1, 0]
[0,-1, 1]
[-1, 0, 1]

x=(1,4,7)ᵀ

e₁=4-1=3
e₂=7-4=3
e₃=7-1=6

e=Ax=(3,3,6)ᵀ

A weighted triangular network

Fix edges $1 \to 2, 2 \to 3, 1 \to 3$, with conductances $2, 3, 1$. Each incidence row represents one oriented edge.

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ -1 & 0 & 1 \end{bmatrix}.$$

For $e_i = (Ax)_i = x_{\text{head}} - x_{\text{tail}}$ and $x = (1, 4, 7)^T$, the rows give

$$Ax = \begin{bmatrix} 4 - 1 \\ 7 - 4 \\ 7 - 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 6 \end{bmatrix} = e.$$

Multiply by $C = \operatorname{diag}(2, 3, 1)$.

$$y = Ce = (6, 9, 6)^T.$$

The physical currents along the arrows are $j = (-6, -9, -6)^T$, all flowing in reverse. Multiplication by A^T gives the external injections at nodes $1, 2, 3$.

A weighted triangular network (continued)

$$f = A^T y = \begin{bmatrix} -6 - 6 \\ 6 - 9 \\ 9 + 6 \end{bmatrix} = \begin{bmatrix} -12 \\ -3 \\ 15 \end{bmatrix}.$$

Their sum is $-12 - 3 + 15 = 0$, as conservation requires. The weighted Laplacian is

$$L = A^T C A = \begin{bmatrix} 3 & -2 & -1 \\ -2 & 5 & -3 \\ -1 & -3 & 4 \end{bmatrix}, \qquad Lx = f, \quad L\mathbf{1} = 0.$$

The incidence rank is 2 and its kernel consists of constants. Grounding node 1 at zero and deleting its row and column gives $x_2 = 3, x_3 = 6$. Adding the common offset 1 recovers $x = (1, 4, 7)^T$. Changing the reference potential preserves differences and flows.

A weighted triangular network (continued)

Currents and injections: y=CAx, j=−y, f=−A^Tj=A^Ty

Weighted differences

C=diag(2,3,1)

e=Ax=$(3,3,6)^T$

y=Ce=$(6,9,6)^T$

Current along arrows: j=−y

Accumulate at nodes

f=A^Ty=−A^Tj

node 1 : -6-6=-12

node 2 : 6-9= -3

node 3 : 9+6= 15

$\Sigma f_i=0$

Laplacian

L=[[3,-2,-1],
[-2,5,-3],
[-1,-3,4]]

Lx=f · L1=0

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Springs and energy forms

Let u denote displacements and $e = Au$ spring elongations. The constitutive relation is

$$y = Ce, \qquad C = \operatorname{diag}(c_i).$$

Equilibrium gives

$$A^T y = f.$$

Combining gives

$$A^T C A u = f.$$

The stiffness matrix is symmetric positive semidefinite for positive spring constants. It is positive definite when the constraints remove all zero-energy motions. The strain energy is

$$\frac{1}{2} u^T K u = \frac{1}{2} \sum_i c_i e_i^2.$$

For a free connected one-dimensional chain, constant translation stretches no spring and lies in the nullspace. Boundary constraints or a displacement reference remove this mode.

Springs and energy forms (continued)

Finite-element assembly uses the same pattern, adding local matrices $A_e^T C_e A_e$ into a global stiffness matrix.

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Second differences and boundaries

For N interior nodes with both endpoints fixed, the $(N+1) \times N$ difference matrix maps u to $(u_1, u_2 - u_1, \dots, u_N - u_{N-1}, -u_N)^T$. It gives

$$K = A^T A = \begin{bmatrix} 2 & -1 & 0 & \cdots \\ -1 & 2 & -1 & \ddots \\ 0 & -1 & 2 & \ddots \end{bmatrix}.$$

Before dividing by h^2,

$$(Ku)_i = 2u_i - u_{i-1} - u_{i+1} \approx -h^2 u''(x_i).$$

Known boundary values move to the right-hand side. A free right endpoint removes the last spring contribution, changing the final diagonal entry. Boundary conditions determine the matrix itself.

A free-free chain has a constant-displacement nullspace and singular stiffness. The force vector must balance the rigid mode; a reference displacement can then select a representative, with reaction forces included where constraints impose them.

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Engineering model checks

Before solving $Ku = f$, identify the following.

1. Degrees of freedom and units.
2. Displacement, force, and elongation signs.
3. difference matrix A;
4. constitutive diagonal C;
5. Boundary constraints and grounded modes.
6. Symmetry and positive definiteness.
7. Nonnegative energy and rigid modes.
8. Conditioning under mesh refinement.

A missing boundary condition often appears as a zero eigenvalue and translation mode. Thin elements or large coefficient contrasts can also cause poor conditioning in an invertible positive-definite system.

Markov matrices

A column-stochastic matrix satisfies

$$P_{ij} \geq 0, \qquad \sum_i P_{ij} = 1.$$

For a probability vector u_k,

$$u_{k+1} = Pu_k,$$

we have

$$\mathbf{1}^T u_{k+1} = \mathbf{1}^T u_k.$$

so total probability remains 1. Also,

$$\mathbf{1}^T P = \mathbf{1}^T,$$

Thus 1 is a left eigenvalue, and hence an eigenvalue of P, since transposition preserves the characteristic polynomial. In general $P\mathbf{1} \neq \mathbf{1}$. A stationary distribution is a right eigenvector satisfying

$$Pu_\infty = u_\infty.$$

Convergence and spectral radius

If every entry of P is positive, let $\alpha = n \min_{ij} P_{ij} > 0$. For $\alpha < 1$, write $P = \alpha \mathbf{1}\mathbf{1}^T/n + (1-\alpha)R$, with column-stochastic R. For probability vectors u, v, $\|Pu - Pv\|_1 \le (1-\alpha)\|u - v\|_1$. The probability simplex is complete in ℓ_1, so the contraction theorem gives a unique stationary vector and geometric convergence. At $\alpha = 1$, one step produces the uniform distribution. Every stationary component is positive.

With zero entries allowed, the convergence version of Perron–Frobenius requires irreducibility and aperiodicity. These imply a unique positive stationary distribution attracting every initial probability vector. The periodic matrix $P = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ has $P^2 = I$ and period-two cycling. For reducible $P = I$, every standard basis vector is stationary, and the limit depends on the initial state.

If P is diagonalizable, order the eigenvalues as

$$1 = \lambda_1 > |\lambda_2| \ge \cdots,$$

Then

Convergence and spectral radius (continued)

$$u_k = P^k u_0 = \sum_i c_i \lambda_i^k v_i.$$

The second eigenvalue controls mixing through

$$\|u_k - u_\infty\| \le C|\lambda_2|^k.$$

For example,

$$P = \begin{bmatrix} 0.8 & 0.3 \\ 0.2 & 0.7 \end{bmatrix}$$

The eigenvalues are $1, 0.5$, with stationary vector $u_\infty = (0.6, 0.4)^T$. The nonstationary mode halves each step. A nondiagonalizable matrix adds Jordan factors k^{m-1}; spectral radius still controls convergence, but a pure $C|\lambda_2|^k$ bound needs modification.

Population and input-output models

A Leslie matrix separates age classes.

$$n_{k+1} = Ln_k.$$

Its first row contains fertility rates and its subdiagonal survival rates. For an irreducible, aperiodic nonnegative transition graph, Perron–Frobenius gives a unique positive dominant eigenvalue λ. Normalize its eigenvector to obtain the stable age distribution; λ is the long-run growth factor. Without these hypotheses, periodic behavior or isolated cohorts can prevent a single limiting distribution.

The consumption coefficient A_{ij} is input from sector i needed for one unit of sector j's output. Total output p, less internal consumption Ap, leaves

$$(I - A)p = y.$$

If $\rho(A) < 1$, the Neumann series converges.

$$(I - A)^{-1} = I + A + A^2 + \cdots.$$

For a proof, examine the finite sum.

Population and input-output models (continued)

$$(I - A)(I + A + \cdots + A^N) = I - A^{N+1}.$$

Since $\rho(A) < 1$ is equivalent to $A^N \to 0$, the sums tend to $(I - A)^{-1}$. For $A \geq 0$, every term is nonnegative, hence so is the inverse. Nonnegative demand y yields

$$p = (I - A)^{-1}y$$

which accumulates direct and indirect production requirements. If some $|\lambda| \geq 1$, the series fails to converge. The matrix $I - A$ might remain invertible, but that alone does not ensure nonnegative production or a sustainable long-run model.

Linear-program geometry

Standard form:

$$\begin{aligned} &\min\ c^T x, \\ &\text{s.t. } Ax = b, \quad x \geq 0. \end{aligned}$$

For example,

$$\min\ 5x_1 + 3x_2 + 8x_3$$

subject to

$$x_1 + x_2 + 2x_3 = 4, \qquad x_1, x_2, x_3 \geq 0.$$

The equality is a plane. Nonnegativity cuts out a triangle, and objective level sets are parallel planes. A linear objective on this nonempty compact polytope attains an optimum at a vertex.

Assume redundant equalities have been removed from $A \in \mathbb{R}^{m \times n}$, so its rank is m. A basic feasible solution selects m independent columns as square B, sets nonbasic variables to zero, solves $Bx_B = b$, and requires $x_B \geq 0$.

Simplex, duality, and numerical conditions

For basis B, the objective value is

$$c_B^T B^{-1} b.$$

A nonbasic column a_j has reduced cost

$$\bar{c}_j = c_j - c_B^T B^{-1} a_j.$$

If every $\bar{c}_j \geq 0$, the feasible change $x_B \mapsto x_B - B^{-1} a_j t$ changes the objective by $\bar{c}_j t \geq 0$, certifying optimality. For $\bar{c}_j < 0$, let x_j enter and apply the ratio test.

$$t = \min_{i:\,(B^{-1}a_j)_i > 0} \frac{x_{Bi}}{(B^{-1}a_j)_i}$$

Simplex, duality, and numerical conditions (continued)

Among positive denominators, take the smallest nonnegative ratio so the first basic variable reaching zero leaves. A zero basic value must participate and can give a zero step. If all denominators are nonpositive, the entering variable can grow without bound while decreasing the objective, giving an unboundedness certificate. A nondegenerate pivot reaches an adjacent vertex; a degenerate pivot can leave the point fixed. When both primal and dual are strictly feasible, interior-point methods can approach the optimum along the positive central path below. Without strict feasibility, preprocessing or other feasibility treatment is needed.

The primal and dual are

$$\min c^T x \quad \text{s.t. } Ax = b,\ x \geq 0,$$
$$\max b^T y \quad \text{s.t. } A^T y \leq c.$$

Every feasible pair satisfies weak duality because

$$c^T x - b^T y = x^T (c - A^T y) \geq 0.$$

Simplex, duality, and numerical conditions (continued)

Equal objective values for feasible primal and dual points certify optimality. For finite-dimensional LP, feasibility and a finite attained optimum on one side imply an equal attained optimum on the other by strong duality. The explicit matching feasible pair in this example already certifies optimality. Optimal solutions satisfy complementary slackness.

$$x_j (c_j - A_j^T y) = 0.$$

A positive variable has zero reduced cost. Dual variables describe marginal demand prices only while the current optimal basis stays valid.

Degeneracy means a zero basic variable and can give multiple bases at one vertex. A tiny pivot is a separate numerical issue. Interior-point methods use

$$Ax = b, \qquad A^T y + s = c, \qquad x_j s_j = \mu > 0$$

for the central path, solve linearized Newton corrections, and let $\mu \downarrow 0$. Report solver status, primal and dual residuals, duality gap, and conditioning alongside the vector.

A complete pivot and dual certificate

Minimize $5x_1 + 3x_2 + 8x_3$, with $x_1 + x_2 + 2x_3 = 4$, $x \geq 0$. Initially choose x_3 as basic, so $B = [2]$, $c_B = 8$, and

$$x_3 = 2 - \tfrac{1}{2}x_1 - \tfrac{1}{2}x_2, \qquad z = 16 + x_1 - x_2.$$

The reduced cost of x_2 is -1. Its direction denominator is $B^{-1}a_2 = 1/2 > 0$, giving ratio $2/(1/2) = 4$. Increasing x_2 to 4 makes x_3 zero. After the pivot,

$$x_2 = 4 - x_1 - 2x_3, \qquad z = 12 + 2x_1 + 2x_3.$$

all nonbasic coefficients are nonnegative, so $x = (0, 4, 0)^T$, $z = 12$, is optimal. Generally, substitute $x_B = B^{-1}b - \sum_{j \notin B} B^{-1}a_j x_j$ into the objective to get $z = c_B^T B^{-1} b + \sum_{j \notin B} \bar{c}_j x_j$. For $x_j \geq 0, \bar{c}_j \geq 0$, no feasible point has a lower value.

Choose dual $y = 3$. Then $A^T y = (3, 3, 6)^T \leq (5, 3, 8)^T$, with value $4y = 12$. Matching feasible values independently certify optimality. Positive x_2 corresponds to the tight second dual constraint; the other slacks are both 2.

A zero ratio is a valid degenerate pivot

Minimize $-x_1$, subject to $x_1 + s_1 = 0$, $x_1 + s_2 = 1$, and nonnegativity. The initial basis (s_1, s_2) gives $(x_1, s_1, s_2) = (0, 0, 1)$. Increase x_1 by t.

Constraint	Basic variable as a function of t	Positive denominator	Nonnegative ratio
1	$s_1 = 0 - t$	1	0
2	$s_2 = 1 - t$	1	1

The minimum ratio is 0. Variable s_1 leaves and x_1 enters, but the point stays fixed. Excluding zero and taking $t = 1$ gives $s_1 = -1$, violating feasibility.

The new dictionary is $x_1 = -s_1$, $s_2 = 1 + s_1$, with objective $s_1 \geq 0$. Thus value 0 is optimal. Degenerate pivots may not improve the objective and can cycle. Bland's rule chooses the smallest-index improving entering variable, then the smallest-index basic variable among minimum-ratio ties. It avoids cycling without discarding valid zero ratios.

Functions as vectors

For real functions in $L^2([0, 2\pi])$, define

$$\langle f, g \rangle = \int_0^{2\pi} f(x) g(x)\, dx.$$

For complex functions conjugate the first argument. Strictly, L^2 consists of equivalence classes modulo equality almost everywhere; otherwise a function nonzero at only one point could have zero norm. Square integrability requires

$$\|f\|^2 = \int_0^{2\pi} |f(x)|^2\, dx < \infty.$$

The functions $1, \cos nx, \sin nx$, for positive integer n, are mutually orthogonal in the appropriate pairs. For example,

$$\int_0^{2\pi} \sin x \cos x\, dx = 0.$$

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Functions as vectors (continued)

An infinite trigonometric system requires completeness and a convergence notion to be a Hilbert-space basis. Pointwise convergence requires additional conditions; L^2 convergence is the natural projection framework.

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Fourier coefficients as projections

Begin with a finite truncation.

$$f_N(x) = a_0 + \sum_{n=1}^{N} (a_n \cos nx + b_n \sin nx).$$

Take an inner product with each basis function; orthogonality removes the other terms.

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} f(x)\, dx,$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx\, dx,$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx\, dx.$$

These are least-squares projection coefficients in the selected trigonometric subspace. Under 2π-periodic extension, odd functions have zero cosine coefficients and even functions have zero sine coefficients. This parity is about the origin, not merely a shape seen on $[0, 2\pi]$.

Fourier coefficients as projections (continued)

Let $e_N = f - f_N$. Since the residual is orthogonal to every retained basis function,

$$\|f\|^2 = \|f_N\|^2 + \|e_N\|^2.$$

adding an independent orthogonal mode transfers that mode's energy from the residual to the approximation. A zero coefficient leaves the result unchanged. This Bessel/Parseval energy relation alone does not prove pointwise or uniform convergence. Rates depend on smoothness, jumps, and boundary behavior; jumps produce Gibbs oscillations near the discontinuity.

Homogeneous coordinates

An affine map on $\mathbb{R}^3$,

$$y = Rx + t$$

becomes a linear matrix map in homogeneous coordinates.

$$\begin{bmatrix} y \\ 1 \end{bmatrix} = \begin{bmatrix} R & t \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ 1 \end{bmatrix}.$$

Translation:

$$T(t) = \begin{bmatrix} I & t \\ 0 & 1 \end{bmatrix}.$$

Scaling:

$$S(s_x, s_y, s_z) = \begin{bmatrix} \operatorname{diag}(s_x, s_y, s_z) & 0 \\ 0 & 1 \end{bmatrix}.$$

Homogeneous tuples

$$(x_0, x_1, x_2, w) \sim (\lambda x_0, \lambda x_1, \lambda x_2, \lambda w)$$

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Homogeneous coordinates (continued)

represent the same projective point when $\lambda \neq 0$ and the tuple is nonzero. Finite points normalize to $w = 1$. Directions use $w = 0$, so translations leave them unchanged.

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Composing graphics transformations

To rotate around p, translate p to the origin, rotate, then translate back. With column vectors,

$$T(p)R_\theta T(-p)x.$$

the rightmost transformation acts first.

For a plane through the origin with unit normal n, projection and reflection are

$$P = I - nn^T, \qquad M = I - 2nn^T.$$

For an offset plane, translate it to the origin and back. Perspective uses a nonzero homogeneous component followed by division. With the eye at the origin and image plane $z = d > 0$,

$$P_{\text{persp}} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1/d & 0 \end{bmatrix}.$$

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Composing graphics transformations (continued)

a point $[x, y, z, 1]^T$ becomes $[x, y, z, z/d]^T$; division by z/d gives the image coordinates. This projective transformation generally does not preserve affine geometry, since the divisor varies. Clipping and depth conventions depend on the graphics API.

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Finite fields and the Hill cipher

For prime p, the field

$$\mathbb{F}_p = \{0, 1, \ldots, p-1\},$$

uses addition and multiplication modulo p. Every nonzero element has a reciprocal. A square key matrix is invertible exactly when

$$\det K \not\equiv 0 \pmod{p}.$$

Encode:

$$y = Kx \pmod{p}.$$

Decode:

$$x = K^{-1}y \pmod{p}.$$

This defines a Hill cipher. For $p = 29$ and

$$K = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix},$$

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Finite fields and the Hill cipher (continued)

the determinant is $-1 \equiv 28 \pmod{29}$, so the key is invertible. An affine version adds a fixed message-space translation.

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Security and chapter check

A Hill cipher is linear. Enough suitable known plaintext and ciphertext pairs give equations for the key. Small keys can also be searched directly. This is a finite-field teaching example, not a modern secure encryption scheme.

You can now do the following.

- Infer incidence ranks and nullspaces from connectivity.
- Remove the Laplacian's constant mode by grounding.
- Build network and stiffness equations from energy forms.
- Compute stationary distributions and spectral convergence rates.
- Interpret productivity through spectral radius.
- Relate LP geometry, bases, and duality.
- Compute Fourier projection coefficients.
- Maintain point and direction conventions in homogeneous coordinates.
- Check modular invertibility before decoding.

Security and chapter check (continued)

Chapter 11 studies executable methods and their numerical error.

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References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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Three distinct computational goals

Matrix computation must account for the following goals.

1. Speed depends on arithmetic, memory, communication, and parallelism.
2. Accuracy measures distance from the exact solution.
3. Backward stability asks whether rounding effects can be represented by small input perturbations. This differs from the problem's sensitivity to those perturbations.

An exact identity need not give a stable algorithm. Even $\det A = 1$ does not ensure insensitivity to perturbation. A mathematically nonsingular matrix can be poorly conditioned.

Specify the error metric.

$$\text{absolute error } \|x - \widehat{x}\|, \qquad \text{relative error } \frac{\|x - \widehat{x}\|}{\|x\|},$$

$$\text{residual } \|b - A\widehat{x}\|.$$

A claim of small error requires a norm, scale, and tolerance.

Floating point and backward error

The basic floating-point model is

$$\mathrm{fl}(x \circ y) = (x \circ y)(1 + \delta), \qquad |\delta| \le u,$$

Here u is unit roundoff. The model assumes represented inputs, ordinary rounding, and no overflow or underflow effects requiring a separate model. Cancellation need not violate the single-operation relative model, but can expose errors already present in the operands. Growth and conditioning also affect accumulated error.

Subtracting nearly equal values leaves few significant digits relative to the original operand scale.

$$1.0000001 - 1.0000000 = 1 \cdot 10^{-7}.$$

Backward error asks whether the computed $\widehat{x}$ exactly solves a nearby problem.

$$(A + \Delta A)\widehat{x} = b + \Delta b.$$

A backward-stable algorithm produces small relative input perturbations, but an ill-conditioned problem can still have a large forward error.

Elimination multipliers

For pivot d and an entry below it, the multiplier is

$$\ell_{ik} = \frac{a_{ik}}{d}.$$

A small pivot can create a large multiplier and amplify rounding effects. Partial pivoting selects a largest-magnitude available entry in the current column.

$$|d| = \max_{i \geq k} |a_{ik}|.$$

This bounds every elimination multiplier by 1 in magnitude. It controls one source of growth but does not guarantee a small growth factor for every matrix. Other pivoting strategies can be useful for particular problems.

The factorization is

$$PA = LU.$$

Apply row permutations consistently to stored multipliers. Swapping rows of the active upper factor while leaving earlier lower-factor multipliers unpermuted gives the wrong factorization.

Factorization and solution cost

For dense square matrices, the leading costs are

- pivoted LU:$\frac{2}{3}n^3 + O(n^2)$ flops;
- forward solve $Ly = Pb$:$O(n^2)$;
- back solve $Ux = y$:$O(n^2)$.

Reuse a factorization for multiple right-hand sides instead of repeating cubic work.

A solve can follow this sequence.

1. Factor with suitable pivoting.
2. Estimate matrix norms and conditioning.
3. solve;
4. Compute the true residual in the original equation.
5. Apply iterative refinement if needed.
6. Record an error estimate and its scale assumptions.

Factorization and solution cost (continued)

A nonzero pivot or determinant is insufficient as an accuracy check. Determinants can overflow or underflow, and their magnitudes are not scale-invariant condition measures.

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Sparse matrices and fill-in

Large engineering matrices are often sparse. Storing n^2 zero entries wastes memory and multiplication work.

Whether elimination preserves zeros depends on ordering. Eliminating a star's center first connects all leaves into a clique; eliminating leaves first reduces fill-in.

Read the sparsity graph as follows.

- Variables are vertices.
- Nonzero off-diagonal entries are edges.
- Eliminating a vertex creates fill edges between its remaining neighbors.

Minimum-degree ordering, nested dissection, and reverse Cuthill–McKee can reduce fill or bandwidth. Parallel systems also depend on communication and load balance.

Fix the small numerical example

$$A = \begin{bmatrix} 0.001 & 1 \\ 1 & 1 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

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Sparse matrices and fill-in (continued)

Without a swap, the first multiplier is $1/0.001 = 1000$, giving

$$U = \begin{bmatrix} 0.001 & 1 \\ 0 & -999 \end{bmatrix}.$$

After swapping, the multiplier is only 0.001, giving

$$U = \begin{bmatrix} 1 & 1 \\ 0 & 0.999 \end{bmatrix}.$$

Both exact-arithmetic paths give the same solution

$$x = \left(\frac{1000}{999}, \frac{998}{999} \right)^T, \qquad Ax - b = 0.$$

The star graph separately illustrates ordering. Center-first elimination creates three fill edges between the leaves; leaves-first elimination creates none. Numerical growth and sparse fill are distinct costs.

Sparse matrices and fill-in (continued)

With lower and upper bandwidths p, q, band storage uses $O(n(p+q))$ memory, no-pivot band elimination uses $O(npq)$ work, and triangular solves use $O(n(p+q))$. Tridiagonal matrices have $p = q = 1$, so factorization and solution cost $O(n)$. These bounds assume pivoting does not destroy the bandwidth; otherwise account for permutations and the resulting bandwidth.

Pivoting: multiplier size and the same solution

No row exchange

A=[[0.001,1],[1,1]]

ℓ_{21}=1000

U=[[0.001,1],[0,-999]]

x=(1000/999,998/999) · Ax-b=0

Partial pivoting

Swap the two rows first

ℓ_{21}=0.001

U=[[1,1],[0,0.999]]

Sparse matrices and fill-in (continued)

Elimination ordering: fill edges in a sparse star



Positive-definite elimination and Cholesky

For symmetric positive-definite $A = A^T$, symmetric elimination gives

$$A = LDL^T,$$

with unit lower-triangular L and positive diagonal D. Set

$$R = LD^{1/2}, \qquad A = RR^T$$

to obtain Cholesky. Storing one triangle requires about $n(n+1)/2$ values, and the leading cost is $n^3/3$, half that of general LU. Existence also follows inductively. The first pivot $a_{11} > 0$, and elimination leaves

$$S = A_{22} - \frac{1}{a_{11}} a_{21} a_{21}^T$$

as a positive-definite Schur complement. For any nonzero y, substitute $[-(a_{21}^T y)/a_{11};\ y]$ into $x^T A x > 0$, obtaining $y^T S y > 0$. Induction gives positive pivots at every step.

Positive pivots also test positive definiteness. Forming $A^T A$ for least squares squares the condition number, $\kappa_2(A^T A) = \kappa_2(A)^2$, and may overflow. Prefer stable QR directly on A.

Stable QR and Householder reflections

A Householder reflector is

$$H = I - 2\frac{vv^T}{v^Tv}.$$

For nonzero v, it is symmetric orthogonal, so $H^TH = I$. For nonzero x, choose $\alpha = -\operatorname{sign}(x_1)\|x\|$, with sign 1 at $x_1 = 0$, and $v = x - \alpha e_1$. Since $v^Tx = \|x\|^2 - \alpha x_1$, $v^Tv = 2(\|x\|^2 - \alpha x_1)$, we obtain $Hx = \alpha e_1$. This sign avoids cancellation in v_1. Successive reflectors give

$$R = H_k \cdots H_1 A, \qquad Q = H_1 \cdots H_k,$$

and hence $A = QR$.

Householder QR generally preserves orthogonality better than classical Gram–Schmidt and is standard for dense least squares.

For full-column-rank $A \in \mathbb{R}^{m\times n}$, $m \geq n$, use the full orthogonal $Q \in \mathbb{R}^{m\times m}$.

$$Q^TA = \begin{bmatrix} R \\ 0 \end{bmatrix}, \qquad Q^Tb = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix},$$

Stable QR and Householder reflections (continued)

Here R is invertible upper triangular of size $n \times n$. Solve the first n equations.

$$Rx = c_1.$$

For reduced QR, $A = Q_1R$ with $Q_1 \in \mathbb{R}^{m\times n}$, the formula is $Rx = Q_1^Tb$. Full Q^Tb has m components and cannot be the right-hand side of n equations without taking its first block.

The minimum residual squared is $\|c_2\|_2^2$. Do not form A^TA or an explicit R^{-1}. Dense Householder QR costs about $2mn^2 - \frac{2}{3}n^3$ flops. Forming Q adds storage and work; a solve can apply the stored reflectors directly.

A Givens rotation equals the identity except in a 2×2 block on coordinates i, j, and can annihilate one selected entry.

$$G_{ij} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}, \qquad c^2 + s^2 = 1.$$

Stable QR and Householder reflections (continued)

It preserves orthogonality and permits local parallel operations, useful for sparse or banded structures. Dense methods usually favor Householder reflectors, which clear a column segment at once.

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Power iteration and QR eigenvalue iteration

Dense nonsymmetric eigenvalue solvers usually use QR iteration rather than roots of a high-degree characteristic polynomial. First consider power iteration when

$$A = X\,\mathrm{diag}(\lambda_1, \ldots, \lambda_n)X^{-1}, \qquad |\lambda_1| > |\lambda_2| \geq \cdots,$$

and the initial vector has a nonzero coefficient in the leading eigendirection. Then $A^k v_0$ approaches that direction at a rate controlled by $|\lambda_2/\lambda_1|^k$. Normalize each step to prevent overflow or underflow. Tied largest moduli, as in a rotation, need not yield a unique limiting direction.

Inverse iteration solves

$$(A - \mu I)w_{k+1} = w_k$$

and normalizes, amplifying the eigendirection nearest the shift μ. The Rayleigh quotient

$$\mu_k = \frac{w_k^T A w_k}{w_k^T w_k}$$

often supplies the next shift. A nearly singular $A - \mu I$ requires checking solver status; breakdown is not proof of exact convergence.

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Power iteration and QR eigenvalue iteration (continued)

This direction-convergence claim assumes diagonalizability, a shift that is not an eigenvalue, a unique eigenvalue nearest μ, and a nonzero initial coefficient in that eigendirection. The transformed eigenvalues $(\lambda_i - \mu)^{-1}$ then satisfy the power method's unique-largest-modulus condition. Equal distances do not guarantee a unique direction. Use real Rayleigh quotients for real symmetric problems and $w^H Aw/(w^H w)$ for complex ones.

The unshifted QR step is

$$A_k = Q_k R_k, \qquad A_{k+1} = R_k Q_k.$$

Since

$$A_{k+1} = Q_k^T A_k Q_k,$$

so every step is a similarity preserving eigenvalues. Suitable shifts and deflation lead to Schur form. Complex Schur form is triangular; real Schur form can have 2×2 blocks for conjugate pairs, whose eigenvalues cannot be read individually from the diagonal. Unshifted QR need not triangularize every matrix; plane rotation is a counterexample.

Power iteration and QR eigenvalue iteration (continued)

Shifted QR:

$$A_k - \mu_k I = Q_k R_k, \qquad A_{k+1} = R_k Q_k + \mu_k I$$

can accelerate convergence. Implementations first reduce to Hessenberg form, or tridiagonal form for symmetric matrices, then use implicit QR. Hessenberg structure reduces each QR step from $O(n^3)$ to $O(n^2)$; symmetric tridiagonal iteration is cheaper. Large sparse problems often use Lanczos for Hermitian matrices or Arnoldi generally, solving a small projected eigenproblem. For a few extreme eigenvalues this can cost less than a full factorization.

Match the method to the matrix

Structure, conditioning, and the requested output guide algorithm choice.

- general dense $Ax = b$:pivoted LU;
- symmetric positive definite:Cholesky;
- overdetermined least squares:Householder QR;
- Large sparse SPD systems can use preconditioned CG.
- Large nonsymmetric sparse systems can use GMRES or related Krylov methods.
- Rectangular low-rank analysis uses SVD.
- dense nonsymmetric eigenproblem:Hessenberg QR iteration;
- Symmetric eigenproblems use tridiagonal reduction and a suitable structured eigensolver.
- singular values:bidiagonal SVD algorithms.

No algorithm is best for every problem. Hardware and blocking change constants, while mathematical structure determines which methods apply.

Vector and matrix norms

Common vector norms are

$$\|x\|_1 = \sum_i |x_i|, \qquad \|x\|_2 = \left(\sum_i |x_i|^2 \right)^{1/2}, \qquad \|x\|_\infty = \max_i |x_i|.$$

Induced matrix norm:

$$\|A\| = \max_{x \neq 0} \frac{\|Ax\|}{\|x\|}.$$

For the induced Euclidean norm,

$$\|A\|_2 = \sigma_{\max}(A).$$

Other induced norms satisfy

$$\|A\|_1 = \max_j \sum_i |a_{ij}|, \qquad \|A\|_\infty = \max_i \sum_j |a_{ij}|.$$

Vector and matrix norms (continued)

Frobenius norm:

$$\|A\|_F = \left(\sum_{ij} |a_{ij}|^2 \right)^{1/2}.$$

The Frobenius norm treats all entries as one long vector. It is generally not an induced operator norm.

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Condition numbers

For invertible A and a consistent induced norm, define

$$\kappa(A) = \|A\|\|A^{-1}\|.$$

Right-hand side perturbation:

$$\frac{\|\Delta x\|}{\|x\|} \leq \kappa(A) \frac{\|\Delta b\|}{\|b\|}.$$

A first-order matrix-perturbation estimate is

$$\frac{\|\Delta x\|}{\|x\|} \lesssim \kappa(A) \frac{\|\Delta A\|}{\|A\|}.$$

In the Euclidean norm,

$$\kappa_2(A) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

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Condition numbers (continued)

For nonzero b, the right-hand-side bound follows from $\Delta x = A^{-1}\Delta b$ and $\|b\| \le \|A\|\|x\|$. For matrix-only perturbations, set $\epsilon_A = \|\Delta A\|/\|A\|$ and assume $\kappa(A)\epsilon_A < 1$. The perturbed equation gives $(I + A^{-1}\Delta A)\Delta x = -A^{-1}\Delta A x$. The Neumann-series inverse bound yields $\|\Delta x\|/\|x\| \le \kappa(A)\epsilon_A/[1 - \kappa(A)\epsilon_A]$. A large condition number describes problem sensitivity rather than an implementation defect. Scaling changes the perturbation metric being measured.

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Conditioning of inversion and an exact perturbation bound

Let A be invertible and $B = A + E$, using a compatible induced norm. If $\nu = \|A^{-1}E\| < 1$, the Neumann series inverts $I + A^{-1}E$. Thus B is invertible and

$$B^{-1} = (I + A^{-1}E)^{-1}A^{-1}, \qquad B^{-1} - A^{-1} = -B^{-1}EA^{-1}.$$

Using $\|B^{-1}\| \le \|A^{-1}\|/(1 - \nu)$, let $\varepsilon = \|E\|/\|A\|$. If $\kappa(A)\varepsilon < 1$,

$$\frac{\|B^{-1} - A^{-1}\|}{\|A^{-1}\|} \le \frac{\kappa(A)\varepsilon}{1 - \kappa(A)\varepsilon}.$$

The derivative of inversion is $D\,\mathrm{inv}_A[E] = -A^{-1}EA^{-1}$, not entrywise reciprocation. In the 2-norm, its relative condition number equals $\kappa_2(A)$. To attain the bound, write $A = U\Sigma V^T$ and choose $E = -\delta u_n v_n^T$, $0 < \delta < \sigma_n$. Only the smallest singular direction changes, from σ_n to $\sigma_n - \delta$, giving

$$\frac{\|(A + E)^{-1} - A^{-1}\|_2}{\|A^{-1}\|_2} = \frac{\delta}{\sigma_n - \delta} = \frac{\kappa_2(A)(\delta/\sigma_1)}{1 - \kappa_2(A)(\delta/\sigma_1)}.$$

Conditioning of inversion and an exact perturbation bound (continued)

Equality holds in the bound, and $\delta \uparrow \sigma_n$ approaches singularity. Sensitivity of a fixed-right-hand-side solution, sensitivity of the entire inverse, and backward stability of an algorithm are separate claims.

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Residual and forward error

Let x be the exact solution and $\widehat{x}$ the computed one.

$$e = x - \widehat{x}, \qquad r = b - A\widehat{x} = Ae.$$

Therefore

$$e = A^{-1}r, \qquad \|e\| \le \|A^{-1}\|\|r\|.$$

A small absolute residual guarantees a small absolute error only when $\|A^{-1}\|$ is controlled.

For scalar $A = 10^{-12}$, exact $x = 1$, and computed $\widehat{x} = 1 + \delta$,

$$r = -10^{-12}\delta$$

can be tiny even when the relative solution error $|\delta|$ is substantial.

Normalized backward error:

$$\eta = \frac{\|r\|}{\|A\|\|\widehat{x}\| + \|b\|}.$$

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Residual and forward error (continued)

A small η certifies small normwise backward error for this result. One such result alone does not establish backward stability of an algorithm on all inputs. The denominator uses the computed vector rather than the unknown exact solution. Forward error still depends on conditioning.

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Spectral sensitivity

If

$$A = X\Lambda X^{-1},$$

eigenvalue perturbations can be amplified by $\kappa(X)$. A normal matrix has a unitary eigenbasis, avoiding that particular basis-conditioning factor.

Nonnormal matrices can have sensitive eigenvectors and large transients even with closely grouped eigenvalues.

Singular values satisfy the absolute perturbation bound

$$|\sigma_i(A+E) - \sigma_i(A)| \le \|E\|_2.$$

Individual singular vectors can still be sensitive near repeated values. Numerical rank therefore requires a tolerance tied to the application.

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Numerical rank and truncated pseudoinverses

If the singular values have a separation

$$\sigma_1 \geq \cdots \geq \sigma_r \gg \sigma_{r+1} \geq \cdots,$$

a common relative test retains

$$\sigma_i > \tau \sigma_1$$

as numerically nonzero. Choose τ with machine precision, dimension, and noise in mind.

Truncated pseudoinverse:

$$A_\tau^+ = V_r \operatorname{diag}(\sigma_1^{-1}, \ldots, \sigma_r^{-1}) U_r^T.$$

This is the pseudoinverse of the truncated model, not an exact inverse of the original full matrix. Report the threshold, retained rank, residual, and error metric.

Stationary splittings

Choose

$$A = S - T,$$

with invertible S that is inexpensive to solve. Then

$$Sx = Tx + b.$$

Iteration:

$$Sx_{k+1} = Tx_k + b.$$

Set

$$G = S^{-1}T, \qquad c = S^{-1}b,$$

to obtain

$$x_{k+1} = Gx_k + c.$$

Subtracting the exact fixed point gives

$$x - x_{k+1} = G^{k+1}(x - x_0).$$

Stationary splittings (continued)

For a unique exact solution, convergence from every initial guess is equivalent to

$$\rho(G) < 1.$$

If every initial error decays, eigenvector errors force each eigenvalue modulus below 1. Conversely, powers of a Jordan block contain polynomial factors multiplying λ^k, which tend to zero when $|\lambda| < 1$; a zero-eigenvalue block is nilpotent. Thus $G^k \to 0$. Entrywise comparisons of the splitting do not replace this spectral test.

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Jacobi, Gauss-Seidel, and SOR

Use the convention $A = D + L + U$, with strictly lower and upper parts taken directly from A, and assume nonzero diagonal entries. Jacobi uses

$$S = D, \qquad T = -(L + U),$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right).$$

Every update uses the old iterate, allowing parallel computation.

Gauss-Seidel:

$$S = D + L, \qquad T = -U,$$

Gauss-Seidel immediately uses newly computed lower-index components.

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right).$$

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Jacobi, Gauss-Seidel, and SOR (continued)

Strict row diagonal dominance gives $\|G_J\|_\infty < 1$ and also guarantees Gauss-Seidel convergence. SPD guarantees Gauss-Seidel convergence, but not Jacobi convergence. A 3×3 matrix with diagonal 1 and off-diagonal entries 0.9 is SPD, yet its Jacobi matrix has eigenvalue -1.8 and diverges for some initial errors.

SOR relaxes each component using the already relaxed components of the current sweep.

$$x_i^{(k+1)} = (1-\omega)x_i^{(k)} + \frac{\omega}{a_{ii}}\left(b_i - \sum_{j<i} a_{ij}x_j^{(k+1)} - \sum_{j>i} a_{ij}x_j^{(k)}\right).$$

For SPD matrices, $0 < \omega < 2$ ensures convergence, and $\omega = 1$ gives Gauss-Seidel. Averaging a completed unrelaxed GS vector with the old vector gives a different iteration. A coordinate relaxation changes energy by $-\omega(2-\omega)r_i^2/(2a_{ii})$. Each full sweep strictly reduces error energy unless the residual is zero. On the compact unit error-energy sphere, the continuous sweep map has a contraction factor strictly below 1, proving convergence to the unique minimizer.

Convergence and failure examples

For

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix},$$

the Jacobi matrix is

$$G_J = -D^{-1}(L+U) = \begin{bmatrix} 0 & -1/4 \\ -1/3 & 0 \end{bmatrix}.$$

Since

$$G_J^2 = \begin{bmatrix} 1/12 & 0 \\ 0 & 1/12 \end{bmatrix},$$

so

$$\rho(G_J) = 1/\sqrt{12} < 1,$$

so Jacobi converges.

A matrix with zero diagonal,

Convergence and failure examples (continued)

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

cannot use Jacobi directly because of division by zero. Reordering variables can sometimes repair the diagonal, but a nonzero diagonal alone does not imply convergence.

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Conjugate gradients for real SPD systems

For real symmetric positive-definite A, define the A-inner product

$$\langle u, v \rangle_A = u^T A v.$$

and quadratic energy

$$\phi(x) = \frac{1}{2} x^T A x - b^T x.$$

whose unique minimum solves $Ax = b$. CG chooses A-orthogonal search directions in Krylov spaces. In exact arithmetic, it terminates in at most n steps.

Specify an initial vector x_0, defaulting to zero, nonnegative $\mathrm{atol}, \mathrm{rtol}$, and an iteration limit. Set $\tau = \max(\mathrm{atol}, \mathrm{rtol} \|b\|_2)$, and compute $r_0 = b - A x_0$. If $\|r_0\|_2 \le \tau$, return x_0 immediately without computing any step length. Otherwise set $p_0 = r_0$.

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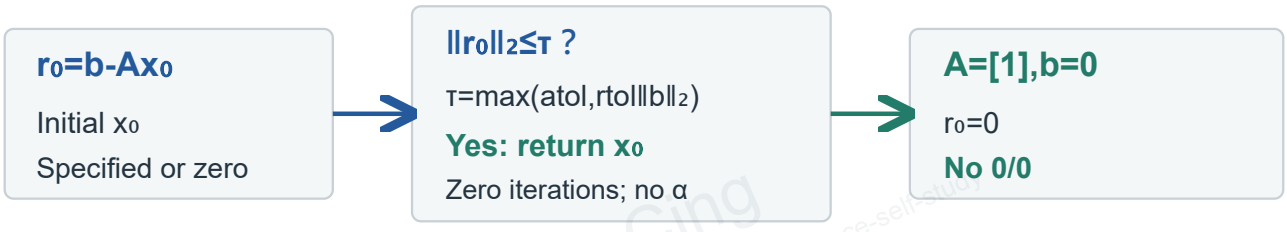
Conjugate gradients for real SPD systems (continued)

At iteration k, compute $q_k = Ap_k$, $\alpha_k = (r_k^T r_k)/(p_k^T q_k)$, then $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k - \alpha_k q_k$. If the new residual passes tolerance, recompute $b - Ax_{k+1}$ before returning. Otherwise compute $\beta_k = (r_{k+1}^T r_{k+1})/(r_k^T r_k)$, $p_{k+1} = r_{k+1} + \beta_k p_k$. In exact arithmetic, every nonzero direction has $p_k^T A p_k > 0$. A nonpositive or nonfinite denominator in floating point signals numerical failure. If the recursive residual passes but the true residual fails, replace it with the true residual and restart with $p = r$; do not reuse a nearly zero old residual. At the iteration limit, report nonconvergence if the true residual still fails.

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Conjugate gradients for real SPD systems (continued)

CG initialization: compute r_0 before iterating

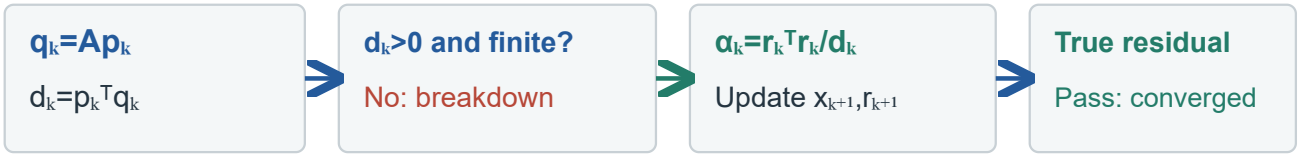


Only if tolerance fails, set $p_0 = r_0$ and begin iteration.

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Conjugate gradients for real SPD systems (continued)

CG iteration: denominator, updates, and true residual



True pass: converged; failed at limit: iteration_limit.

Recursive pass, true fail: r ← b−Ax; p ← r (restart).

Otherwise compute β, update p, and repeat from q=Ap.

For $A = [1], b = [0], x_0 = [0]$, the initial residual is zero, so return after zero iterations; a premature step calculation gives $0/0$. For $A = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix}$, $b = (1, 2)^T$, $x_0 = 0$, start with $r_0 = p_0 = (1, 2)^T$, $Ap_0 = (6, 7)^T$. Then $\alpha_0 = 5/20 = 1/4$, $x_1 = (1/4, 1/2)^T$, $r_1 = (-1/2, 1/4)^T$. Next $\beta_0 = 1/16$, $p_1 = (-7/16, 3/8)^T$, $\alpha_1 = 4/11$, and $x_2 = (1/11, 7/11)^T$, which satisfies $Ax_2 = b$.

Conjugate gradients for real SPD systems (continued)

Exact line searches and this β_k make residuals mutually orthogonal and directions mutually A-conjugate. Nonzero conjugate directions are independent, so at most n occur. Floating point can lose orthogonality, so finite termination does not replace residual checks. Each iteration uses one matrix-vector product and a fixed number of inner products, costing $O(\mathrm{nnz}(A) + n)$ with a fixed number of length-n work vectors. No inverse is formed.

In the energy norm, the error satisfies

$$\|x - x_k\|_A \leq 2 \left(\frac{\sqrt{\kappa_2(A)} - 1}{\sqrt{\kappa_2(A)} + 1} \right)^k \|x - x_0\|_A.$$

Ill conditioning can slow convergence, while clustered eigenvalues often give faster convergence than this condition-number bound predicts. Standard CG requires an SPD operator and an SPD preconditioner used in a symmetry-preserving formulation.

CG correctness through Krylov spaces and energy minimization

Work in exact arithmetic. Let x_* be the solution, and define $\mathcal{K}_0 = \{0\}$ and

$$\mathcal{K}_k = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1}r_0\}.$$

At iteration k, maintain that $p_0, \ldots, p_{k-1}$ span $\mathcal{K}_k$ and are mutually A-conjugate, $x_k \in x_0 + \mathcal{K}_k$, and $r_k = b - Ax_k \perp \mathcal{K}_k$. Then x_k uniquely minimizes energy in this affine space, because for $z \in \mathcal{K}_k$,

$$\phi(x_k + z) - \phi(x_k) = -z^T r_k + \tfrac{1}{2} z^T A z = \tfrac{1}{2} z^T A z,$$

which is strictly positive for nonzero z. The invariant holds at $k = 0$. Also $r_k = r_0 - A(x_k - x_0) \in \mathcal{K}_{k+1}$. A nonzero residual is orthogonal to $\mathcal{K}_k$ and adds an independent direction.

Set $p_0 = r_0$. For $k \geq 1$, let $p_k = r_k + \beta_{k-1} p_{k-1}$. For $i \leq k-2$, $Ap_i \in \mathcal{K}_{i+2} \subseteq \mathcal{K}_k$, so $p_i^T A r_k = (Ap_i)^T r_k = 0$. Existing conjugacy leaves only the preceding direction to cancel.

$$\beta_{k-1} = -\frac{p_{k-1}^T A r_k}{p_{k-1}^T A p_{k-1}}.$$

Since $r_k = r_{k-1} - \alpha_{k-1} A p_{k-1}$, with $r_{k-1} \in \mathcal{K}_k$,

CG correctness through Krylov spaces and energy minimization (continued)

$$\|r_k\|^2 = -\alpha_{k-1} r_k^T A p_{k-1}.$$

substituting $\alpha_{k-1} = \|r_{k-1}\|^2 / (p_{k-1}^T A p_{k-1})$ gives $\beta_{k-1} = \|r_k\|^2 / \|r_{k-1}\|^2$. Thus the short recurrence enforces conjugacy to every earlier direction. The new direction differs from the nonzero residual only by an old direction, so it extends the span exactly to $\mathcal{K}_{k+1}$.

Line minimization gives derivative $-p_k^T r_k + \alpha p_k^T A p_k$. Since $p_k^T r_k = \|r_k\|^2$, this yields the stated α_k. After updating, $p_k^T r_{k+1} = 0$. For $i < k$,

$$p_i^T r_{k+1} = p_i^T r_k - \alpha_k p_i^T A p_k = 0.$$

so $r_{k+1} \perp \mathcal{K}_{k+1}$, preserving the invariant. Since $r_i \in \mathcal{K}_{i+1}$, residuals are mutually orthogonal. If $\sum c_i p_i = 0$, multiply by $p_j^T A$ to get $c_j p_j^T A p_j = 0$. Nonzero conjugate directions are independent, so there are at most n. If not terminated earlier, the residual at step n is orthogonal to $\mathbb{R}^n$ and must vanish.

Deriving the CG condition-number bound

Let $e_k = x_* - x_k$. Energy minimization makes CG minimize $\|e_k\|_A$ in $x_0 + \mathcal{K}_k$. Since $r_0 = Ae_0$, every available error is $p(A)e_0$ with $\deg p \le k, p(0) = 1$. Expanding in eigen-coordinates of $A = Q\Lambda Q^T$ gives

$$\|e_k\|_A \le \min_{\deg p \le k,\, p(0)=1} \max_i |p(\lambda_i)| \, \|e_0\|_A.$$

Let $a = \lambda_{\min} > 0, b = \lambda_{\max} > a$. Chebyshev polynomials satisfy $T_k(\cos\theta) = \cos(k\theta)$, hence $|T_k| \le 1$ on $[-1, 1]$. Choose

$$p(t) = \frac{T_k((b + a - 2t)/(b - a))}{T_k((b + a)/(b - a))}.$$

so $p(0) = 1$ and $[a, b]$ maps into $[-1, 1]$ in the numerator. Cosine addition gives $T_{k+1}(z) = 2zT_k(z) - T_{k-1}(z)$, with $T_0 = 1, T_1 = z$. This verifies $T_k(z) = (\rho^k + \rho^{-k})/2$ for $z > 1$, where $\rho = z + \sqrt{z^2 - 1}$. With $\kappa = b/a$, substituting $z = (b + a)/(b - a)$ gives $\rho = (\sqrt{\kappa} + 1)/(\sqrt{\kappa} - 1)$, so

$$\|e_k\|_A \le 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|e_0\|_A.$$

Deriving the CG condition-number bound (continued)

If $a = b$, then $A = aI$; a single step of length $1/a$ solves any nonzero residual. Treat this case separately rather than dividing by zero. Eigenvalue clusters permit better polynomials, so the interval bound need not be tight. The proof uses exact orthogonality; floating-point acceptance still requires the true residual.

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CG for complex Hermitian positive-definite systems

This extension uses the complex inner product. The real SPD formulas with $r^T r$ must not be applied unchanged to complex matrices. If

$$A^H = A, \qquad z^H A z > 0 \quad (z \neq 0),$$

then A is Hermitian positive definite. With inner product $u^H v$, replace the real formulas by

$$r_k = b - A x_k, \qquad \alpha_k = \frac{r_k^H r_k}{p_k^H A p_k},$$

$$x_{k+1} = x_k + \alpha_k p_k, \qquad r_{k+1} = r_k - \alpha_k A p_k,$$

$$\beta_k = \frac{r_{k+1}^H r_{k+1}}{r_k^H r_k}, \qquad p_{k+1} = r_{k+1} + \beta_k p_k.$$

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CG for complex Hermitian positive-definite systems (continued)

Hermitian symmetry makes $p_k^H A p_k$ real, and positive definiteness makes numerator and denominator positive for nonzero residual and direction. Use $\|r_k\|_2 = \sqrt{r_k^H r_k}$ and check r_0 first. Return x_0 immediately if it is zero. A preconditioner M must also be Hermitian positive definite and enter through a symmetry-preserving change of variables. The real proof then carries over with conjugate inner products and real positive energies and step lengths. Checking $A^T = A$ is insufficient; complex symmetric [i] has no positive real energy.

Real SPD matrices are the special case where conjugation leaves entries unchanged.

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GMRES for nonsymmetric systems

First take $A \in \mathbb{R}^{n \times n}$; complex Arnoldi uses conjugate inner products. The following Arnoldi and least-squares construction provides the algorithmic details. GMRES works in the Krylov space

$$\mathcal{K}_k(A, r_0) = \operatorname{span}\{r_0, Ar_0, \ldots, A^{k-1} r_0\}$$

and minimizes the residual.

$$\min_{x \in x_0 + \mathcal{K}_k} \|b - Ax\|_2.$$

Arnoldi starts with $v_1 = r_0 / \|r_0\|$. For $w = Av_j$, subtract $h_{ij} v_i$ successively, with $h_{ij} = v_i^T w$, and normalize using $h_{j+1,j} = \|w\|$. This yields $AV_k = V_{k+1} \overline{H}_k$. For $x = x_0 + V_k y$, the residual norm equals $\|\|r_0\| e_1 - \overline{H}_k y\|_2$; solve this small least-squares problem for the iterate. Zero initial residual finishes immediately. A zero Arnoldi remainder requires distinguishing convergence from breakdown. Restarting limits storage but can cause stagnation by discarding old directions.

GMRES for nonsymmetric systems (continued)

For nonnormal matrices, eigenvalues alone do not predict GMRES transients. The field of values, pseudospectra, and actual residual history add information. Monitor the true residual. Each unrestarted Arnoldi step uses a matrix-vector product and orthogonalization against all old directions. After $k \leq n$ steps, extra storage is $O(kn)$, and work is $O(k \operatorname{nnz}(A) + k^2 n)$. Dense matrix-vector products alone cost $O(kn^2)$. Restarted GMRES(m) limits storage to $O(mn)$, with possible stagnation.

GMRES step one: solve the reduced least-squares problem

Use the nonsymmetric invertible matrix and initial guess

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x_0 = 0.$$

The initial residual is $r_0 = b$, with $\beta = \|r_0\|_2 = 1$, so $v_1 = e_2 = (0,1)^T$. The first Arnoldi step gives

$$w = Av_1 = (1,1)^T, \quad h_{11} = v_1^T w = 1,$$
$$w \leftarrow w - h_{11} v_1 = (1,0)^T, \quad h_{21} = 1, \quad v_2 = e_1.$$

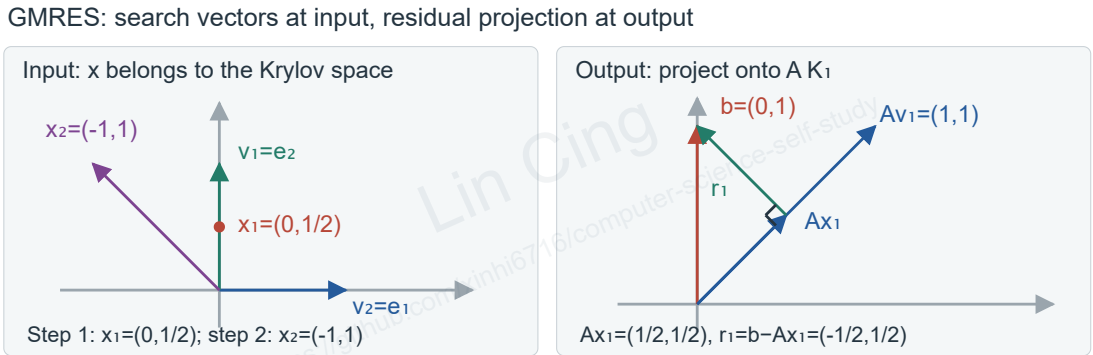
Thus $\overline{H}_1 = (1,1)^T$. Solve the one-variable least-squares problem:

$$\min_y \left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} y \right\|_2^2 = \min_y \left((1-y)^2 + y^2 \right).$$

Its derivative is $4y - 2$, giving $y = 1/2$. Reconstruct $x_1 = V_1 y = (0, 1/2)^T$. The true residual is $r_1 = b - Ax_1 = (-1/2, 1/2)^T$, with norm $1/\sqrt{2}$. Check $(Av_1)^T r_1 = 0$: the residual is perpendicular to the **output space** $A\mathcal{K}_1$, not necessarily to the input search space $\mathcal{K}_1$.

GMRES: input search and output projection

Both spaces happen to be $\mathbb{R}^2$, but they have different roles. The left panel tracks the unknown x. The right projects b onto $A\mathcal{K}_1 = \text{span}\{(1,1)^T\}$. Each panel uses equal coordinate scales. The right-angle marker represents $(Av_1)^T r_1 = 0$.



r₁ is perpendicular to A K₁, not necessarily K₁

GMRES step two: zero Arnoldi remainder and exact termination

Now $V_2 = [v_1, v_2] = [e_2, e_1]$. Start from $w = Av_2 = e_1$:

Operation	Numerical result
$h_{12} = v_1^T w$	0; w is unchanged
$h_{22} = v_2^T w = 1$	Subtracting $h_{22}v_2$ leaves $w = (0, 0)^T$
$h_{32} = \|w\|_2$	0; do not divide by it to form v_3

The Krylov space is the whole $\mathbb{R}^2$, and $AV_2 = V_2 H_2$, where

$$H_2 = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \qquad H_2 y = \beta e_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

The rows give $y_1 = 1$, then $y_1 + y_2 = 0$, so $y = (1, -1)^T$. Reconstruct

$$x_2 = V_2 y = e_2 - e_1 = (-1, 1)^T, \qquad Ax_2 = (0, 1)^T = b.$$

Zero true residual makes this a happy breakdown. A zero Arnoldi remainder alone does not certify convergence; verify the original residual.

Preconditioners

A useful $M \approx A$ is cheap to apply, supports parallel computation, and makes the transformed problem easier to iterate. Left preconditioning is

$$M^{-1}Ax = M^{-1}b.$$

For right preconditioning, set

$$AM^{-1}y = b, \qquad x = M^{-1}y$$

which changes variables while retaining the original residual interpretation.

Common choices include

Preconditioners (continued)

- diagonal/Jacobi;
- block Jacobi;
- incomplete LU or incomplete Cholesky;
- algebraic multigrid;
- domain decomposition;
- physics-informed approximation.

Compare setup, iteration, and communication costs, not only iteration counts. The left-preconditioned residual is $M^{-1}(b - Ax)$, so also compute the true residual. CG requires a symmetry-preserving equivalent formulation; an arbitrary nonsymmetric $M^{-1}A$ cannot simply be treated as SPD.

Stopping rules and reproducibility

Record at least the following for a solve.

1. initial residual;
2. relative residual reduction;
3. Norm and scaling conventions.
4. Maximum iterations and time limit.
5. library convergence/breakdown status;
6. final true residual;
7. Condition or forward-error estimate.
8. Deterministic settings or seed where relevant, plus a repeat check.

Evaluate backward error relative to $\|A\|\|\widehat{x}\| + \|b\|$. Rounding can make a recursively updated residual appear small while the true residual remains larger.

Chapter check

You can now do the following.

- Explain partial pivoting and separate multiplier growth from conditioning.
- Estimate factorization costs and reason about sparse fill.
- Derive perturbation bounds and distinguish residual from forward error.
- Compute norms and interpret conditioning through SVD.
- Test stationary iterations by spectral radius.
- State correct Jacobi, Gauss-Seidel, and component SOR formulas and assumptions.
- Explain CG, Arnoldi-GMRES, restart, and preconditioning.
- Report numerical rank tolerances, residuals, status, and assumptions.

Chapter 12 connects covariance, Gaussian models, and weighted least squares.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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Samples and expectations

Observed sample values $x_1, \ldots, x_N$ are numbers, with $N \geq 1$. Expectations here require finite first absolute moments. Variance, covariance, and the associated transformation formulas require finite second moments. The sample mean is

$$\overline{x} = \frac{1}{N} \sum_{i=1}^{N} x_i.$$

A random variable's model mean is

$$\mu = \mathbb{E}[X].$$

For discrete values a_i with probabilities p_i,

$$\mathbb{E}[X] = \sum_{i=1}^{k} p_i a_i.$$

Continuous density f_X:

$$\mathbb{E}[X] = \int x f_X(x)\, dx.$$

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Samples and expectations (continued)

After observation, the sample mean is a numerical summary. Before observation, it is a random statistic. Expectation belongs to the probability model. Similar-looking formulas have different interpretations.

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Variance and the N−1 denominator

For fixed observed samples,

$$\overline{x} = \frac{1}{N}\sum_{i=1}^{N} x_i.$$

the usual unbiased sample variance is

$$s^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \overline{x})^2.$$

The population variance is

$$\operatorname{Var}(X) = \mathbb{E}[(X-\mu)^2] = \mathbb{E}[X^2] - \mu^2.$$

The $N-1$ denominator requires $N \geq 2$. For IID samples with finite variance, use $\sum_i (X_i - \overline{X})^2 = \sum_i (X_i - \mu)^2 - N(\overline{X} - \mu)^2$. Expectations give $N\sigma^2 - N(\sigma^2/N) = (N-1)\sigma^2$. Therefore

$$\mathbb{E}[s^2] = \sigma^2.$$

Variance and the N−1 denominator (continued)

For the mean squared deviation of an entire finite population, division by N may instead be appropriate. Distinguish a descriptive statistic from an estimator.

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Means and variances under affine changes

For constants a, b,

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b,$$
$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

A shift changes the mean but not spread. Negative scaling reverses direction while multiplying variance by the same positive factor a^2.

Independent X, Y:

$$\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y.$$

Without independence, the general identity is

$$\operatorname{Var}(X + Y) = \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X, Y).$$

Independence with finite second moments implies zero covariance. The converse generally fails.

Densities, CDFs, and a uniform example

For an absolutely continuous random variable, the CDF has a derivative equal to its density almost everywhere. A continuous CDF alone does not guarantee a density. In the absolutely continuous case,

$$F_X(x) = \mathbb{P}(X \le x), \qquad f_X(x) = F_X'(x).$$

Interval probabilities satisfy

$$\mathbb{P}(a \le X \le b) = F_X(b) - F_X(a) = \int_a^b f_X(x)\,dx.$$

For $X \sim U(0, L)$, with $L > 0$, the density is $1/L$ on the interval. Its mean is

$$\mathbb{E}[X] = \frac{L}{2}.$$

Variance:

$$\operatorname{Var}(X) = \int_0^L \left(x - \frac{L}{2}\right)^2 \frac{1}{L}\,dx = \frac{L^2}{12}.$$

Densities, CDFs, and a uniform example (continued)

For $L = 3$, the mean is 1.5 and variance is $3/4$. Translating the interval by a adds a to the mean and leaves variance unchanged.

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Gaussian standardization and the CLT

Standard normal:

$$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

$X \sim N(\mu, \sigma^2)$:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardization:

$$Z = \frac{X - \mu}{\sigma}$$

gives mean zero and variance one, requiring $\sigma > 0$.

For i.i.d. observations with $0 < \sigma^2 < \infty$, the central limit theorem states

$$\frac{\sqrt{N}(\overline{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

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Gaussian standardization and the CLT (continued)

This describes the limiting distribution of the standardized sample mean, not conversion of the raw data into Gaussian observations. Dependence or infinite-variance tails can violate these assumptions. The next section proves this IID finite-variance form using Lévy's continuity theorem.

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The CLT through independence and a second-order expansion

This proof uses Lévy's continuity theorem for the distributional convergence step; the general probability-theory proof of that theorem is outside this chapter. Standardize by setting $U_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}[U_i] = 0$, $\mathrm{Var}(U_i) = 1$, and

$$Z_N = \frac{\sqrt{N}(\overline{X} - \mu)}{\sigma} = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} U_i.$$

Let $\varphi_N(t) = \mathbb{E}[e^{itZ_N}]$. Independence multiplies characteristic functions, and identical distributions make the factors equal.

$$\varphi_N(t) = \left[\varphi_U\left(\frac{t}{\sqrt{N}}\right)\right]^N.$$

A finite second moment gives a second-order expansion at zero.

$$\varphi_U(s) = 1 - \frac{1}{2}s^2 + o(s^2).$$

The CLT through independence and a second-order expansion (continued)

No third moment is needed. Write $e^{iy} = 1 + iy - y^2/2 + R(y)$, with $R(y)/y^2 \to 0$ and a constant C such that $|R(y)| \le Cy^2$. Thus $R(sU)/s^2 \to 0$ almost surely and is dominated by integrable CU^2. Dominated convergence gives $\mathbb{E}[R(sU)] = o(s^2)$; linearity and $\mathbb{E}[U] = 0, \mathbb{E}[U^2] = 1$ give the expansion.

Consequently,

$$N \log \varphi_U(t/\sqrt{N}) = N \log \left(1 - \frac{t^2}{2N} + o(1/N)\right) \longrightarrow -\frac{t^2}{2}.$$

Hence $\varphi_N(t) \to e^{-t^2/2}$, the characteristic function of $N(0,1)$. Lévy's continuity theorem states that pointwise convergence to a characteristic function continuous at zero implies convergence in distribution. Thus Z_N converges to the standard normal. Uniqueness of characteristic functions alone would not establish convergence. The conclusion concerns standardized averages, not arbitrary large samples becoming normally distributed.

The CLT through independence and a second-order expansion (continued)

For fixed t and sufficiently large N, choose one logarithm branch near 1 and exponentiate the limit. Arbitrary principal complex logarithms do not satisfy an unrestricted product-addition rule. Finite-variance CLT gives no distribution-uniform finite-sample error guarantee.

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Standardized binomial distributions

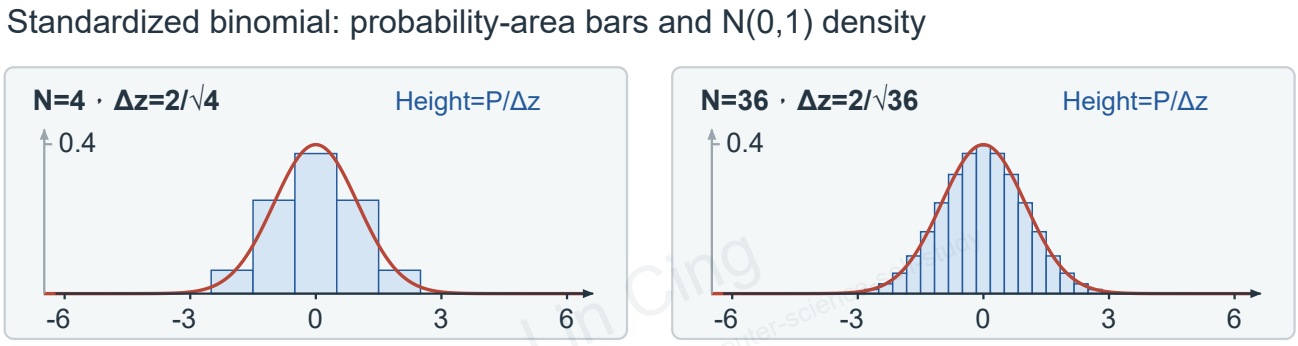
Let B_i be independent fair 0/1 observations, with $S_N = \sum_i B_i$. Then

$$P(S_N = k) = \binom{N}{k} 2^{-N}, \quad \mathbb{E}[S_N] = N/2, \quad \mathrm{Var}(S_N) = N/4.$$

The standardized $Z_N = (S_N - N/2)/(\sqrt{N}/2)$ has spacing $\Delta z = 2/\sqrt{N}$. At $N = 4$, its values are $-2, -1, 0, 1, 2$, with probabilities $(1, 4, 6, 4, 1)/16$. Its mean is zero and variance one, but it remains discrete.

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Standardized binomial distributions (continued)



Each bar has width Δz and height $P(S_N = k)/\Delta z$, so its area equals probability and total area is 1. The red curve is the standard normal density. Point probability itself cannot be compared with density height. These two finite examples illustrate the general theorem without proving it.

Error of a Monte Carlo average

For IID simulation outputs $Y_1, \dots, Y_N$ with mean μ and finite variance τ^2, the estimator $\widehat{\mu} = N^{-1} \sum_i Y_i$ satisfies

$$\mathbb{E}[\widehat{\mu}] = \mu, \qquad \mathrm{Var}(\widehat{\mu}) = \frac{\tau^2}{N}, \qquad P(|\widehat{\mu} - \mu| \geq \varepsilon) \leq \frac{\tau^2}{N\varepsilon^2}.$$

The last step is Chebyshev's inequality. Standard error scales as $N^{-1/2}$, a probabilistic sampling statement rather than a deterministic error bound on each run.

For $X \sim U(-1, 1)$, $Y = X^2$, integration gives $\mathbb{E}[Y] = 1/3$, $\mathbb{E}[Y^2] = 1/5$, hence $\tau^2 = 4/45$. At $N = 1000$, standard error is $\sqrt{4/45000} \approx 0.00943$, and $P(|\widehat{\mu} - 1/3| \geq 0.1) \leq 4/450 \approx 0.00889$. Correlated simulations require all cross-covariance terms; τ^2/N no longer applies unchanged.

A joint probability table

For discrete variables, define joint mass

$$P_{ij} = \mathbb{P}(X = x_i, Y = y_j),$$
$$P_{ij} \geq 0, \qquad \sum_{i,j} P_{ij} = 1.$$

Marginals:

$$\mathbb{P}(X = x_i) = \sum_j P_{ij}, \qquad \mathbb{P}(Y = y_j) = \sum_i P_{ij}.$$

Independence requires every pair to satisfy

$$P_{ij} = \mathbb{P}(X = x_i)\mathbb{P}(Y = y_j).$$

Marginals alone generally do not determine the joint distribution. Means or one covariance value cannot establish independence either.

A joint probability table (continued)

Joint probability table: marginals and covariance

X \ Y	Y=0	Y=1
X=0	0.2	0.1
X=1	0.3	0.4
PX=(0.3,0.7)	PY=(0.5,0.5)	

Total probability=0.2+0.1+0.3+0.4=1

E(X)=0.7 · E(Y)=0.5

E(XY)=0.4

Cov=0.4-0.7×0.5=0.05

VarX=0.21 · VarY=0.25

ρ=0.05/√0.0525=1/√21

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Conditional probability, Bayes, and zero correlation

If $P(X = x_i) = p_i > 0$, define

$$P(Y = y_j \mid X = x_i) = \frac{P_{ij}}{p_i}.$$

Using $P = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.4 \end{bmatrix}$, with rows $X = 0, 1$ and columns $Y = 0, 1$,

$$P(Y = 1 \mid X = 0) = \frac{1}{3}, \quad P(Y = 1 \mid X = 1) = \frac{4}{7}.$$

total probability gives $P(Y = 1) = (1/3)(0.3) + (4/7)(0.7) = 0.5$. Bayes' formula reverses the conditioning.

$$P(X = 1 \mid Y = 1) = \frac{P(Y = 1 \mid X = 1)P(X = 1)}{P(Y = 1)} = \frac{(4/7)(0.7)}{0.5} = 0.8.$$

Zero covariance does not imply independence. Let X be uniform on $-1, 0, 1$ and $Y = X^2$. Then $\mathbb{E}[X] = \mathbb{E}[XY] = 0$, so $\mathrm{Cov}(X, Y) = 0$, but $P(Y = 0 \mid X = 0) = 1 \neq P(Y = 0) = 1/3$. Joint normality supplies the additional zero-covariance-implies-independence result.

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Covariance and correlation

Correlation summarizes linear joint variation. Conditional probabilities from the joint distribution distinguish dependence from correlation.

$$\mathrm{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

For a discrete joint distribution,

$$\mathrm{Cov}(X, Y) = \sum_{i,j} P_{ij}(x_i - \mu_X)(y_j - \mu_Y).$$

Covariance depends on units; correlation removes scale.

$$\rho_{XY} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}.$$

With both variances positive,

$$-1 \leq \rho_{XY} \leq 1.$$

To prove the lower or upper bound, use

$$\mathrm{Var}\left(\frac{X}{\sigma_X} - \frac{Y}{\sigma_Y}\right) = 2(1 - \rho_{XY}) \geq 0,$$

Covariance and correlation (continued)

Replacing the minus by plus gives $1 + \rho_{XY} \geq 0$. Zero correlation generally does not imply independence.

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Covariance matrices and positive semidefiniteness

For a random column vector,

$$\mu = \mathbb{E}[X], \qquad V = \mathbb{E}[(X-\mu)(X-\mu)^T].$$

$V_{ij} = \sigma_{ij}$; the diagonal contains variances, and $V = V^T$.

For every coefficient vector c,

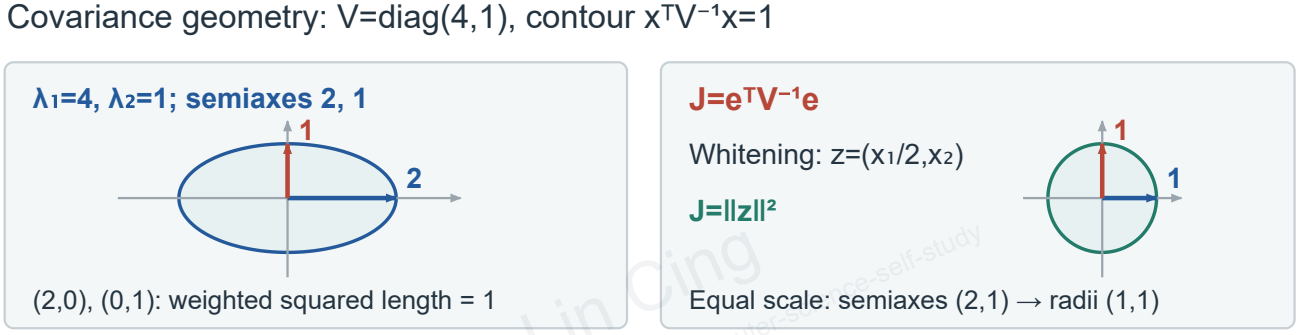
$$c^T V c = \mathrm{Var}(c^T X) = \mathbb{E}[(c^T(X-\mu))^2] \geq 0.$$

so V is positive semidefinite. A discrete joint distribution also gives the constructive factorization

$$V = \sum_{i,j} P_{ij} u_{ij} u_{ij}^T, \qquad u_{ij} = \begin{bmatrix} x_i \\ y_j \end{bmatrix} - \mu.$$

Positive definiteness means no nonzero combination $c^T X$ is almost surely constant. Deficient rank means an exact linear dependence among variable directions.

Covariance matrices and positive semidefiniteness (continued)



The diagram uses a joint normal distribution with $V = \mathrm{diag}(4,1)$, $\mu = 0$. Its equal-density contour $x^T V^{-1} x = 1$ is $x_1^2/4 + x_2^2 = 1$, with semiaxes 2 and 1. For general distributions, covariance alone need not determine elliptical density contours. Joint normality makes this quadratic form determine the density.

Linear transformation of a random vector

Let

$$Y = AX + b.$$

Mean:

$$\mu_Y = A\mu_X + b.$$

Covariance:

$$V_Y = \mathbb{E}[(AX + b - \mu_Y)(\cdots)^T] = AV_XA^T.$$

The shift cancels under centering. The transformed covariance may be singular if the transformation has redundant output directions, such as repeated rows, or the original covariance already has null directions.

The variance of a^TX is a^TVa, and the covariance of a^TX, b^TX is a^TVb. Thus covariance defines a symmetric bilinear form on coefficient vectors.

Sample covariance

Place samples in columns.

$$X_{\text{data}} \in \mathbb{R}^{M\times n}.$$

Row means:

$$\widehat{\mu}_j = \frac{1}{n}\sum_{k=1}^{n}(X_{\text{data}})_{jk}.$$

Center each row to obtain X_c. The usual sample covariance is

$$\widehat{V}_{n-1} = \frac{1}{n-1}X_cX_c^T.$$

Population/descriptive version:

$$\widehat{V}_n = \frac{1}{n}X_cX_c^T.$$

The two denominators give identical eigenvectors and eigenvalues differing by $n/(n-1)$. State the convention in a PCA report.

Sample covariance (continued)

Unbiasedness requires $n \geq 2$, i.i.d. observations, and finite second moments. Centering gives $X_c\mathbf{1} = 0$, so rank is at most $\min(M, n-1)$. The nonzero covariance eigenvalues are scaled squared singular values, squaring the condition ratio on the nonzero spectrum. For deficient rank, the ordinary inverse condition number is already infinite. SVD avoids explicitly forming the Gram matrix.

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Whitening noise

For SPD error covariance V, V^{-1} weights directions inversely to their variances. Whitening gives

$$W = V^{-1/2}E.$$

Then

$$\mathrm{Cov}(W) = V^{-1/2}VV^{-1/2} = I.$$

The Mahalanobis length is

$$\|e\|_{V^{-1}}^2 = e^T V^{-1} e.$$

For diagonal V, divide each residual by its standard deviation. When covariance is estimated with more dimensions than samples, it can be singular; use shrinkage, dimension reduction, or a structural model.

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Multivariate Gaussian density

For $V \succ 0$, the density is

$$p(x) = \frac{1}{(2\pi)^{M/2}\det(V)^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^T V^{-1}(x-\mu)\right].$$

Its exponent has level sets

$$(x-\mu)^T V^{-1}(x-\mu) = r^2$$

which are ellipsoids. Write the spectral decomposition

$$V = Q\Lambda Q^T, \qquad \Lambda = \operatorname{diag}(\lambda_i).$$

and set

$$Y = Q^T(X-\mu).$$

The transformed covariance is diagonal. For jointly Gaussian variables, the resulting uncorrelated components are independent. The Gaussian density factors in these coordinates, giving the statistical meaning of the spectral transformation.

Gaussian normalization

Using

$$V = Q\Lambda Q^T$$

and orthogonality gives

$$(x-\mu)^T V^{-1}(x-\mu) = \sum_i \frac{y_i^2}{\lambda_i}, \qquad y = Q^T(x-\mu).$$

The orthogonal coordinate change has absolute determinant 1, so $dx = dy$. The positive integrand factors into one-dimensional Gaussian integrals.

$$\int_{\mathbb{R}^M} \exp\left(-\frac{1}{2}\sum_i \frac{y_i^2}{\lambda_i}\right) dy = (2\pi)^{M/2} \prod_i \lambda_i^{1/2}.$$

Also,

$$\prod_i \lambda_i^{1/2} = \det(V)^{1/2},$$

Gaussian normalization (continued)

The one-dimensional constant follows by squaring the integral. If $I = \int_{\mathbb{R}} e^{-t^2/2} dt$, then polar coordinates give $I^2 = \int_0^{2\pi} \int_0^{\infty} e^{-r^2/2} r \, dr \, d\theta = 2\pi$, so $I = \sqrt{2\pi}$. Scaling each coordinate proves the stated normalization. A singular covariance instead gives a Gaussian supported on a lower-dimensional affine subspace, without an ordinary density relative to full-dimensional Lebesgue measure.

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Weighted least squares

For $A \in \mathbb{R}^{m \times n}$, $b, e \in \mathbb{R}^m$, $x \in \mathbb{R}^n$, use the model

$$b = Ax + e, \qquad \mathbb{E}[e] = 0, \qquad \operatorname{Cov}(e) = V \succ 0.$$

Ordinary least squares minimizes

$$\|b - Ax\|_2^2,$$

With fixed design and zero-mean errors, OLS can remain unbiased. Equal variances and uncorrelated errors, covariance $\sigma^2 I$, suffice for BLUE; independence is unnecessary. For general known SPD covariance, minimize the Mahalanobis residual.

$$J(x) = (b - Ax)^T V^{-1} (b - Ax).$$

The gradient is

$$\nabla J = -2A^T V^{-1} (b - Ax).$$

Setting it to zero gives the weighted normal equations.

$$A^T V^{-1} A x = A^T V^{-1} b.$$

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Weighted least squares (continued)

At full column rank,

$$\widehat{x} = (A^T V^{-1} A)^{-1} A^T V^{-1} b.$$

this stationary point is the unique global minimum. For nonzero h, expand

$$J(x+h) - J(x) = -2h^T A^T V^{-1}(b - Ax) + h^T A^T V^{-1} A h.$$

At $\widehat{x}$, the first term vanishes. Full column rank gives $Ah \neq 0$, and $V \succ 0$ gives $h^T A^T V^{-1} A h > 0$. Every nonzero displacement increases the objective.

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Three heteroscedastic measurements

Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$, $b = (2, 5, 4)^T$, $V = \operatorname{diag}(1, 4, 9)$. The whitened equations are

$$x_1 \approx 2, \quad \tfrac{1}{2}x_1 + \tfrac{1}{2}x_2 \approx \tfrac{5}{2}, \quad \tfrac{1}{3}x_2 \approx \tfrac{4}{3}.$$

Consequently,

$$A^T V^{-1} A = \begin{bmatrix} 5/4 & 1/4 \\ 1/4 & 13/36 \end{bmatrix}, \quad A^T V^{-1} b = \begin{bmatrix} 13/4 \\ 61/36 \end{bmatrix}.$$

Multiply the first normal equation by 4 to get $5x_1 + x_2 = 13$, and the second by 36 to get $9x_1 + 13x_2 = 61$. Substitute $x_2 = 13 - 5x_1$, obtaining $-56x_1 = -108$. Thus

$$\widehat{x} = (27/14, 47/14)^T, \qquad A\widehat{x} = (27/14, 37/7, 47/14)^T.$$

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Three heteroscedastic measurements (continued)

Measurement	Residual $e_i = b_i - (A\widehat{x})_i$	Weighted square e_i^2/σ_i^2
1	$1/14$	$1/196$
2	$-2/7$	$4/196$
3	$9/14$	$9/196$

The sum is $J = 14/196 = 1/14$. Since $V^{-1}e = (1/14, -1/14, 1/14)^T$, $A^TV^{-1}e = 0$. The ordinary residual need not satisfy $A^Te = 0$; the whitened residual is orthogonal to the whitened design.

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Uncorrelated errors with unequal variances

For finite positive variances and mutually uncorrelated errors, with independence sufficient as well,

$$V = \operatorname{diag}(\sigma_1^2, \dots, \sigma_m^2),$$

minimize

$$J(x) = \sum_{i=1}^{m} \frac{(b_i - a_i^T x)^2}{\sigma_i^2}.$$

High-variance measurements receive less weight.

For repeated measurements of one scalar x,

$$A = (1, 1)^T, \qquad b = (b_1, b_2)^T.$$

the normal equation is

$$\left(\frac{1}{\sigma_1^2} + \frac{1}{\sigma_2^2}\right) x = \frac{b_1}{\sigma_1^2} + \frac{b_2}{\sigma_2^2}.$$

so

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Uncorrelated errors with unequal variances (continued)

$$\widehat{x} = \frac{b_1/\sigma_1^2 + b_2/\sigma_2^2}{1/\sigma_1^2 + 1/\sigma_2^2},$$

giving the precision-weighted average.

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Covariance of the estimator

The estimator is linear.

$$\widehat{x} = Gb, \qquad G = (A^T V^{-1} A)^{-1} A^T V^{-1}.$$

If b has covariance V,

$$\mathrm{Cov}(\widehat{x}) = G V G^T.$$

substituting G gives

$$\mathrm{Cov}(\widehat{x}) = (A^T V^{-1} A)^{-1}.$$

Diagonal entries are estimator variances; off-diagonal entries are covariances, not correlations until divided by the relevant standard deviations. The formula assumes fixed full-column-rank A and fixed known SPD V. Estimating V from the same data makes G random, so the plug-in inverse is not automatically the exact sampling covariance.

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Covariance of the estimator (continued)

For BLUE, $GA = I$ proves $\mathbb{E}[Gb] = x$. Any other linear estimator unbiased for every parameter is $Hb = (G + D)b$, with $DA = 0$. Since $GV = (A^T V^{-1} A)^{-1} A^T$, $GVD^T = 0$, hence $HVH^T - GVG^T = DVD^T \succeq 0$. Minimum covariance means PSD order, not entrywise order. Normality is unnecessary; exact confidence intervals and tests require additional distributional assumptions.

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Conditioning of weighted least squares

Avoid explicitly inverting V. For uncorrelated errors, whiten equation by equation.

$$\tilde{A} = V^{-1/2} A, \qquad \tilde{b} = V^{-1/2} b.$$

Then apply ordinary least squares to $\tilde{A}, \tilde{b}$.

Forming $\tilde{A}^T \tilde{A}$ squares the 2-norm condition number; inversion itself is not the source of the squaring. For general SPD $V = LL^T$, use triangular solves to form $\tilde{A} = L^{-1} A, \tilde{b} = L^{-1} b$, then QR or SVD. Large sparse problems may use matrix-free operations.

Shrinkage, regularization, or dimension reduction can handle uncertain or singular covariance estimates, but their changes to the statistical model must be stated.

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Recursive least squares with new observations

Let the old WLS estimate be $\widehat{x}_0$, with error covariance $W_0 \succ 0$. New data satisfy $b_1 = Hx + \eta$, with $\mathbb{E}[\eta] = 0$, $\mathrm{Cov}(\eta) = R \succ 0$, and new errors uncorrelated with the old data errors. Add the two weighted normal equations.

$$W_1^{-1} = W_0^{-1} + H^T R^{-1} H,$$
$$W_1^{-1}\widehat{x}_1 = W_0^{-1}\widehat{x}_0 + H^T R^{-1} b_1.$$

Subtract the first coefficient equation times $\widehat{x}_0$ from the second to obtain

$$\widehat{x}_1 = \widehat{x}_0 + K(b_1 - H\widehat{x}_0), \quad K = W_1 H^T R^{-1}.$$

The innovation $b_1 - H\widehat{x}_0$ is the difference between new observations and old predictions. To avoid repeated inversions, compute

$$S = R + HW_0H^T, \quad K = W_0H^TS^{-1}, \quad W_1 = W_0 - W_0H^TS^{-1}HW_0.$$

Recursive least squares with new observations (continued)

To check the gain identity, $(W_0^{-1} + H^T R^{-1} H)W_0 H^T S^{-1} = H^T R^{-1}(R + HW_0H^T)S^{-1} = H^T R^{-1}$. Multiplication by W_1 gives $K = W_1 H^T R^{-1}$. Multiplying the final covariance formula by W_1^{-1} verifies it similarly. The subtracted matrix is PSD, so $W_1 \preceq W_0$; under this model, new data do not increase linear-estimation uncertainty.

This is fixed-state recursive least squares and a simplified Kalman update. Dynamic filtering additionally needs state transitions, process errors, and a prediction step.

Checking a recursive update numerically

Estimate a constant x. Old observation $b_0 = 10$ has variance 1, giving $\widehat{x}_0 = 10, W_0 = 1$. With $H = 1$, add two mutually uncorrelated new observations in sequence.

Step	New observation and variance	$S = R + W$	$K = W/S$	New estimate	New variance
1	$b_1 = 13, R_1 = 9$	10	$1/10$	$10 + (1/10)(3) = 103/10$	$9/10$
2	$b_2 = 9, R_2 = 1$	$19/10$	$9/19$	$103/10 + (9/19)(9 - 103/10) = 184/19$	$9/19$

The batch precision-weighted average is

$$\widehat{x} = \frac{10 + 13/9 + 9}{1 + 1/9 + 1} = \frac{184}{19}, \quad \operatorname{Var}(\widehat{x}) = \frac{1}{1 + 1/9 + 1} = \frac{9}{19}.$$

Both procedures agree exactly. A copy of an old observation has correlated error; treating it as independent counts the same information twice and understates uncertainty.

Robustness and model assumptions

WLS depends on its error model. Its assumptions include

- known or zero error means;
- known or credibly estimated covariance V;
- a correct linear conditional-mean model Ax;
- a joint covariance capturing dependence;
- full column rank for parameter identification.

Heavy tails without finite second moments invalidate covariance-based analysis. Outliers or omitted nonlinear terms can also make WLS unsuitable. Correlated errors are permitted when modeled correctly in V. An L_1 loss penalizes large residuals linearly and reduces sensitivity to vertical outliers, but does not automatically resist high-leverage design points. Linear least absolute deviations can be formulated as an LP.

Time-series errors often have nondiagonal covariance. Replacing it by $V = I$ changes both weights and uncertainty. Correlation is not a causal effect.

Chapter check

You can now do the following.

- Separate observed statistics from model expectations and densities.
- Derive variance identities and the unbiased sample denominator.
- Compute marginals, covariance, and correlation.
- Prove covariance positivity through variances of linear combinations.
- Derive transformed covariance.
- Normalize Gaussian density using spectral coordinates.
- Derive weighted normal equations.
- Derive estimator covariance and prove BLUE without assuming Gaussianity.
- Use stable whitening, QR, or SVD and report conditioning.

The course now connects vector spaces, projections, factorizations, applications, numerical reliability, and statistical estimation.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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